

MTEP22

MTEP22 REPORT

Highlights

- 382 Projects with an estimated \$4.3 billion investment
- \$31 billion in projects constructed in the MISO region since 2003
- Generator Interconnection Queue cycle added a record 956 applications submitted for nearly 171 GW of potential installed capacity

MTEP22 Table of Contents

CHAPTER 1: MTEP OVERVIEW	4
1.1 MISO Overview	4
1.2 MISO Transmission Planning Process	10
1.3 MTEP22 Investment Summary	16
Chapter 2: Portfolio Evolution	22
2.1 A Call to Action	22
2.2 Retirement Trends and Future Outlook	23
2.3 Current State of the Queue	25
2.4 Resource Outlook	27
2.5 Planning for the Future	32
Chapter 3: Regional and Interregional Planning Studies	33
3.1 Long Range Transmission Planning – Tranche 2 Introduction	33
3.2 Interregional Studies	36
3.2.1 MISO-SPP Joint Targeted Interconnection Queue (JTIQ) Study Results	36
3.2.2 MISO-SPP Coordinated System Planning	39
3.2.3 MISO-PJM Coordinated System Planning	40
Chapter 4: Reliability Studies	42
4.1 Reliability Assessment and Compliance	42
4.2 Project Justifications – Central Region	45
Central Region Overview	45
4.2.1 American Electric Power Service Corporation (AEP)	49
4.2.2 Ameren Illinois	49
4.2.3 Ameren Missouri	52
4.2.4 Big Rivers Electric Corporation (BREC)	57
4.2.5 City of Columbia, MO (CWLD)	58
4.2.6 City of Springfield, IL (CWLP)	59
4.2.7 Duke Energy Corporation (DEI)	59
4.2.8 GridLiance Heartland LLC (GLH)	68
4.2.9 Henderson Municipal Power & Light (HMPL)	69
4.2.10 Hoosier Energy REC, Inc. (HE)	69

4.2.11 Indianapolis Power & Light Company (IPL)	70
4.2.12 Northern Indiana Public Service Company (NIPSCO)	71
4.2.13 Pioneer Transmission, LLC	73
4.2.14 Prairie Power, Inc. (PPI)	73
4.2.15 Republic Transmission, LLC (RTx)	77
4.2.16 Southern Indiana Gas & Electric Company (SIGE)	78
4.2.17 Southern Illinois Power Cooperative (SIPC)	79
4.2.18 Wabash Valley Power Association, Inc. (WVPA)	79
4.3 Project Justifications – East Region	82
East Region Overview	82
4.3.1 ITC Transmission (ITCT)	85
4.3.2 Michigan Electric Transmission Co. (METC)	91
4.3.4 Wolverine Power Supply Cooperative Inc. (WPSC)	98
4.3.5 Michigan Public Power Agency (MPPA)	99
4.3.6 Lansing Board of Water and Light (LBWL)	99
4.3.7 Michigan South Central Power Agency (MSCPA)	99
4.4 Project Justifications – South Region	100
South Region Overview	100
4.4.1 Arkansas Electric Cooperative Corporation (AECC)	103
4.4.2 City of Alexandria (AXLA)	104
4.4.3 CLECO Power LLC (CLEC)	104
4.4.4 Cooperative Energy (SMEPA)	106
4.4.5 East Texas Electric Cooperative (ETEC)	106
4.4.6 Entergy Arkansas LLC (EAL)	106
4.4.7 Entergy Louisiana LLC (ELL)	108
4.4.8 Entergy Mississippi LLC (EML)	112
4.4.9 Entergy New Orleans LLC (ENO)	114
4.4.10 Entergy Texas, Inc. (ETI)	114
4.4.11 Lafayette Utilities System (LAFA)	117
4.5 Project Justifications – West Region	118
West Region Overview	118
4.5.1 American Transmission Company (ATC)	121
4.5.2 Cedar Falls Utilities (CFU)	134
4.5.3 Central Minnesota Municipal Power Agency (CMMPA)	134

4.5.4 City of Ames, IA (COA) _____	135
4.5.5 Dairyland Power Cooperative (DPC) _____	135
4.5.6 Great River Energy (GRE) _____	135
4.5.7 ITC Midwest (ITCM) _____	144
4.5.8 MidAmerican Energy Company (MEC) _____	152
4.5.9 Minnesota Municipal Power Agency (MMPA) _____	158
4.5.10 Minnesota Power (MP) _____	159
4.5.11 Missouri River Energy Services (MRES) _____	164
4.5.12 Montana-Dakota Utilities Co. (MDU) _____	165
4.5.13 Muscatine Power and Water (MPW) _____	166
4.5.14 Northwestern Wisconsin Electric (NWECE) _____	166
4.5.15 Otter Tail Power Company (OTP) _____	166
4.5.16 Rochester Public Utilities (RPU) _____	167
4.5.17 Southern Minnesota Municipal Power Agency (SMMMPA) _____	167
4.5.18 Wilmar Municipal Utilities (WMU) _____	168
4.5.19 WPPI Energy (WPPI) _____	168
4.5.20 Xcel Energy (Northern States Power) _____	168
Legal Disclaimer	173

CHAPTER 1: MTEP OVERVIEW

1.1 MISO Overview

Midcontinent Independent System Operator (MISO) is an independent, not-for-profit organization that delivers safe, cost-effective electric power across 15 U.S. states and the Canadian province of Manitoba. Forty-two million people depend on MISO to generate and transmit the right amount of electricity every minute of every day. MISO is committed to delivering electricity reliably, dependably, and efficiently. In addition to managing the power grid within our region, MISO administers the buying and selling of electricity, and partners with members and stakeholders to plan the grid of the future.

Scope of Operations

Generation Capacity

- 189,421 MW (market)
- 205,177 MW (reliability)

Historic Summer Peak Load (set July 20, 2011)

- 127,125 MW (market)
- 130,917 MW (reliability)

Historic Winter Peak Load (set January 6, 2014)

- 109,336 MW (market)
- 117,903 MW (reliability)

Transmission

- Approximately 72,000 miles
(Reliability Footprint)

Balancing Authorities

- 38 Local Balancing Authorities in MISO

Network Model

- 296,915 SCADA data points
- 9,852 generating units

Registered Wind	29,552 MW
Registered In-Service Wind Generation Capacity	28,646 MW
Registered Solar Generation Capacity	3,638 MW
Registered In-Service Solar Generation Capacity	1,872 MW

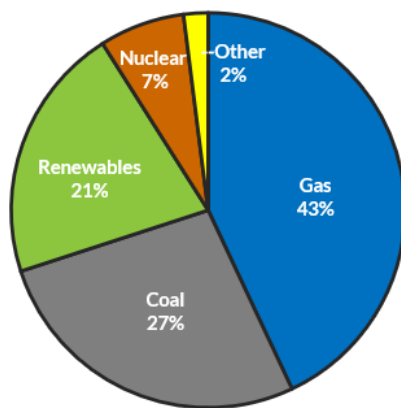
Markets Overview

MISO manages one of the world's largest energy and operating reserve markets using security-constrained economic commitment and dispatch of generation.

The Energy and Operating Reserves Market includes a Day-Ahead Market, a Real-Time Market, and a Financial Transmission Rights Market. These markets are operated and settled separately.

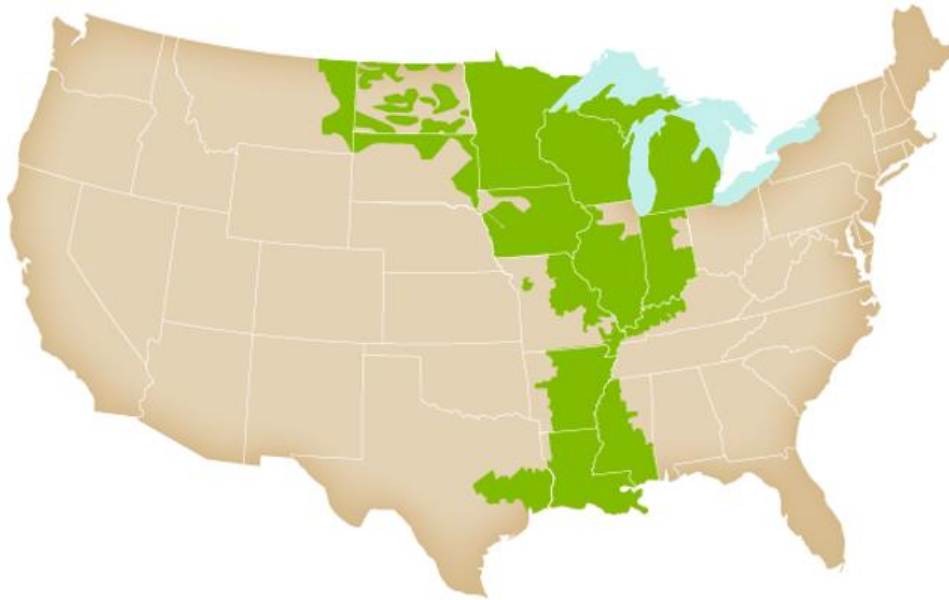
- \$40 billion annual gross market charges
- 510 Market Participants that serve approximately 42 million people

Market Capacity

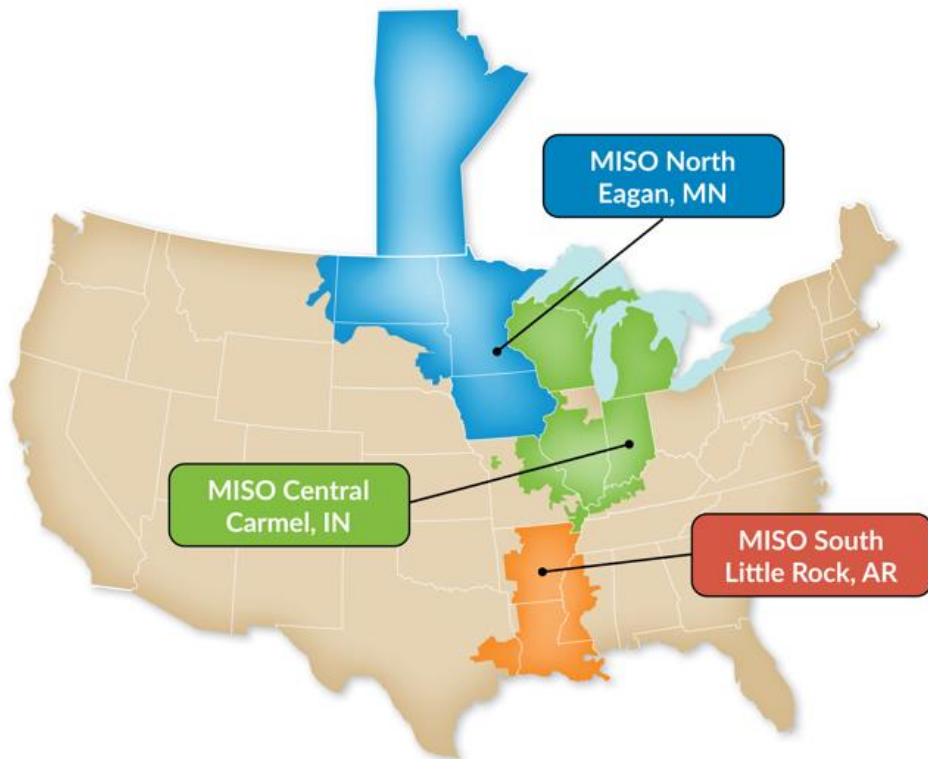


[Corporate data](#) as of June 2022

MISO Market Footprint



MISO Reliability Footprint and location of regional control centers



MISO Transmission Infrastructure Investment

MISO is a not-for-profit organization; it does not own any generation or transmission facilities. MISO strictly manages the generation and flow of electricity across the high-voltage lines within its territory. Through the collaborative efforts of a diverse group of industry participants, MISO manages approximately 72,000 miles of transmission lines across 15 states and the Canadian Province of Manitoba. There are currently 58 registered transmission owning members and 134 registered non-transmission owning members in MISO.

This iteration of the MTEP report, MTEP22, builds and expands on the 18 prior years of projects since 2003 for a total of over \$54.3 billion of investment in the United States (Figure 1.1-1). MISO's proposed new projects for this MTEP cycle are detailed in Section 1.3, Chapter 4, and Appendix A of this report.

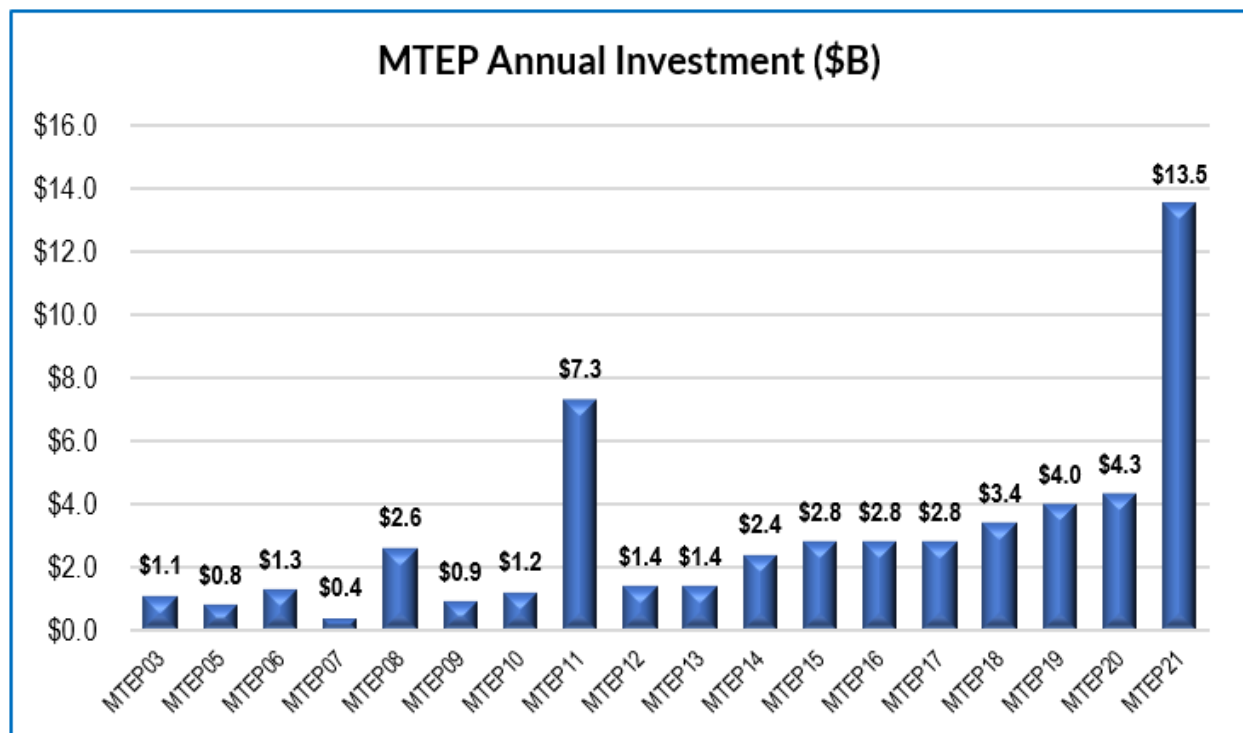


Figure 1.1-1: MTEP annual investment

Highlights in prior MTEP cycles include:

- MTEP06 and MTEP07 were approved in the same calendar year, which accounts for the comparatively small incremental value of projects in MTEP07.
- MTEP11 reflects the approval of the Multi-Value Project portfolio, which accounts for the significantly higher investment totals compared to other MTEPs.
- MTEP14 reflects the inclusion of the new MISO South region projects. A single transmission delivery service project accounts for around 25 percent of the total MTEP14 investment.
- MTEP21 reflects the MTEP21 Addendum approval of the LRTP Tranche 1 portfolio, which accounts for \$10.3 billion.

Consistent growth in infrastructure investment in the MISO footprint over the last several years reflects the predominance in Transmission Owner-driven upgrades to improve efficiency, reliability, and safety in system designs and replace aging assets to make the system more resilient.

MISO's transmission planning responsibilities include the monitoring of previously approved Appendix A projects. MISO surveys all Transmission Owners and Selected Developers every quarter to determine the progress of each project. These [status updates](#) are reported to the MISO Board of Directors and posted quarterly to the MISO Transmission Expansion Plan page at [misoenergy.org](https://www.misoenergy.org/planning/planning/)¹.

MTEP Approved Projects Status

Since MTEP03, \$31.2 billion of investment has gone into service and \$23.1 billion of approved projects are yet to be fully placed into service (Figure 1.1-2).

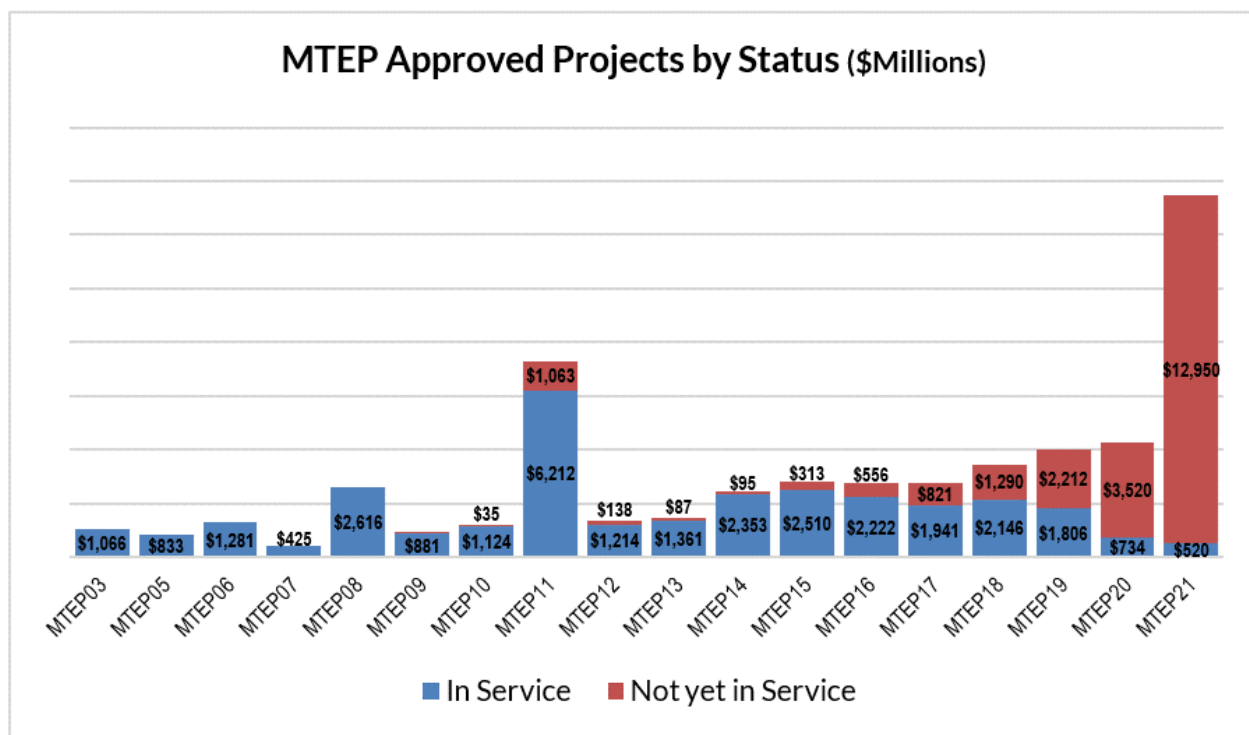


Figure 1.1-2: Appendix A project status (excluding withdrawn)

Active (Not yet in Service) Appendix A Projects - Future Line Miles

There are approximately 6,229 circuit-miles of planned new or upgraded transmission lines projected in the 10-year planning horizon in Appendix A (Figure 1.1-3).

- 4,113 circuit-miles of upgraded transmission line on existing corridors are planned of which 62% are ≤ 230 kV and 38% are ≥ 345 kV.
- 2,116 circuit-miles of new transmission line on new corridors are planned of which 39% are ≤ 230 kV and 61% are ≥ 345 kV.

¹ MISO Transmission Expansion Plan website address: <https://www.misoenergy.org/planning/planning/>

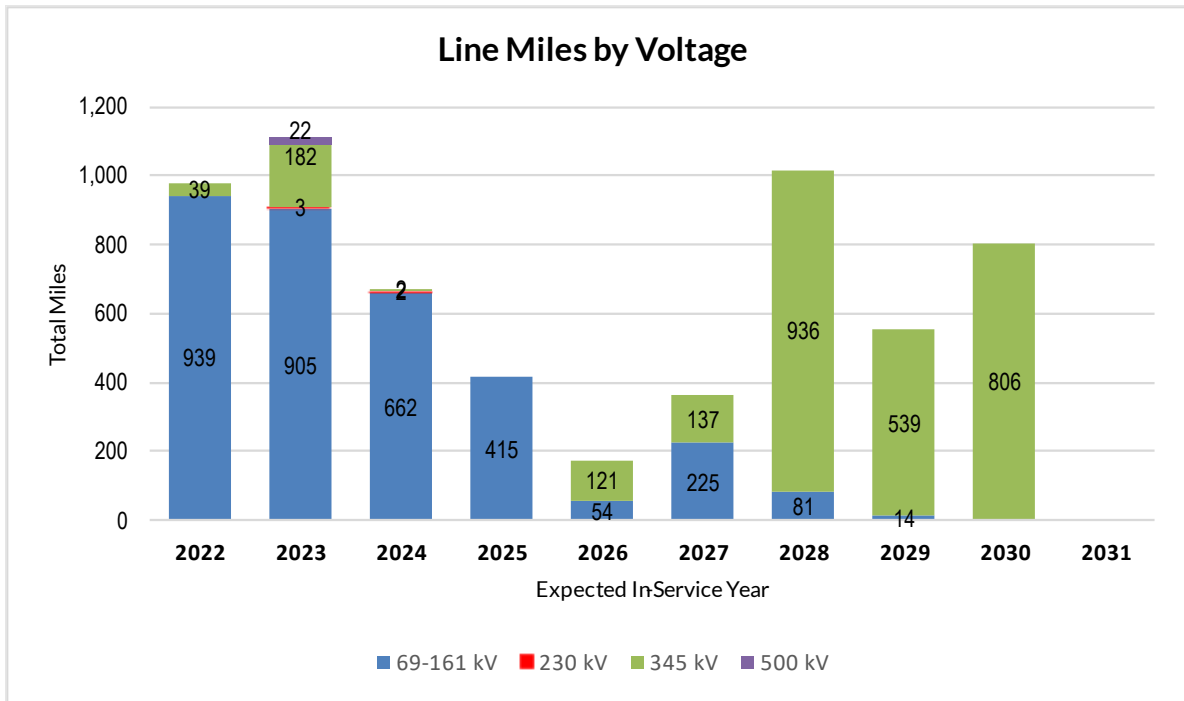


Figure 1.1-3: Active Project circuit line miles by voltage and expected in-service year.

Existing Line Miles Summary

MISO has approximately 68,000 circuit-miles of transferred functional control transmission lines serving as the backbone of the footprint (Figure 1.1-4) in the United States. Currently, the West region holds 40% of total footprint line miles, the South region holds 24%, the Central region holds 21%, and the East holds 14%.

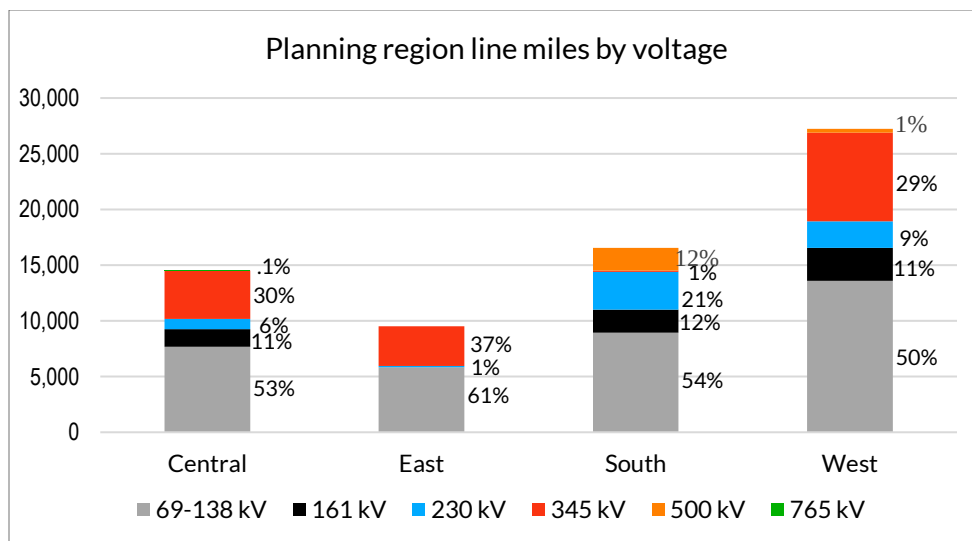


Figure 1.1-4: 2022 in-service transferred circuit miles by voltage class

Transmission Facility Investment

Of the over \$54.3 billion of approved investment that remain active or in-service, \$28 billion of that investment, or 52%, has occurred in the last five MTEP cycles. The MTEP21 LRTP Tranche 1 portfolio accounts for \$10.3 billion of this recent investment. In the first 13 MTEP cycles, the predominant investment was in new line assets at 56% or \$13B cost. There was a shift in investment in the last 5 cycles with the leading investments showing in Substation and Line Upgrades, and new lines representing 16% of the total investment. With the approval of the Tranche 1 portfolio of projects, that investment has now adjusted to 39% of total investment with Line Upgrades at 27% and Substation investment at 26%. Figure 1.1-5 below reflects current asset investment.

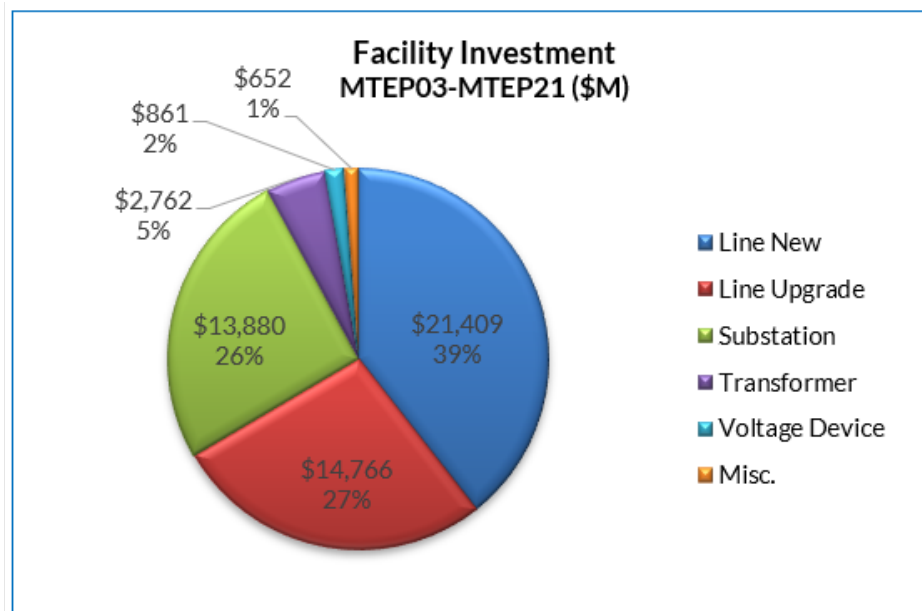


Figure 1.1-5: Appendix A project facility investment dollars in all MTEP cycles.

Full archived files of [previous MTEP Reports](#) can be accessed via the MISO Transmission Expansion Plan page at misoenergy.org.

Planning Guiding Principles

- Develop transmission plans that will ensure a reliable and resilient transmission system that can respond to the operational needs of the MISO region.
- Make the benefits of an economically efficient electricity market available to customers by identifying solutions to transmission issues that are informed by near-term and long-range needs and provide reliable access to electricity at the lowest total electric system cost.
- Support federal, state, and local energy policy and member goals by planning for access to a changing resource mix.
- Provide an appropriate cost allocation mechanism that ensures that costs of transmission projects are allocated in a manner roughly commensurate with the projected benefits of those projects.
- Analyze system scenarios and make the results available to federal, state, and local energy policy makers and other stakeholders to provide context and to inform choices.
- Coordinate planning processes with neighbors and work to eliminate barriers to reliable and efficient operations.

1.2 MISO Transmission Planning Process

A goal of the MTEP report is to satisfy the regulatory requirements as specified in Federal Energy Regulatory Commission Orders and the ISO Agreement. The report provides an annual snapshot with results and recommendations from the continuous planning processes MISO undertakes. MISO's planning process follows established guiding principles to ensure reliability, support policy requirements, and enable a competitive market to benefit all customers (see Figure 1.2-1 planning regions).

MISO will continue to follow federal and state policy as well as monitor fuel prices, plant retirements, and announced member plans for any changing industry trends. The ability not only to meet peak demand, but to move bulk power from resource areas to load centers across the footprint in all hours of the day will be needed to maintain system reliability and improve efficiency with the evolving resource fleet. Regional planning solutions will play an essential role in optimizing the natural and geographic diversity of these resources.

Periodically, the System Planning Committee of the Board of Directors provides input into MISO's Planning Guiding Principles. The most recent review and approval by the Board of Directors occurred in June 2021.

Planning Functions

The planning process includes these functions, which are described in detail in the [Transmission Planning Business Practices Manual](https://www.misoenergy.org/legal/business-practice-manuals/)².

- Model Development
- Generator Interconnection Planning
- Transmission Service Planning
- Cyclical regional expansion planning activities
- Interregional coordination with neighboring transmission planning regions
- System Support Resource studies for unit suspension or retirement
- Transmission-to-Transmission Interconnection
- Load Interconnections
- Focus Studies

MISO addresses current dramatic changes in the projected resource mix in its current strategic vision, which focuses on a dramatic shift in member utilities' resource mix, increased electrification and increased incidence of severe weather events. The regional

² <https://www.misoenergy.org/legal/business-practice-manuals/>

Reliability Imperative is the shared responsibility of MISO, states, and utilities to address these issues in “minding the gap” between today’s and the future resource mix which is driven by economics, state and utility policies, our members’ stated carbon-free goals, and consumer preferences.

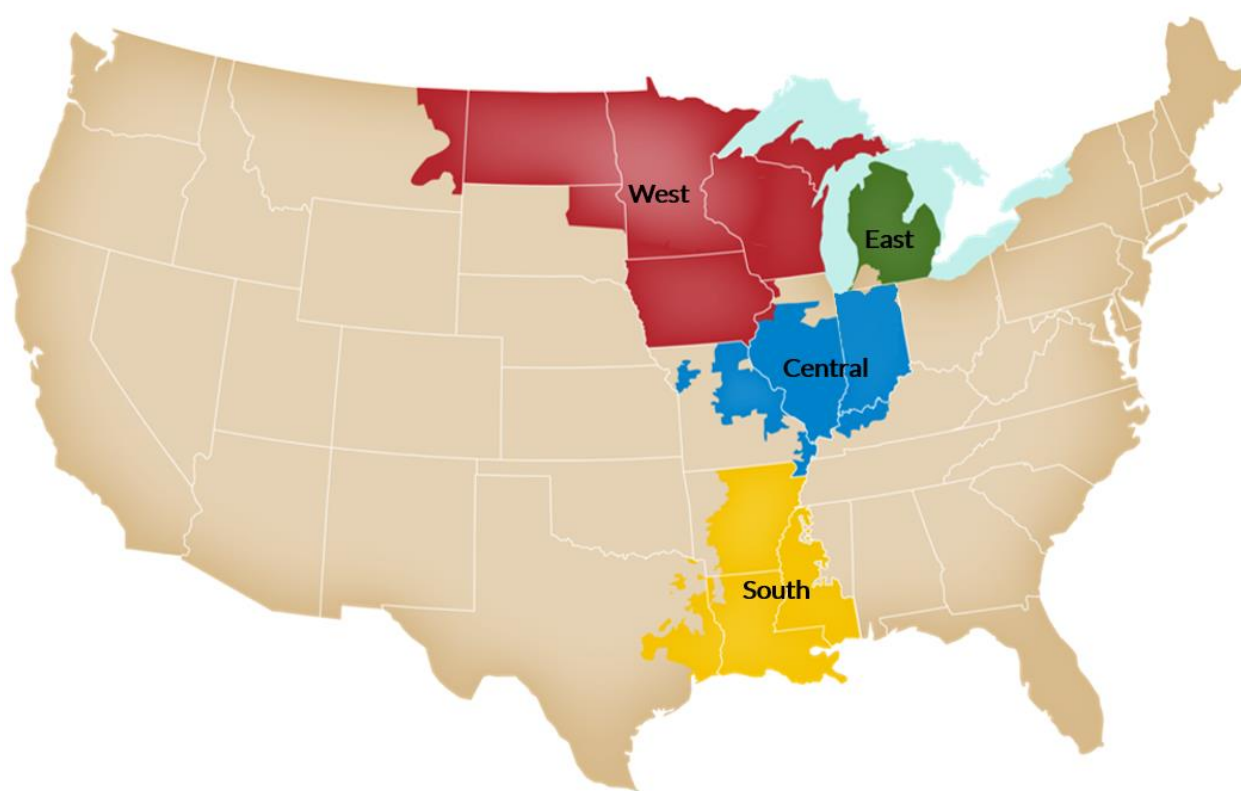


Figure 1.2-1: MISO footprint planning regions

Project Input and Stakeholder Coordination

Each planning cycle commences with regional model development; identification of potential expansions from the local planning processes of the Transmission Owners; identification and selection of transmission needs driven by public policy requirements to be included as transmission issues; and identification by stakeholders or MISO staff of potential expansions that address the transmission issues. Each cycle concludes with recommendations to the MISO Board of recommended solutions to the transmission issues evaluated.

Transmission Owner plans developed through local planning processes are included in the beginning of each regional planning cycle as potential solutions to local transmission issues identified by the Transmission Owners.

MISO’s regional planning process makes evaluations – with stakeholder input from the Sub-regional Planning Meetings, the Planning Subcommittee, and the Planning Advisory Committee – throughout the cycle to develop expansion plans to meet the needs of the system. This multi-party collaborative process allows analysis of all projects with regional and inter-regional impact for their combined effects on the

Transmission System. Moreover, the design of this collaborative process ensures that the MTEP addresses transmission issues within the applicable planning horizon in an efficient and cost-effective manner, while considering the input of stakeholders.

MISO's planning process ensures local needs are integrated with regional requirements

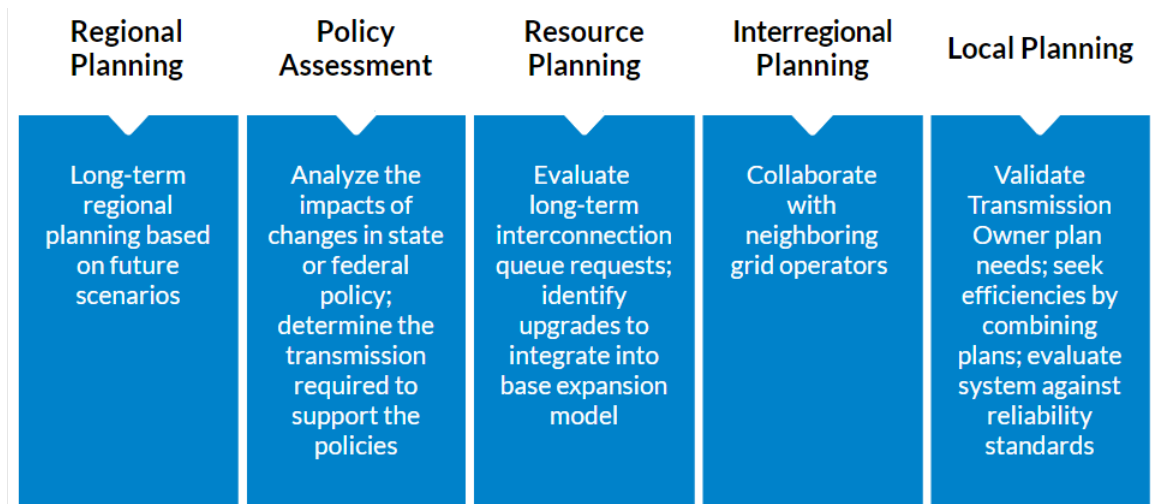


Figure 1.2-2: The MISO Value-Based Planning approach

Key Planning Cycle Milestones

While following the MISO value-based planning approach (Figure 1.2-2), key milestones in the typical MTEP development process include requirements and timelines for data submittal, review, and comment at each of these milestone points as described in the [Transmission Planning Business Practices Manual](#) posted on the MISO public website.

- Model development
- Identification of transmission issues
- Testing models against applicable planning criteria
- Development of possible solutions to identified transmission issues
- Selection of preferred solution
- Determination of funding and cost responsibility
- Monitoring progress on solution implementation

Planning Analysis Methods

Planning analyses performed by MISO test the transmission system under a wide variety of conditions using standard industry applications to model steady state power flow, angular and voltage stability, short-circuit, and economic parameters, as determined appropriate by MISO to be compliant with applicable criteria and the Tariff. MISO collaborates with Transmission Owners, other transmission providers, transmission customers, and other stakeholders to develop appropriate planning models that reflect expected system conditions for the planning horizon.

[Models](#) are available to stakeholders with security measures as provided for in the Transmission Planning Business Practices Manual. MISO provides opportunity for stakeholders to review and comment on the posted models before commencing planning studies.

Long Range Transmission Planning

Long Range Transmission Planning (LRTP) is an essential element of planning the regional grid to be reliable and efficient over short and long-range planning horizons. LRTP efforts are launched periodically when needed to address significant changes to future conditions that the grid must be prepared to address. LRTP integrates with other shorter term planning processes to provide for the long-range facility development needs (Figure 1.2-3). Each process feeds the others to cover the entire planning horizon. The processes are sequential and the output of one becomes the baseline to the other. Long Range Transmission Planning results in projects that are backbone facilities needed to move bulk power between areas of the region. While they provide for a reliable and efficient grid based on forecast resource developments, they are not intended to resolve all connection issues associated with precise siting of future generation.

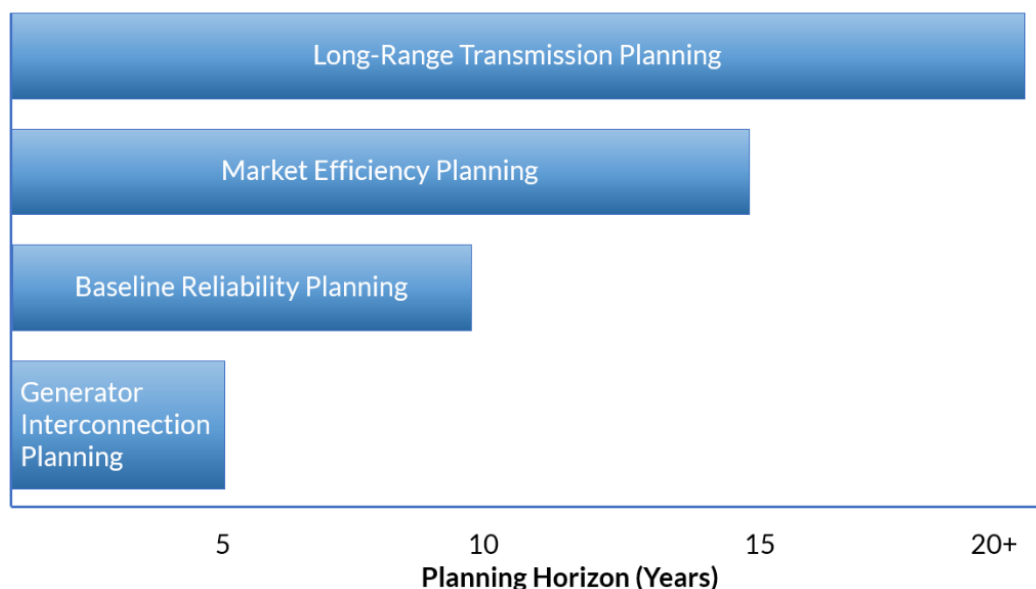


Figure 1.2-3: Long Range Transmission Planning target compared to other planning objectives

MISO staff work to ensure that the future regional transmission system will meet demand levels ‘in all hours’ under conditions defined by MISO Futures. The current iteration of the LRTP efforts is based on the most recent set of Futures developed with stakeholders. Futures assumptions are provided in Appendix E of this MTEP22 Report. In addition, the [MISO Futures Report](#) was published in 2021 and available on misoenergy.org after an 18-month collaboration between MISO and its stakeholders.

Long Range Transmission Planning follows MISO's well-established 7 step value-based planning process:

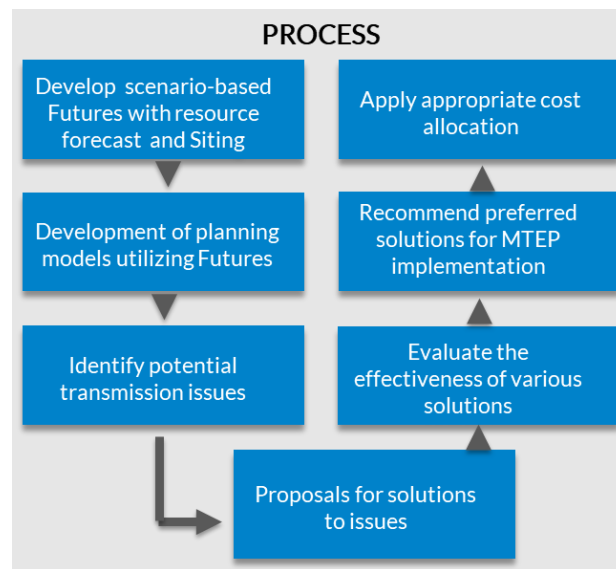


Figure 1.2-4: MISO's 7-step planning process

MISO is working to identify potential grid needs in support of the resource transformation underway and as contemplated under the MISO Futures. This extensive stakeholder process includes monthly workshops, periodic discussions at the Planning Advisory Committee, plus additional stakeholder meetings addressing cost allocation through the Regional Expansion Criteria and Benefits Working Group. Project recommendations resulting from this process will be then presented for Board of Director review and approval over several MTEP cycles as analyses proceed and recommendations are developed.

Details of MISO's Long Range Transmission Planning study progress are summarized in Section 3.1 of the MTEP22 Report.

Project Approval

MISO staff formally recommends a set of projects to the MISO Board of Directors for review and approval after MISO completes an independent review of all proposed projects and addresses any stakeholder feedback received. These projects make up Appendix A of the MTEP report and represent the preferred solutions to the identified transmission needs of the MISO transmission planning process.

Proposed transmission upgrades with sufficient lead times are included in Appendix B for further review in future planning cycles.

Details of the project proposal process and transmission projects reviewed this cycle are summarized in Section 1.3 and Chapter 4 of the MTEP22 report.

Interregional Coordination and Planning Studies

On an annual basis MISO works with the neighboring transmission planning regions, Southwest Power Pool (SPP) and PJM Interconnection (PJM), to identify issues on the seams, perform studies, and jointly evaluate transmission solutions that may be more efficient or cost effective than a corresponding regional solution. While MISO has a separate Joint Operating Agreement (JOA) with both SPP and PJM that details specific processes and criteria, the high-level interregional coordination activities are similar on each seam:

- 1) Exchange modeling data and other system information (typically performed in Q4).
- 2) Review identified issues on the seam (typically performed in Q1).
- 3) Evaluate whether to perform an interregional study based on the identified issues.

MISO performs joint coordinated system plan (CSP) studies with SPP and PJM on a regular basis, in accordance with the timelines and frequencies dictated in their respective JOAs. A CSP study may have a targeted scope or a more complex scope requiring a longer study period, and can include reliability, economic and/or public policy issues. All interregional issues and CSP study efforts are coordinated through a public Interregional Planning Stakeholder Advisory Committee (ISPAC) consisting of representatives and interested parties from each RTO community.

In addition to the joint study efforts with SPP and PJM, MISO performs studies as needed with neighboring entities of the Southeastern Regional Transmission Planning (SERTP) group and the Independent Electricity System Operator of Ontario (IESO). While the study process is less formal, MISO and these entities still meet regularly to review interregional issues and possible areas of collaboration.

Details on planning procedures, on-going studies and stakeholder meetings can be found on the [Interregional Coordination](https://www.misoenergy.org/interregional-coordination) page of the MISO public website ([misoenergy.org](https://www.misoenergy.org)).

1.3 MTEP22 Investment Summary

The MTEP22 cycle proposes 382 new Appendix A projects as justified in this MISO Transmission Expansion Plan and represents \$4.3 billion in transmission infrastructure investment for the MISO region.

Overview of Tariff-defined Project Types

- **Baseline Reliability Project (BRP)** - Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable National Electric Reliability Organization reliability standards and reliability standards adopted by Regional Reliability Organizations, and applicable within the Transmission Provider Region. Baseline Reliability Project costs are allocated to the local Transmission Pricing Zone(s) and recovered through Attachment O by the Transmission Owner(s) developing the projects.
- **Generator Interconnection Project (GIP)** - Projects are New Transmission Access Projects that are associated with interconnection of new generation or the capacity modification of existing generation. Costs are primarily paid for by the interconnection customers with certain exceptions as specified in Attachment FF. Costs of network upgrades rated at 345 kV and above are eligible for 10 percent cost recovery on a system-wide basis.
- **Market Efficiency Project (MEP)** - Projects meet Attachment FF requirements for reduction in market congestion and are eligible for regional cost allocation. Projects qualify as Market Efficiency Projects based on cost and voltage thresholds and are developed to produce a benefit-to-cost ratio of 1.25 or greater. Costs are distributed to benefiting pricing zones, in accordance with Attachment FF of the Tariff.
- **Targeted Market Efficiency Project (TMEP)** - Projects are designed to alleviate historical market-to-market congestion between MISO and PJM Interconnection, while meeting certain cost and construction requirements. The costs of Targeted Market Efficiency Projects are allocated first between MISO and PJM Interconnection by the ratio of each RTO's Day-Ahead and Excess Congestion Fund congestion, offset by historical market-to-market payments. The MISO share of costs for the project is then allocated to beneficiaries using historical nodal load congestion contribution data.
- **Multi-Value Project (MVP)** - Projects meet Attachment FF requirements to provide regional or sub-regional public policy, economic and/or reliability benefits. Costs are shared with loads and export transactions in proportion to metered MWh consumption or export schedules.
- **Other** - Projects included in MTEP22 which do not qualify as Baseline Reliability Projects, New Transmission Access Projects, Targeted Market Efficiency Projects, Market Efficiency Projects, or Multi-Value Projects.
- **Transmission Delivery Service Project (TDSP)** - Projects are required to satisfy a transmission service request. The costs are generally assigned to the requestor.
- **Market Participant Funded Project (MPFP)** - Projects are defined as Network Upgrades fully funded by one or more market participants but owned and operated by an incumbent Transmission Owner.

Overview of MTEP22 Projects

MTEP22 Appendix A Projects

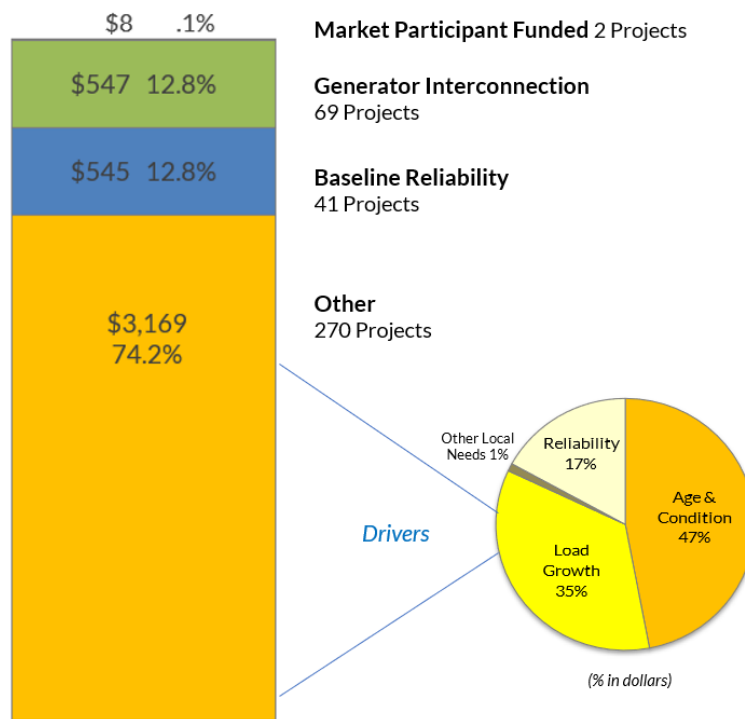


Figure 1.3-1: Appendix A New Project Investment (data as of 9-15-2022)

Of the 382 new Appendix A projects proposed in MTEP22 (Figure 1.3-1), 2 are Market Participant Funded Project, 69 are Generator Interconnection, 41 are Baseline Reliability, and 270 are in the Other project category. These percentages are similar to MTEP21 (without LRTP Tranche 1).

The majority of Other projects address localized reliability issues that are due to aging transmission infrastructure, load serving needs, and local non-baseline reliability needs that are not dictated by NERC and regional reliability standards.

The new projects recommended for approval in MTEP22 Appendix A are broken down by region and project type (Table 1.3-1 and Figure 1.3-3).

Planning Region	BRP	GIP	MPFP	Other	Total
Central	\$131 M	\$270 M		\$1,111 M	\$1,512 M
East	24 M	\$90 M		\$325 M	\$439 M
South	\$72 M	\$117 M		\$817 M	\$1,006 M
West	317 M	\$69 M	\$8 M	\$916 M	\$1,310 M
Total	\$545 M	\$547 M	\$8 M	\$3,169 M	\$4,268 M

Table 1.3-1: MTEP22 Appendix A new project investment by category and planning region

MTEP22 Regional Investment by Project Category

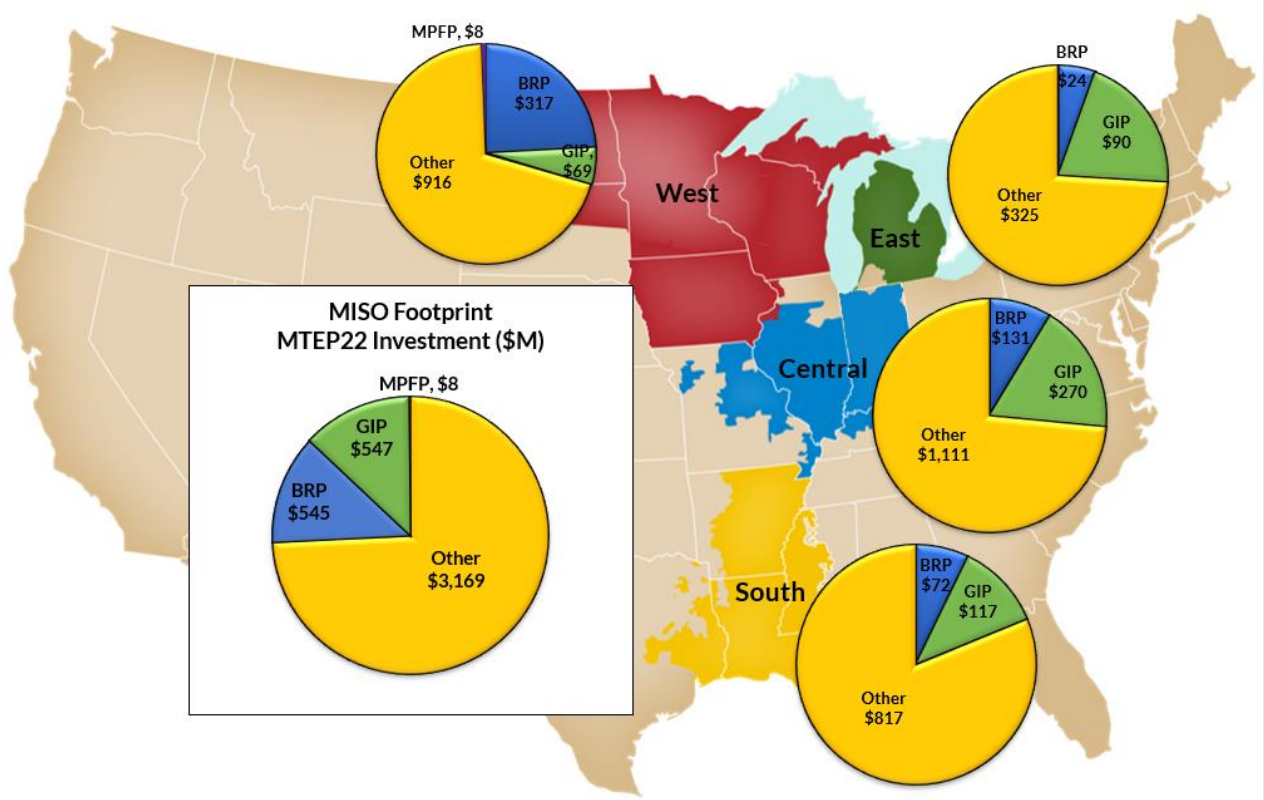


Figure 1.3-3: Regional MTEP22 investment by project category (data as of 9-15-22)

Of the 382 projects proposed for MTEP22, 72% percent are projected to go into service within the next three years (Figure 1.3-4).

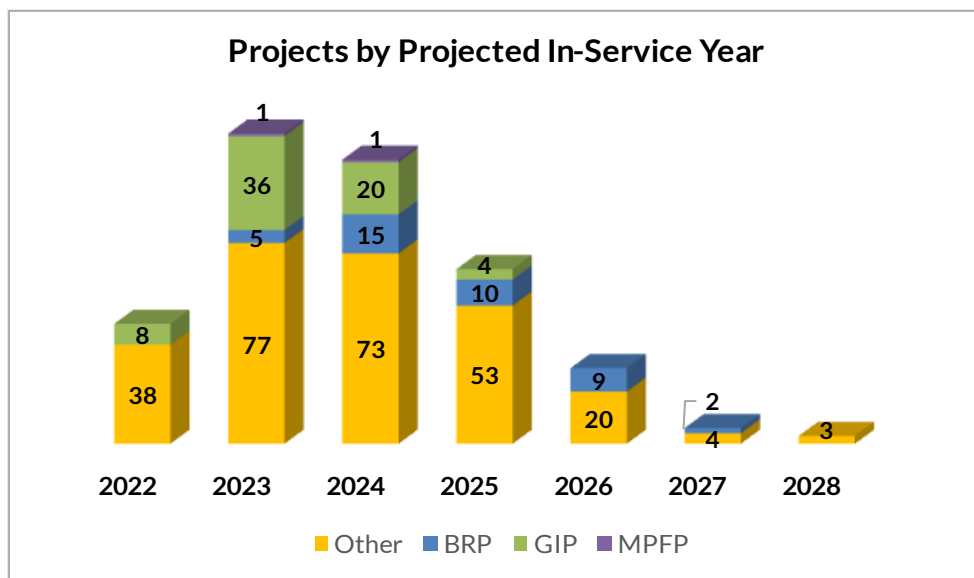


Figure 1.3-4: MTEP22 Projects by In-Service Year (data as of 9-15-22)

New Appendix A projects are spread over 14 states, with eight states scheduled for more than \$200 million in new investment (Figure 1.3-5). These geographic trends vary greatly year to year as local planning dictates blanket asset renewal programs or as existing transmission capacity in other parts of the system is consumed and new build becomes necessary.

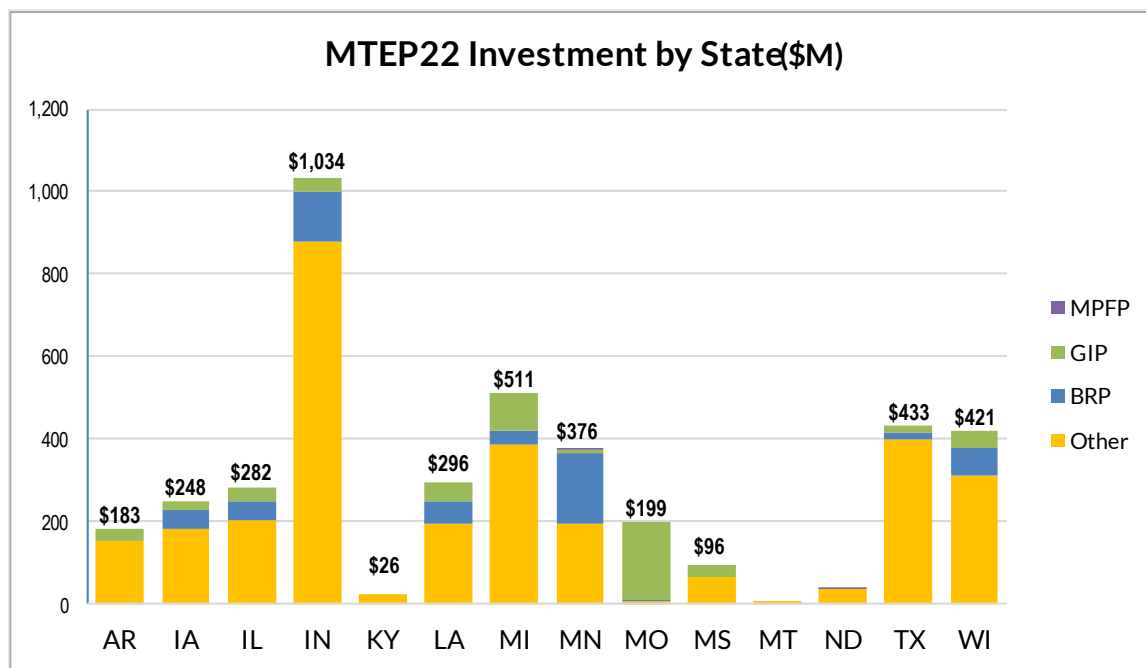


Figure 1.3-5: MTEP22 Appendix A investment categorized by state, excludes multi-state blanket programs, (data as of 9-15-22)

Facility Type

Each MTEP project is composed of one or more facilities, where each facility represents an individual element of the project. Examples of facilities include substations, transformers, voltage devices, circuit breakers or various types of transmission lines (Figure 1.3-7).

The largest share of facility investment in the MTEP22 cycle, 48% percent is dedicated to substation or switching station related construction and maintenance. This includes completely new substations as well as terminal equipment work, circuit breaker additions and replacements. Thirty-five percent is dedicated to line upgrades which includes rebuilds, conversions, and relocations. Ten percent of facility costs are dedicated to new lines on new right-of-way across the MISO footprint.

MTEP22 Transmission Investment by Facility Type

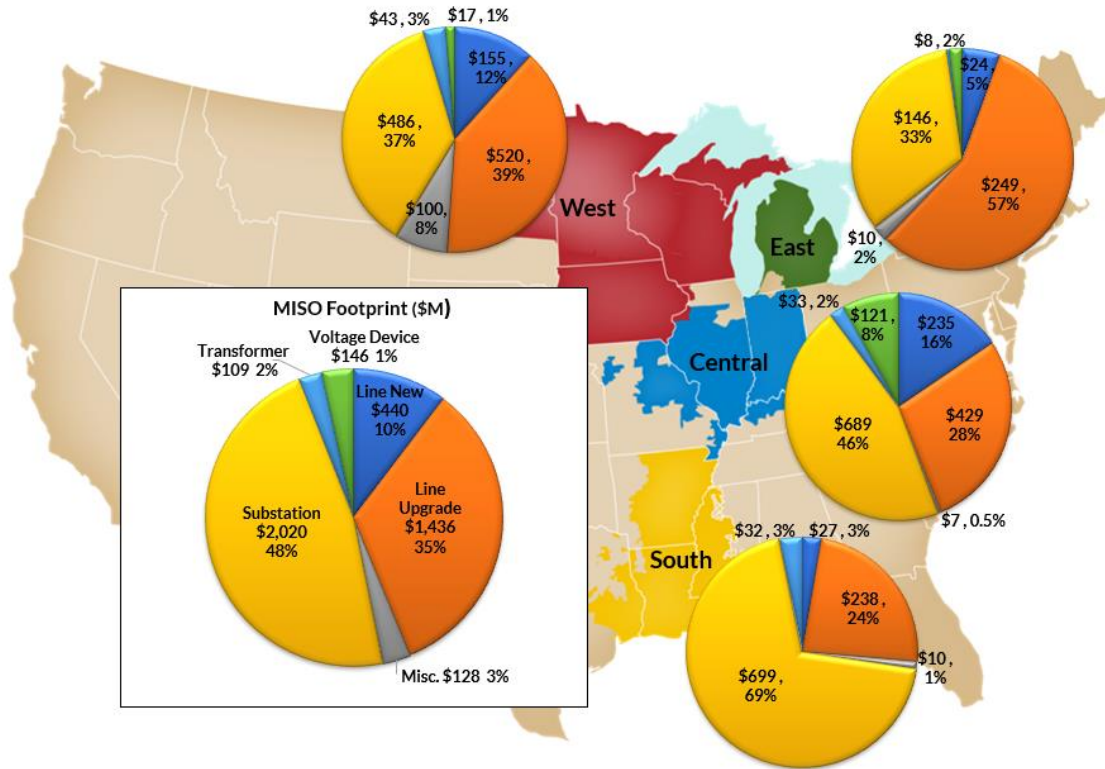


Figure 1.3-7: Facility type for new MTEP22 Appendix A projects by planning region

MTEP22 New and Upgraded Line Miles

MTEP22 Appendix A projects anticipates almost 838 miles of new or upgraded line to go into service through the next six years. Of those, 312 miles are new line mile construction and 526 miles are line upgrades. Eighty-two percent of the total line miles are 161 kV or below. There are 85 new line miles projected at 230 kV or above for the planning horizon shown in Figure 1.3-8.

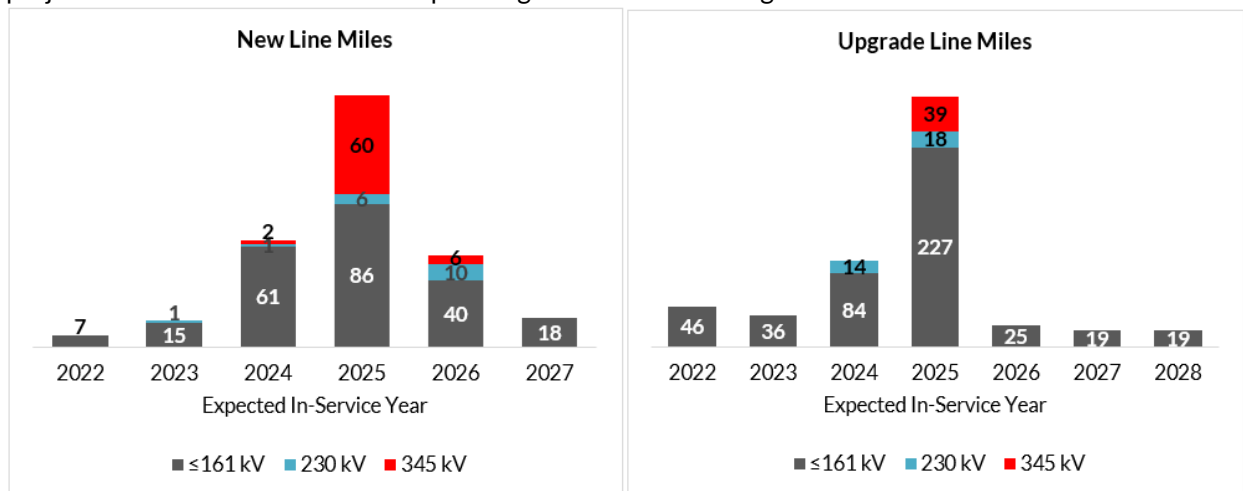


Figure 1.3-8: New and upgraded line miles proposed in MTEP22 Appendix A

Allocation of Costs

MTEP22 includes a total of 33 new cost-share eligible Generator Interconnection Projects (GIPs) for Appendix A. GIP costs are primarily paid for by the interconnecting customer (generator), however, a portion of the costs for certain network upgrades are eligible for regional cost allocation under Attachment FF of the MISO Tariff. Detailed allocations by pricing zone are provided in Appendix A1.

Indicative rates related to past MTEP cost-shared projects are calculated on an annual basis. Please refer to the reports [\(indicative forecasts of annual charges\)](#) posted on the MISO public website³.

MTEP Appendix B

MTEP Appendix B contains all projects that have been validated by MISO as the preferred solution to address an identified system need based on current information and forecasts, but where it is prudent to defer the final recommendation of a solution to a subsequent MTEP cycle.

This generally occurs when the preferred project does not yet need a commitment based on anticipated lead-time and there is still some uncertainty as to the prudence of selecting this project over an alternative project given potential changes in projected future conditions. MTEP Appendix B is limited to Baseline Reliability Projects and Other Projects and will be reviewed by MISO in subsequent cycles.

³ Cost Allocation updates web address: <https://www.misoenergy.org/planning/planning/schedule-26-and-26a-indicative-reports/>

Chapter 2: Portfolio Evolution

2.1 A Call to Action

The role and importance of electricity in people's lives has increased significantly. While people have always relied on the grid to keep them comfortable and safe during heat waves and frigid winter weather, newer factors such as the proliferation of electronic devices, the growth of electric vehicles, and the work-from-home phenomenon ushered in by the COVID-19 pandemic have made the electric system even more important.

At the same time, ensuring uninterrupted delivery of electricity to consumers has become more complex. The industry is experiencing an ongoing evolution of the resource portfolio and more frequent extreme weather events, which can have a significant impact on the grid. The former is not only bringing about a much different fleet makeup, but also a lower reserve margin to manage and mitigate the growing risk.

While the region's generation fleet has been evolving for a number of years, the results of MISO's 2022-2023 Planning Resource Auction (PRA) indicated the change is accelerating. The PRA was expected to demonstrate that the region had sufficient, albeit tight, capacity resources to meet its Planning Reserve Margin Requirement. Instead, four of MISO's 10 Local Resource Zones had capacity shortfalls, which caused auction clearing prices in MISO's North and Central subregions to increase to the maximum value. Shortly following the PRA, the 2023 OMS-MISO Survey, which gathers Load-Serving Entity supply and demand expectations for the forward five-year period, indicated the shortfall would not only continue, but increase each year.

MISO members and states are actively building the grid of the future by putting a large amount of "steel in the ground" in the form of new wind and solar resources. Because of that, the region's level of installed, or "nameplate," capacity has actually *increased* in recent years. So, from that perspective, it may appear as if the grid will have sufficient resources to serve all expected load going forward.

But the crucial question is whether the region will have enough resources with the capabilities and attributes that are needed to keep the lights on during the most challenging times, such as when load ramps up rapidly when the sun is not shining or the wind is not blowing, or when a large fossil power plant unexpectedly goes down. Those capabilities are measured in terms of "accredited" capacity, which reflects when resources are actually expected to be available, as well as how flexible they are to respond to fast-changing grid conditions. And in the MISO Region, accredited capacity is declining because conventional resources that are highly available and flexible are retiring, and the renewables that are being built to replace them have significantly lower accreditation values.

These challenges could be mitigated by building more new resources with high degrees of availability and flexibility such as natural gas power plants and energy storage for example. However, efforts to build some such resources, new gas plants in particular, are facing headwinds due to environmental concerns, and long-range storage is still an emerging technology. Moreover, regulations issued by the U.S. Environmental Protection Agency are contributing to the challenges the region faces by accelerating planned coal and gas retirements and driving new retirements. To exacerbate the situation, load is growing as the economy continues to rebound from COVID, with long-term growth expected from deepening electrification in the transportation, buildings, and industrial sectors. Meanwhile, MISO's Generator Interconnection Queue

currently contains very few resources with the kinds of capabilities and attributes that are needed to address challenges related to availability and flexibility.

In light of these and other complex challenges, it is no longer prudent for the region's utilities to make forward-looking resource decisions in an isolated, siloed manner, assuming they will be able to "lean" on their neighbors if need be. Similarly, MISO cannot solve these problems on its own. Utilities, states, and MISO must all work together in a transparent fashion to ensure the region has sufficient resources with the necessary capabilities and attributes to keep the lights on as the resource mix continues to evolve. One such collaborative effort is the [Regional Resource Assessment \(RRA\) study](#), which aggregates Members' publicly announced resource plans, decarbonization goals, and other information to illustrate what the region's resource mix could look like for 20 years going forward. MISO appreciates the support that its Members and states have shown for the RRA.

The remainder of these chapters provides greater detail on the risks and challenges that the region is facing, and their implications, to help inform the power industry and stakeholders during this dynamic but critical time.

2.2 Retirement Trends and Future Outlook

One aspect of the resource evolution that MISO assists its membership in managing is the retirement of generation facilities, to ensure that the broader MISO footprint and markets remain reliable after resources are removed from service. Through the process articulated in Attachment Y of the MISO Tariff, resource owners submit a request to retire generation resources for MISO approval, which triggers an assessment into the impact that the requested resource would cause once it is retired from service. As a result of these analyses, any reliability issues are addressed through transmission reinforcements or other needed mitigation measures. If the reliability issue cannot be addressed prior to the planned retirement date, MISO may require the resource to remain in service as a system support resource (SSR) until the upgrade is complete, or mitigation is available. As the generation mix continues to evolve, more generating resources are expected to retire, increasing the number of Attachment Y requests MISO receives. This may increase the need for SSR-designated units. As of September 2022, MISO had established one SSR unit to maintain reliability of the region.

In 2022, MISO proposed improvements to the Attachment Y process. These improvements, if accepted, would extend the advance notice timeline from 26 to 52 weeks using a quarterly approach to allow MISO more time to process the increased number of Attachment Y requests. MISO also proposed a quarterly study period. This would better allow for forecasting workload internally. MISO made other proposals around the studies included in the retirement process and the mitigations used in the studies. While MISO has not proposed any new studies for the base reliability study process, the extended advance notice timeline will allow for additional studies as situationally necessary. MISO has proposed to reduce reliance on load shed as a mitigation practice during the studies. Lastly, MISO appreciates the need for greater transparency into the retirement process while maintaining a great deal of confidentiality for its members. As part of the Attachment Y improvements, MISO hopes to communicate the number of requests received by quarter and the number of megawatts requesting suspension or retirement.

Aging coal-fired generating resources have experienced increased retirements in recent years due to cost pressures of operation and competition from gas-fired generation. Renewable generating resources have

become more economically and environmentally attractive sources of generation in recent years, putting further pressure on carbon-based generation. Since 2010, MISO has experienced the retirement of 28.3 GW, of which 20.3 GW was coal-based (Figure 2.2-1). The age of generating facilities retired in 2021 declined to an average of 32 years compared to an average of 44 years in 2011, displaced largely by the interconnection of renewable resources that are more economically viable. Advancements in technology and interest in renewables are expected to continue the current trend.

A trend since 2020 is the utilization of the new Generating Facility Replacement process. This process was approved by FERC in 2019 to allow the owners of an existing facility to use their existing interconnection service to replace the existing generator with a new generating facility at the same injection point without going through the MISO Generator Interconnection queue. Since 2020, MISO has received 23 generator replacement requests to replace a total of 4,259 MW of existing generation, which otherwise would have been retired through the traditional resource retirement process.

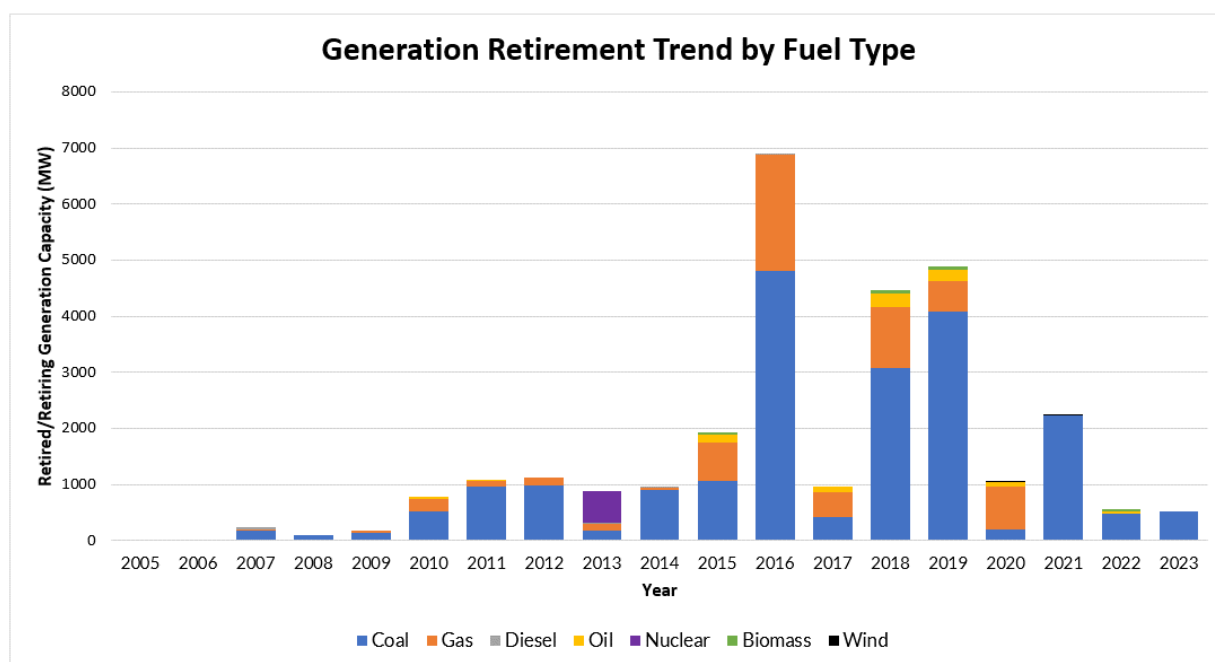


Figure 2.2-1: MW Generation Retirement by Fuel Type

2.3 Current State of the Queue

The MISO Generator Interconnection (GI) queue provides an active and competitive mechanism to enable resource interconnections that will serve future energy and capacity needs. Projects submitted in the annual queue cycle are evaluated by MISO through an iterative study process to determine the reliability impacts and to identify transmission upgrades needed to support resource integration. Project viability is often tied to the costs of network upgrades, with the most viable candidates successfully executing a Generator Interconnection Agreement (GIA).

The Generator Interconnection queue has experienced extremely high volume over the last several years. In 2021, MISO received 487 individual project requests. The preliminary results for 2022 -application period ended on September 15, 2022 – reveal 956 individual project requests at 171GW. Solar, storage, and hybrid applications make up the bulk of the queue.

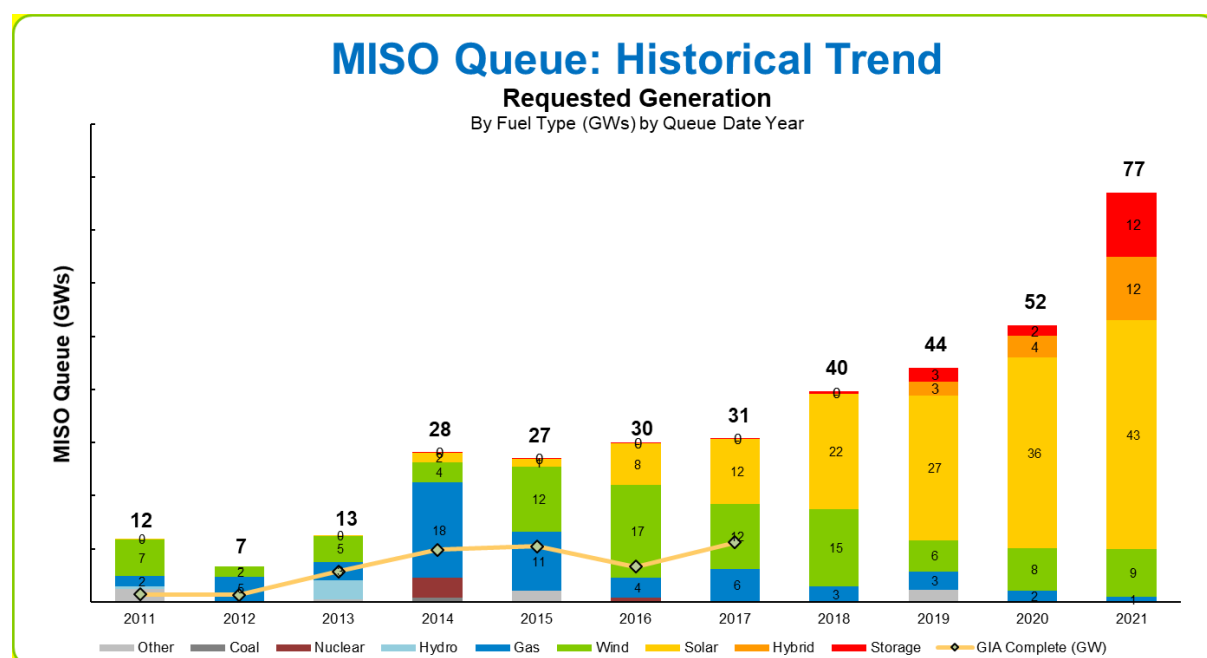


Figure 2.3-1: As of May 1, 2022, the current state of the queue has 806 projects representing 126.3GW of total capacity

The MISO Active Queue by study area and fuel type (Figure 2.3-2) is available on the MISO website under the [GIQ Web Overview](#) link on the [Generator Interconnection Queue](#) page. A list of all active projects can also be reviewed on the page. The five study regions in the GI queue currently have 16 active cycles in various stages of the process from the start of the Definitive Planning Phase (DPP) to GIA negotiations.

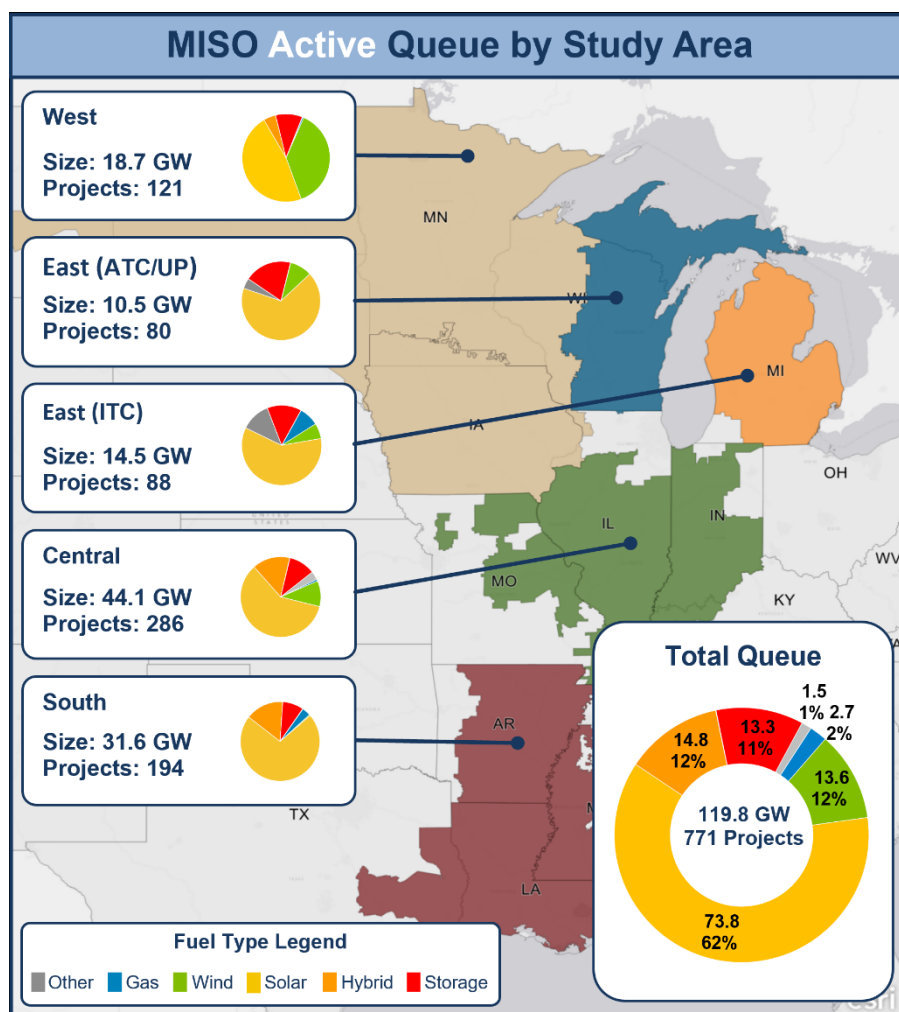


Figure 2.3-2: Active Generation Interconnection Queue by Fuel Type as of September 15, 2022

MISO's 8th Queue Reform Continues to Shrink the Timeline

MISO's queue process is constantly being assessed and refined to make improvements and has undergone eight substantive reforms since being instituted. These reforms have made the queue process quicker, more efficient, and less burdensome to our members. In fact, MISO has the shortest end-to-end queue time among our peer RTOs and ISOs.

In March 2022 FERC approved MISO's latest reform, which is expected to reduce the MISO queue timeline schedule from 505 days to either 373 or 463 days, depending on whether a Network Upgrade Facilities Study (NUFS) is conducted in parallel with or prior to the Generator Interconnection Agreement (GIA) negotiation and execution. In either case, the NUFS must be completed before interconnection takes place. Achieving these timelines is contingent upon neighboring regions completing their Affected System Study on schedule.

In addition to implemented improvements, MISO is tackling several related issues that will further streamline the GI process and MISO's need to bring new resources onto the system quickly:

As of January 1, 2022, fuel changes are not allowed once the Definitive Planning Phase (DPP) I study has begun. This helps to promote projects with more confidence moving forward and reduces the need for model changes during a study period. The change also brings clarity on how to use Surplus Interconnecting Service request and Material Modification changes when considering hybrid units.

Affected Systems Study (AFS) Process: The time between interconnection request and signed Generator Interconnection Agreement is slower than MISO desires, largely due to delays – regularly exceeding 100 days – in completion of the important and necessary Affected Systems Study (AFS) process by MISO's neighbors. MISO is actively working with both SPP and PJM to streamline and speed up the AFS process. Because AFS processes are also critical to maintaining the safety and reliability of the bulk electric system, it is important that the process improvements progress carefully and deliberately.

Joint Operating Agreement reforms with SPP to improve the efficiency and timeliness of the interconnection process by changing the queue prioritization to a “first-ready, first-served” model between MISO and SPP queue resources.

2.4 Resource Outlook

Each year, utilities demonstrate they have sufficient resources to meet projected electricity demand plus reserves during peak times through the Planning Resource Auction (PRA). This year, the MISO North/Central subregion cleared at the highest allowable price (CONE, or Cost of New Entry), for the (Planning Year) PY22-23 due to a lack of resources to meet the regional Planning Reserve Margin Requirement (PRMR). This was driven by a faster-than-expected load recovery post-COVID lockdown and delays in planned generation achieving commercial operation after having an executed GIA. The Planning Reserve Margin, or PRM, was determined to be 8.7% and the actual reserve margin is 7.7% of cleared capacity in the PRA. However additional resources might still be available to contribute to MISO's resource adequacy that did not clear the PRA on account of reasons such as partial availability within the planning year due to a retirement or commencing commercial operations after the start of planning year. Continued action and planning will be critical to ensure resource adequacy into the future years. For 2023, based on the OMS-MISO Survey there is a possibility to have another deficit of committed capacity specifically in the MISO North/Central subregion, while the South is projected to have a surplus. The 2023 peak demand forecast increases at +1GW, the five-year regional demand growth rate remained stable at approximately 0.2% year-over-year. On the supply side, the [2022 OMS Survey](#) continues to indicate that increasing resource adequacy risk can be avoided by firming up the commitments of additional potential resources.

The potential for significant generation fleet transformation has prompted MISO to evaluate how system needs will change and how MISO might adapt its planning, markets, and operations to maintain reliability with aging and retiring units, higher penetration of variable resources, and new load consumption patterns.

MISO is adjusting the Summer only outlook on resource adequacy and is planning on conducting the 2023-24 PY seasonally pending FERC approval. Resource adequacy analysis will need to reflect patterns across the entire year to fully capture the magnitude of risks. Effective dialogue and transparent planning amongst stakeholders will be key to this transformation – identifying needs and working with MISO to develop solutions that work across the footprint. MISO will leverage the forums where discussions are already underway on transmission planning, MISO's resource adequacy construct, and pricing enhancements.

OMS-MISO Survey

In most of the MISO region, load-serving entities with oversight by the applicable state or local jurisdiction are responsible for resource adequacy. The OMS-MISO survey's primary objective is to provide MISO and stakeholders a near to mid-term view of the resource fleet and how it compares to the Planning Reserve Margin Requirement. Market Participants respond to the survey on a generator specific level and designate the UCAP MW amounts for the next 5 years as well as if it is High Certainty(Green) or Low Certainty(Orange). MISO queries the Generator Interconnection queue to add additional projections on Potential New Capacity (Blue). More information can be found in the [2022 OMS-MISO Survey Results](#).

Projected MISO Regional Unforced Capacity (UCAP GW) Position % Above Forecasted Coincident Peak Load against Annual Planning Reserve Margin of ~9%

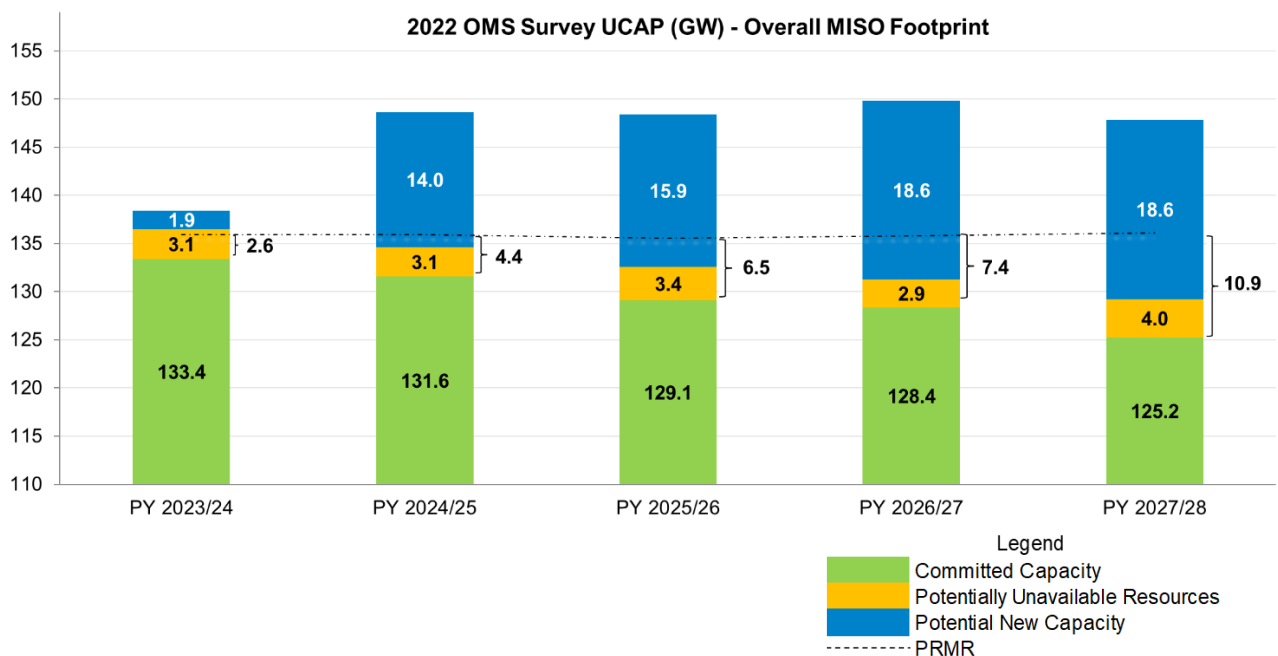


Figure 2.4-1: 2022 OMS-MISO Survey 5-Year Results

This year's OMS-MISO survey shows MISO will have up to a 2.6 GW deficit of committed resources compared to the regional PRMR requirement. If all potential resources materialize in 2023, there could be a surplus of generation.

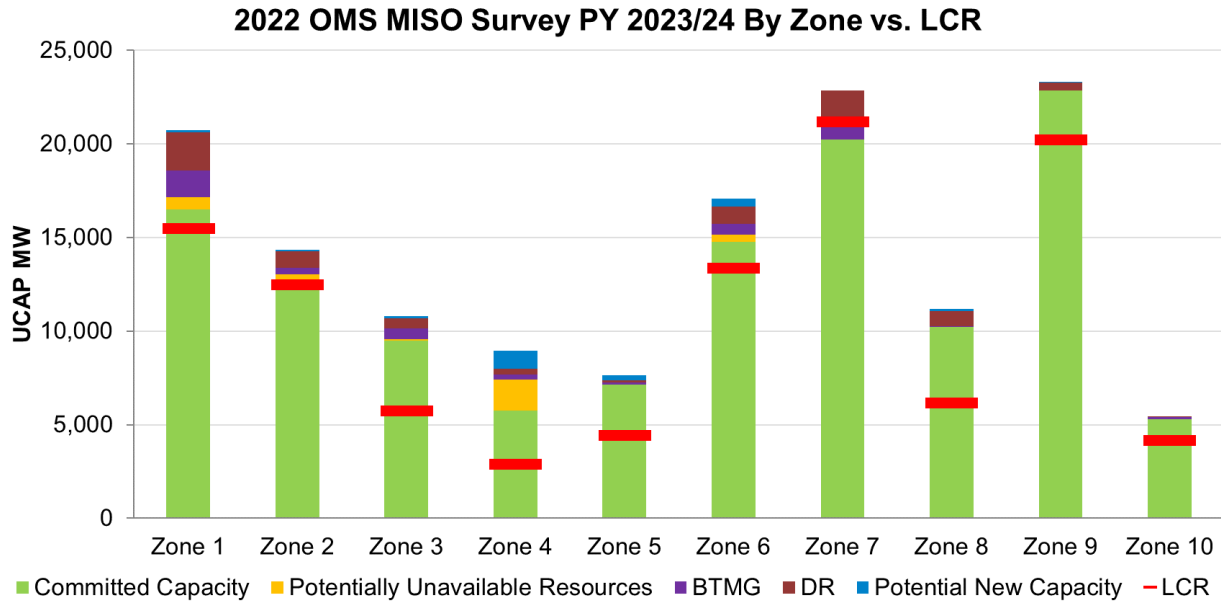


Figure 2.4-2: Potential New Generation and Retirements Within Each Zone, PY23/24

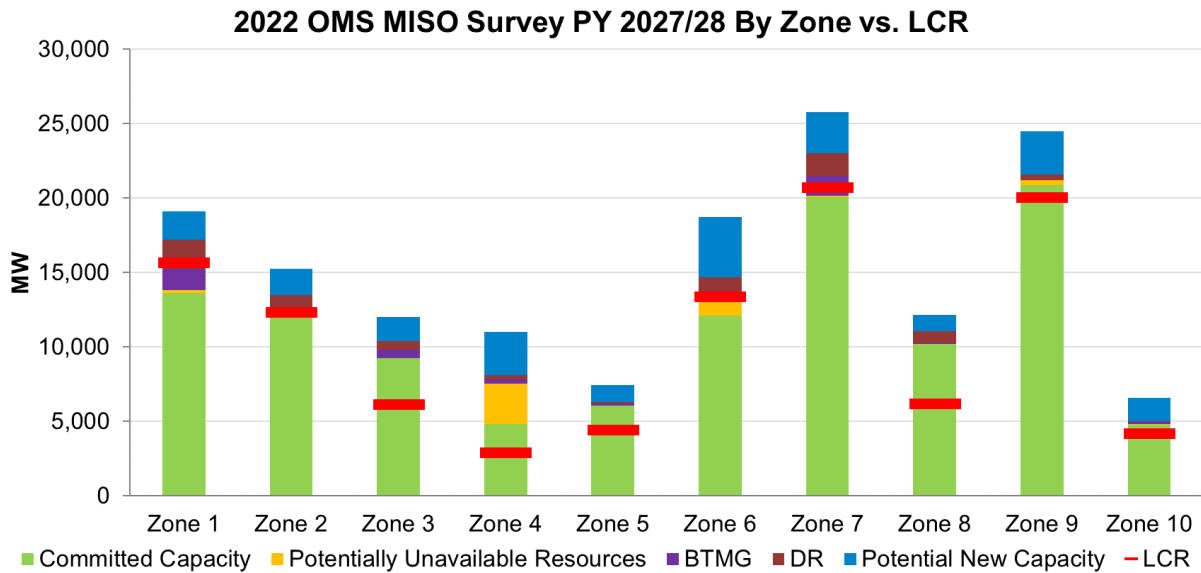


Figure 2.4-3: Potential New Generation and Retirements Within Each Zone, PY27/28

Compared to last year, these committed resource reserve margins are lower for both the first year and the full five-year period of the survey, which translates into increased reliance on less-certain resources to ensure resource adequacy going forward.

The OMS-MISO Survey shows fleet changes in the next 5 years shifting heavily toward renewables. As queue additions are not submitted much beyond 2027, the impacts observed are attributed to increased solar and to unit retirement decisions. Current survey shows a preponderance of solar, however it is anticipated that future surveys will show increases in battery and hybrid units that are nearly non-existent today but projected to grow in the near future.

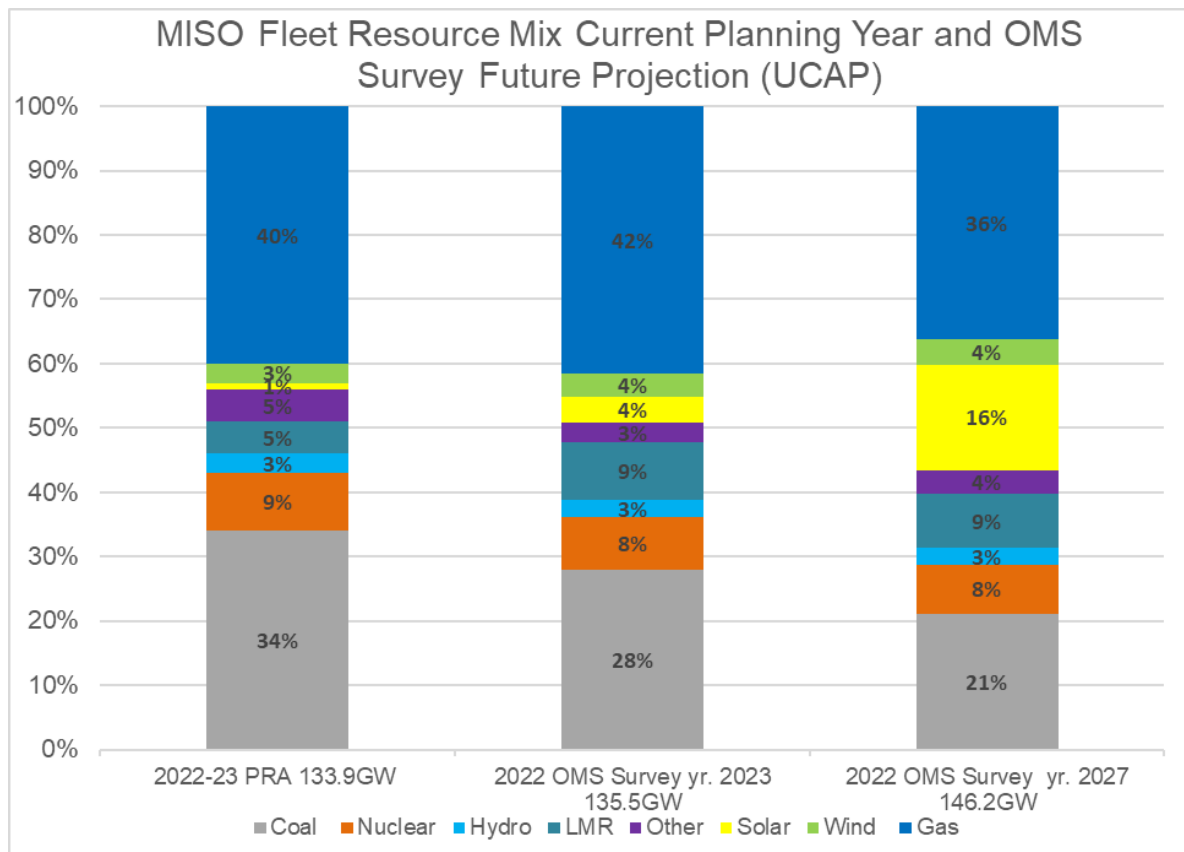


Figure 2.4-5: 2022 PRA Actual, 2023 OMS Survey, and 2027 OMS-MISO Survey Resource Mix

Regional Resource Assessment

In 2021, MISO published the first [Regional Resource Assessment \(RRA\)](#) as a complement to MISO's resource outlook efforts. Primarily focused beyond the time horizon of the OMS-MISO survey, the effort gathers publicly announced plans and goals from utilities across the MISO footprint over the next twenty years and aggregates them to develop an indicative view of how the resource mix may evolve.

As the pace of decarbonization and fleet change accelerates, the annual refresh of the RRA is designed to capture and incorporate year-over-year changes and regularly reflect the potential implications back to utilities and state regulatory decision-makers. One key objective of the assessment is to anticipate potential resource gaps between utility plans and their stated objectives.

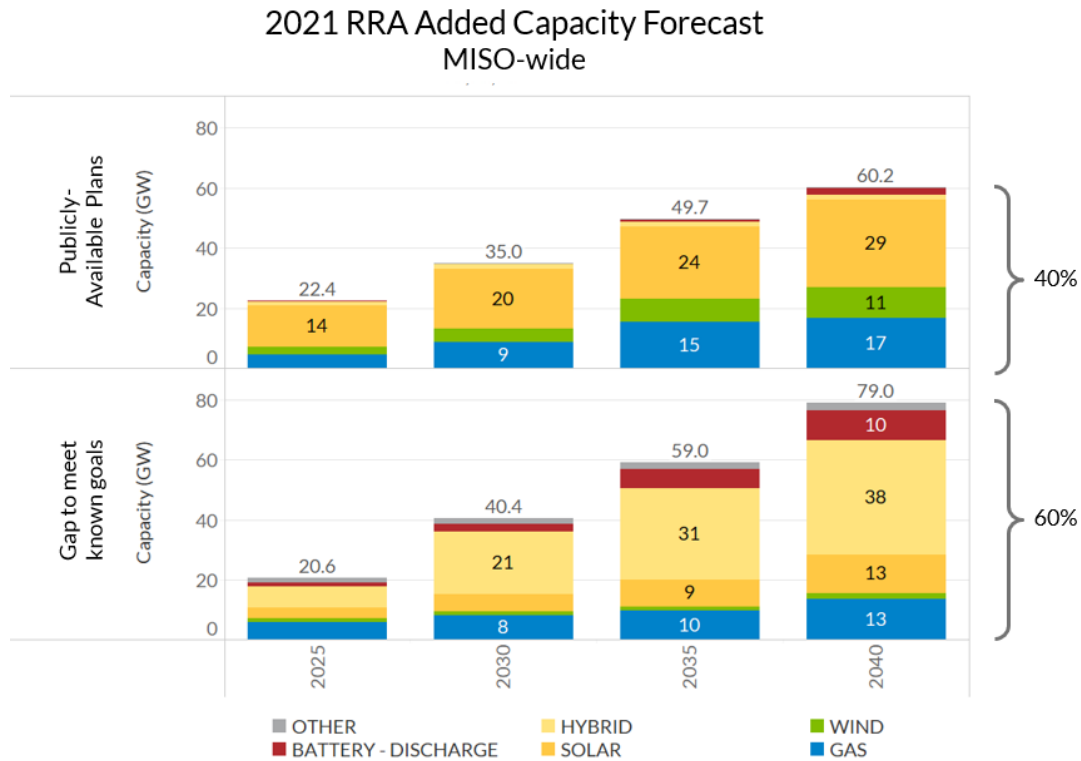


Figure 2.4-6: This figure from the 2021 RRA shows that utility plans include 40% of the added capacity that may be needed to meet stated objectives by the year 2040

In addition to the resource assessment, the RRA includes resource adequacy and flexibility needs assessments—reviewing capacity contributions, quantifying performance at sub-hourly timescales, and assessing risk of capacity and flexibility deficits from operational and market perspectives. The 2022 RRA will be published at the zonal level to provide improved visibility of future zonal resource mixes and attributes, and locational impacts and risks. MISO aims to publish the 2022 RRA report in November.

The RRA is focused solely on the MISO footprint and is not a transmission study. The planning information provided by stakeholders through the RRA Survey will inform many aspects of MISO’s work, including the Reliability Imperative prioritization and updates to the MISO Futures.

2.5 Planning for the Future

To perform analysis on the bulk electric system twenty years into the future, many assumptions must be made to bridge what is known about the system today to what it could be in the future. Complicating matters is the uncertainty of future developments.

MISO has developed a method to address this uncertainty—the use of multiple forward-looking scenarios to provide a range of future outlooks. Within MISO, these forward-looking scenarios are called the “Futures”. These Future scenarios establish different ranges of economic, policy, and technological possibilities—such as load growth, electrification, decarbonization, generator retirements, renewable energy levels, fuel prices, and generation capital costs—over a twenty-year period.

Future Scenarios

The MISO Futures developed in 2020 represent continued collaboration between MISO and its stakeholders. The MISO Futures are the inputs for multiple MTEP cycles, the LRTP initiative, and other planning studies. These Futures form the basis for all components of the Reliability Imperative, such that MISO and its stakeholders can plan to a consistent set of scenarios across transmission, markets, and operations.

Within this context, no new Future scenarios were developed specifically for MTEP22. Additionally, the MISO Futures are intended to apply to multiple planning cycles; therefore, the new MISO Future scenarios developed in 2020 are the Futures used in MTEP21, MTEP22, and Tranche 1 of the LRTP. MISO is working through a Futures Refresh for MTEP23.

Chapter 3: Regional and Interregional Planning Studies

3.1 Long Range Transmission Planning – Tranche 2 Introduction

The Long Range Transmission Planning (LRTP) initiative is MISO's response to the current and future resource evolution that has and continues to affect the bulk electric system. The scale and pace of these changes require prompt attention to develop the most efficient, cost-effective investments that will ensure grid reliability in the future.

MISO is preparing for an unprecedented pace of change in the resource mix. MISO, our members, regulators, and other entities responsible for system reliability all have an obligation to work together to address these challenges. MISO calls this shared responsibility the Reliability Imperative, which consists of several linked initiatives: Market Redefinition, Long Range Transmission Planning (LRTP), Operations of the Future, and Market System Enhancements (refer to Figure 3.1-1).

The Reliability Imperative focuses on preparing the region for industry transformation as the grid evolves toward increased renewable resources. As a critical part of this effort, Long Range Transmission Planning holistically considers needs of the MISO footprint and identifies solutions that will be needed to maintain system reliability.

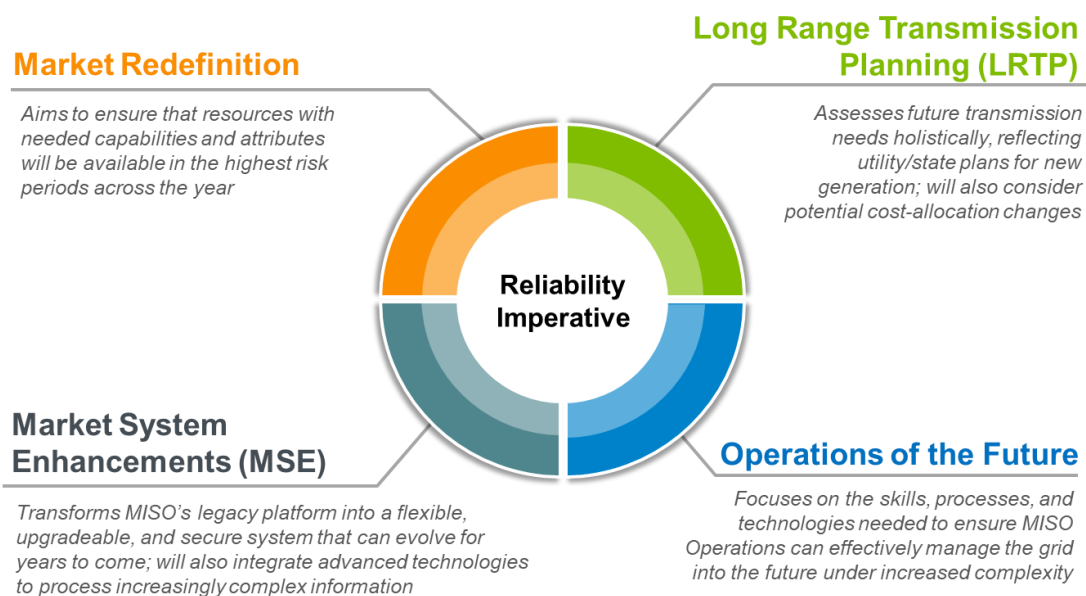


Figure 3.1-1: Reliability Imperative focus areas

In preparation for addressing these transformational needs, MISO has developed an Indicative Transmission Roadmap of potential transmission expansions throughout the region. The Roadmap provides, among other things, an indication of the potential magnitude of transmission expansions that may

be needed to maintain reliable and efficient operations under the expected Futures. This Roadmap was contemplated by MISO planning staff as extensions of the existing grid that would provide for logical connections that could increase connectivity, close gaps between subregions and support a more resilient grid by enabling more transfers of bulk power flows. The Indicative Roadmap is not a plan, nor does it represent any particular study efforts on the part of MISO, but rather provides a basis for considering solutions to Transmission Issues expected in LRTP study efforts. MISO is engaging stakeholders as the planning staff identifies system performance issues under the assumptions projected in the Futures, and as we seek the most cost-effective transmission and non-transmission solutions. The road map shown in Figure 3.1-2 was useful to both MISO planning staff and stakeholder planning staff during the execution of the LRTP Tranche 1 process. The roadmap provided, as expected, direction towards potential solutions to grid issues uncovered in the Tranche 1 study effort. [Tranche 1](#) resulted in a \$10.3 transmission portfolio that was approved by the MISO Board of Directors in July 2022.

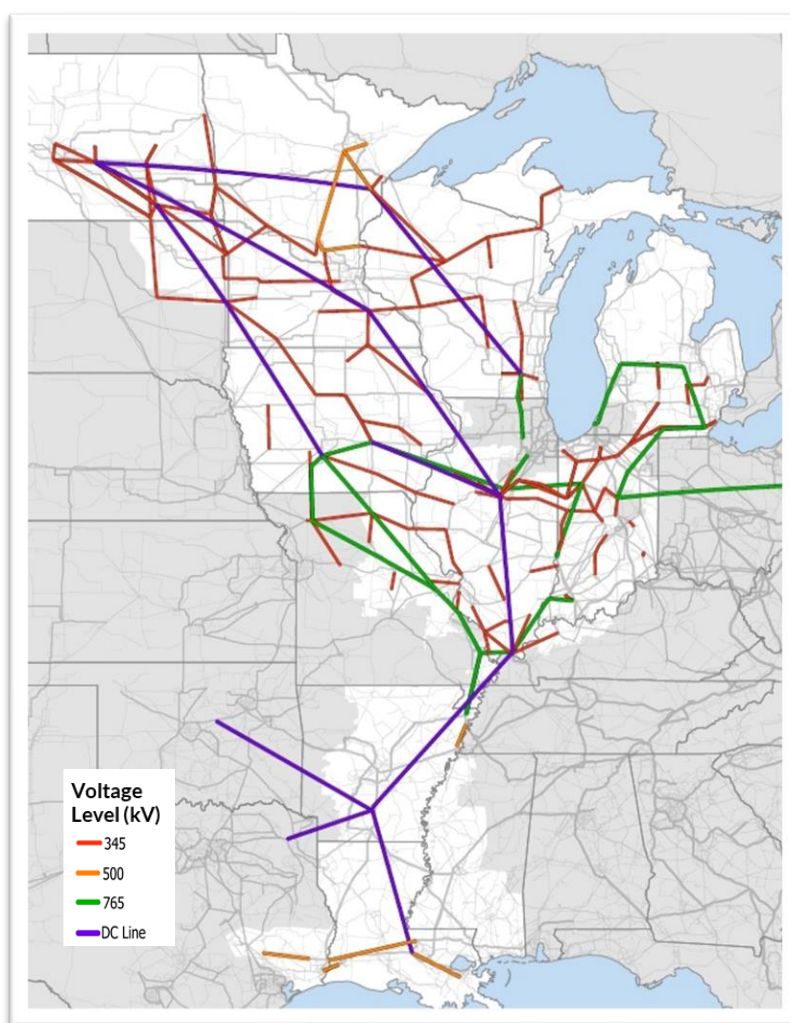


Figure 3.1-2: Futures 1, 2, & 3 Indicative Roadmap

The Long Range Transmission Plan will iterate over time as needs and conditions change. This requires a planning process timeline that differs from what stakeholders are familiar with. Typically, MISO planning studies happen as part of the annual MTEP cycle; they follow the annual reliability process that focuses

primarily on local needs and results in recommendations as part of the MTEP report. In general, this is a very linear process where each step occurs at about the same time every year.

In contrast, Long Range Transmission Planning is a multi-year effort as needed to address the myriad of complex issues associated with the significant resource transformation underway. Because there is urgency to keep pace with this rapid evolution, MISO will seek to recommend projects identified in the LRTP effort over several MTEP cycles as work progresses (Figure 3.1-3).

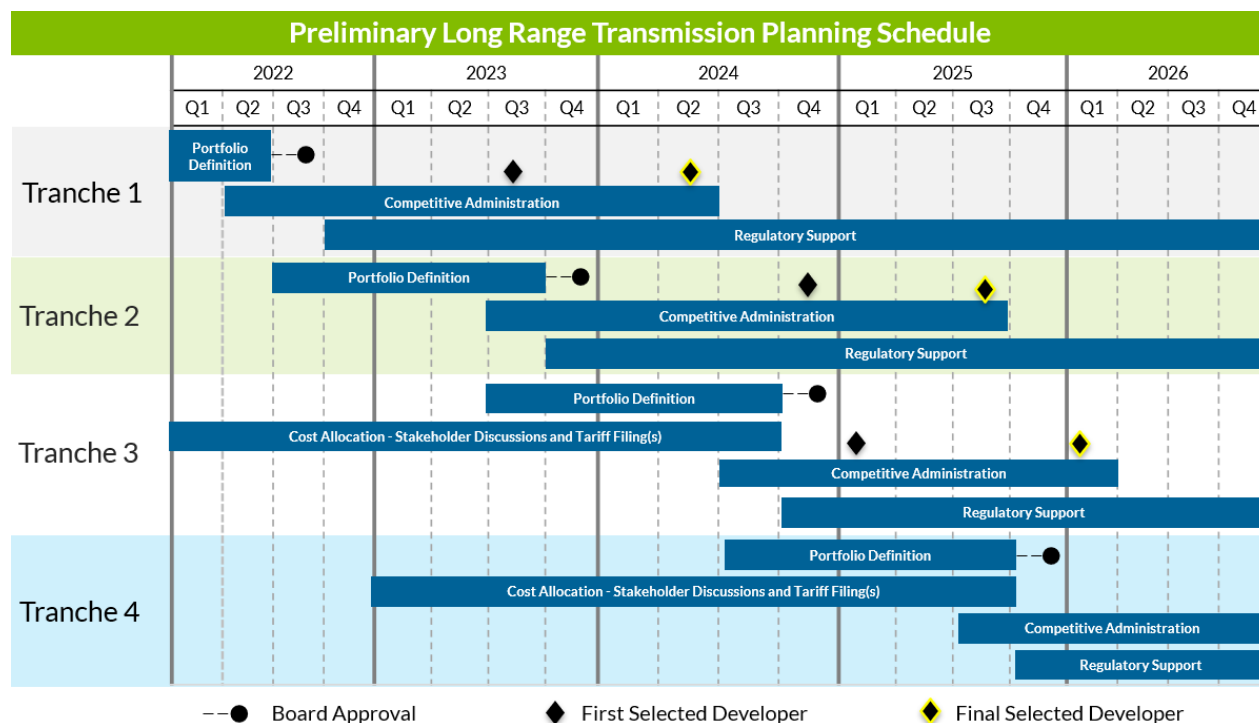


Figure 3.1-3: Long Range Transmission Planning Schedule

MISO believes it is important to move quickly, while ensuring a least regrets plan. Meeting both objectives is critical to enable our member plans, and to ensure reliable power delivery for customers with investment decisions that appropriately balance generation and transmission solutions on a regional scale to ensure the best cost outcomes for customers. Long Range Transmission Planning continues the MISO Value-Based Planning approach to extend value beyond the traditional planning processes to achieve a more efficient comprehensive long-term system plan.

It is also important to note that FERC released a Notice of Proposed Rulemaking (NOPR) in 2022 that is a follow-on to the ANOPR that was published in 2021. In this NOPR, FERC seeks to drive longer-term transmission planning principles throughout the industry. The NOPR provides for much of what MISO is already doing in the LRTP effort. MISO is following the NOPR process closely and will apply as directed once an Order and compliance schedule is released.

3.2 Interregional Studies

3.2.1 MISO-SPP Joint Targeted Interconnection Queue (JTIQ) Study Results

Introduction and Background

The JTIQ study is a result of MISO and SPP's cluster study observations that show that transmission systems at the seams are at capacity and that the necessary network upgrades are too costly for individual or a small group of interconnection projects to proceed. While the addition of renewable resources and transmission along the SPP-MISO seam provides benefits to the markets, current tariff and Joint Operating Agreement (JOA) mechanisms do not provide a cost-sharing approach that can facilitate the construction of the large-scale transmission needed to interconnect expected levels of new generation near the seam. Process, criteria, and schedule differences between the respective RTOs contribute to study delays and introduce questions on study results. The JTIQ Study takes these various barriers into consideration. Full report available using this [link](#).

Study Results

Through collaboration between the MISO and SPP Regional Transmission Organizations (RTOs), the study identified a seven-transmission-project JTIQ portfolio with a planning level estimated cost of \$1.65B required to address the significant transmission limitations restricting the opportunity to interconnect new generating resources near the MISO-SPP seam.

The recommended JTIQ Portfolio is expected to fully address the set of transmission constraints evaluated in the JTIQ Study as being significant barriers to the development of new generation along the MISO-SPP seam. In addition to these substantial reliability benefits, economic analysis conducted by the RTOs show customers can anticipate an Adjusted Production Cost (APC) benefit of \$724 million in the MISO footprint and \$247 million in the SPP region. An estimated 28.7 GW of improved interregional generation enablement would be available to new generator interconnection projects near the seam.

JTIQ Portfolio Update post LRTP Tranche 1 approval

The JTIQ Portfolio noted in Figure 3.2.1-1 and Table 3.2.1-1 is adjusted as first two projects in Table 1 of [study report](#) were included in Tranche 1 of the Long Range Transmission Planning effort that was approved in an addendum to MTEP21. As a result, these two projects were removed from JTIQ Portfolio. The removal of these projects does not reduce the benefit of the remaining JTIQ Portfolio with regard to generation enablement but has impacted the future economic benefit to the JTIQ Portfolio to load. (APC of \$55.7 million in the MISO footprint and \$132.9 million in SPP footprint).



Figure 3.2.1-1: JTIQ Portfolio Map

JTIQ Portfolio	Location by RTO	Cost (\$M)
Bison – Hankinson – Big Stone South 345 kV	MISO	476
Brookings Co – Lakefield 345 kV	MISO	331
Raun – S3452 345 kV	MISO - SPP	144.4
Auburn – Hoyt 345 kV	SPP	90.5
Sibley 345 Bus Reconfiguration	SPP	18.8
Total Cost of Portfolio of Projects	MISO - SPP	1,060.7

Table 3.2.1-1: List of projects comprising the JTIQ Portfolio

Cost Allocation Direction

MISO and SPP are continuously engaging their respective stakeholder communities to refine the JTIQ Affected System Zone Concept that is proposed as a cost allocation method for the current and subsequent JTIQ studies.

JTIQ Affected System Zone Concept

Following the current JTIQ, affected system transmission needs will be identified through biennial future JTIQ studies. JTIQ will be a forward-looking study that will pre-determine transmission needs and Affected System Study costs of future DPP and DISIS study clusters. This will replace the current affected system process between MISO and SPP. JTIQ will assign a predetermined JTIQ Affected System Zonal Charge (\$/MW charge) to applicable interconnection requests in the Affected System Zone. The affected System Zone will be based on affected system study screening criteria of 5% for more distribution factor impacts of new interconnection generation on neighboring RTO's transmission facilities.

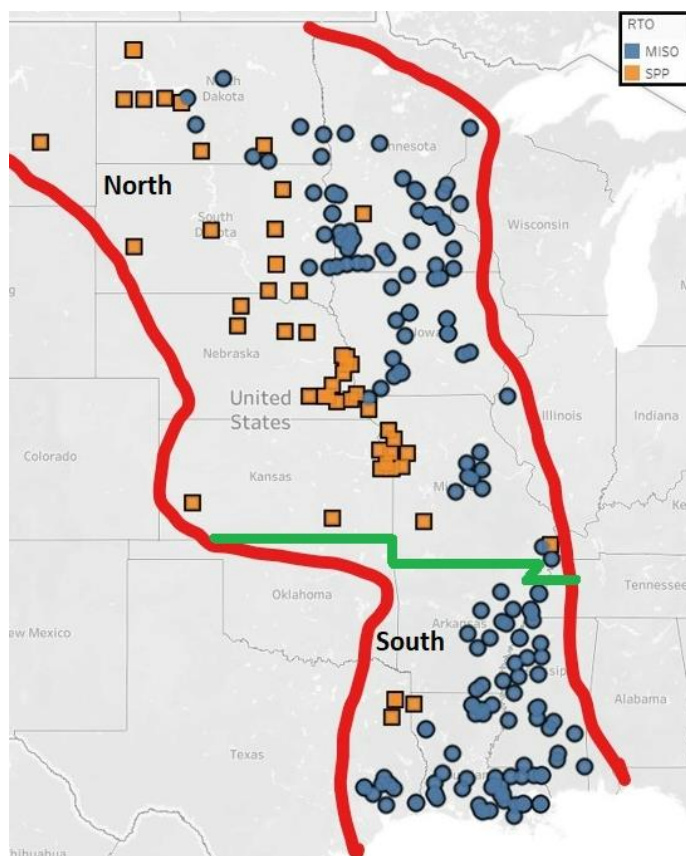


Figure 3.2.1-2 JTIQ Affected System Zone Example Based on Futures Sites

The JTIQ Affected System Zone concept will be documented in the MISO/SPP JOA and is planned to be filed with FERC in Q1 2023. The cost allocation of the JTIQ study and subsequent JTIQ studies will be in line with current Generator Interconnection procedures. JTIQ cost allocation will continue the current cost sharing opportunities between GI customers and load.

This concept aims to provide cost and timing certainty for Generator Interconnection Customers as affected system costs will be known at the beginning of DPP or DISIS in addition to the elimination of Affected System Study need between MISO and SPP. Moreover, this concept will identify more optimized network upgrades as compared to individual Affected System Study clusters as in current process.

3.2.2 MISO-SPP Coordinated System Planning

Introduction

MISO and SPP formally announced the performance of a 2022 Coordinated System Plan (CSP) study at the MISO-SPP Annual Issues Review Interregional Planning Advisory committee (IPSAC) meeting held on February 15, 2022. A CSP study was required in 2022 in accordance with the MISO-SPP Joint Operating Agreement (JOA). After reviewing feedback provided at the issues review meeting, the MISO-SPP Joint Planning Committee (JPC) met on March 17, 2022, and mutually agreed to pursue a new study process modeled around an existing MISO-PJM study process called a Targeted Market Efficiency Project (TMEP) study. The TMEP study is designed to identify potential solutions to historical, persistent congestion issues on the MISO-SPP Seam where a long-term planning horizon study may not effectively capture certain existing day-ahead or real-time market conditions.

Study Objectives

The MISO-SPP CSP study will jointly evaluate historical market-to-market (M2M) congestion issues and seek transmission solutions that efficiently address those issues for the benefit of both MISO and SPP. To accomplish this goal, the study will consist of two primary deliverables:

- Development of a repeatable process to effectively study persistent, historical congestion on the SPP-MISO seam, including a set of appropriate project criteria for inclusion in new JOA language
- Identify valuable transmission upgrades, as applicable, for recommendation using a future, to-be-determined cost allocation methodology

This study process aims to address issues identified between the shorter-term (real-time) planning horizon and the longer-term (5-10+ year) planning horizon. In order to complement the longer-term planning horizon studies that already exist in the MISO-SPP JOA, the process and criteria development for this new study will focus on three guiding principles:

- Target and relieve known M2M issues
- Solutions can be implemented in a relatively short time-frame (e.g., not greenfield)
- Straight-forward benefit determination with quick payback on investment

The JPC, through the IPSAC, will provide stakeholders an open and transparent forum to provide input and review results during the 2022 SPP-MISO CSP study.

Scope

The study scope consists of four primary tasks:

Data gathering and flowgate identification: each RTO will analyze two years of their historical market operations data to determine which M2M constraints are appropriate for further evaluation.

Criteria development: RTO staff, with stakeholder input from the IPSAC, will develop appropriate criteria (e.g., project cost, benefit, in-service time frame, etc.) for proposed transmission upgrades that abide by the study guiding principles.

Transmission solution development: In parallel with the project criteria development, transmission solutions will be identified through coordination with asset owners of the candidate flowgates to determine limiting elements and potential fixes.

Documentation of process and criteria: Staff will work to memorialize process with new MISO-SPP JOA language and prepare for filing the changes at FERC.

As of August 2022, MISO and SPP are in the early phases of seeking stakeholder feedback on proposed project criteria and providing initial results of the historical congestion analysis. Study updates will be provided via IPSAC meetings planned through end of the year with anticipated JOA language development expected in Q1 2023. Once all study work is completed, a full report of the CSP study and exploratory process development will be published for stakeholders in accordance with the MISO-SPP JOA.

3.2.3 MISO-PJM Coordinated System Planning

Introduction

MISO and PJM held their Annual Issues Review Interregional Planning Advisory Committee (IPSAC) meeting on February 17, 2022. After reviewing stakeholder feedback provided at the IPSAC, the MISO-PJM Joint RTO Planning Committee (JRPC) formally announced on March 30, 2022 that the RTOs would pursue a Targeted Market Efficiency Project (TMEP) study for the 2022 Coordinated System Plan (CSP). The TMEP study is designed to identify potential solutions to historical, persistent congestion issues on the MISO-PJM Seam where a long-term planning horizon study may not effectively capture certain existing day-ahead or real-time market conditions.

Scope and Evaluation

MISO and PJM presented the scope of the 2022 TMEP study at the April 25, 2022 IPSAC meeting. At that meeting RTO staff presented a review of the aggregate historical Market-to-Market (M2M) data, which included a preliminary list of 23 potential flowgates totaling \$328 million in congestion costs over the two-year sample period. RTOs then proceeded to further evaluate the candidate flowgates through coordination with operations staff and asset owners. Key evaluation parameters include:

- Determination of whether congestion was expected to persist (e.g., was congestion driven by specific outages or events).
- Determine whether any planned upgrades would already alleviate the identified congestion.
- Determine whether viable TMEP-type solutions exist (e.g., low cost, easy to construct etc.).

After further evaluation, 19 of the 23 preliminary flowgates were eliminated from consideration due to the identification of previously planned projects, non-persistent or specific outage driven congestion, and solutions under evaluation in a different regional planning process. At the July 21, 2022 MISO-PJM IPSAC meeting staff presented the refined TMEP candidate list with four remaining candidates:

Monitored Facility	Flowgate Description	Two-Year Total Congestion
Quad Cities – Rock Creek 345 kV	Quad Cities- Rock Creek 345 kV I/o QuadCities-Sub91 345/161 kV Sub91 XF	\$20.3 M
Mohomet – ChampTP 138 kV	Mohomet – ChampTP 138 kV FLO Clinton – Oreana – Goose Crk 345 kV	\$19.3 M
Chicago – Praxair 13831	Chicago-Praxair 13831 I/o Wilton CenterDumont 765	\$9.6 M
Powerton - Towerline 138 kV	Powerton - Towerline 138 kV I/o Fargo - Sandburg 345 kV	\$4.6 M

MISO and PJM continued the evaluation process by coordinating with asset owners of these candidate flowgates to determine the limiting elements of each circuit and whether any effective solutions might exist that would meet the TMEP cost and in-service criteria. The remaining TMEP study work schedule will include two additional scheduled 2022 IPSAC meetings:

- October 4, 2022: Preliminary results on available solutions for flowgate candidates or the elimination of candidates from further project consideration
- December 15, 2022: Joint review of final results and JRPC recommendations

If the JRPC recommends any TMEP projects they will then move into each RTO's regional process for respective Board of Directors recommendations and approvals. Because the interregional study process is scheduled to conclude in December 2022, any MTEP project recommendations and approvals would occur in Q1 2023.

Chapter 4: Reliability Studies

4.1 Reliability Assessment and Compliance

MISO, in collaboration with its transmission-owning members and stakeholders, performs annual reliability assessments to identify transmission infrastructure upgrades needed to ensure the continued system reliability in compliance with applicable local and regional reliability standards. The reliability assessment process for MTEP22 (shown in Figure 4.1-1) began with a roll-up of issues and potential solutions from the local planning processes of the Transmission Owners (TOs), followed by an independent reliability assessment conducted by MISO to evaluate and integrate the TO local planning information into the development of the overall MISO Transmission Expansion Plan.

MISO closely coordinates the annual reliability assessment with other planning efforts to ensure the transmission expansion plan is identified in an efficient and cost-effective fashion. A variety of factors are considered as part of MISO’s transmission expansion plan development, including but not limited to, urgency of needs, cost effectiveness of solutions, system performance of solution alternatives to address identified transmission issues, and other considerations such as lead time to develop a project, right-of-way (ROW) or substation impacts, expandability, operational flexibility, etc.

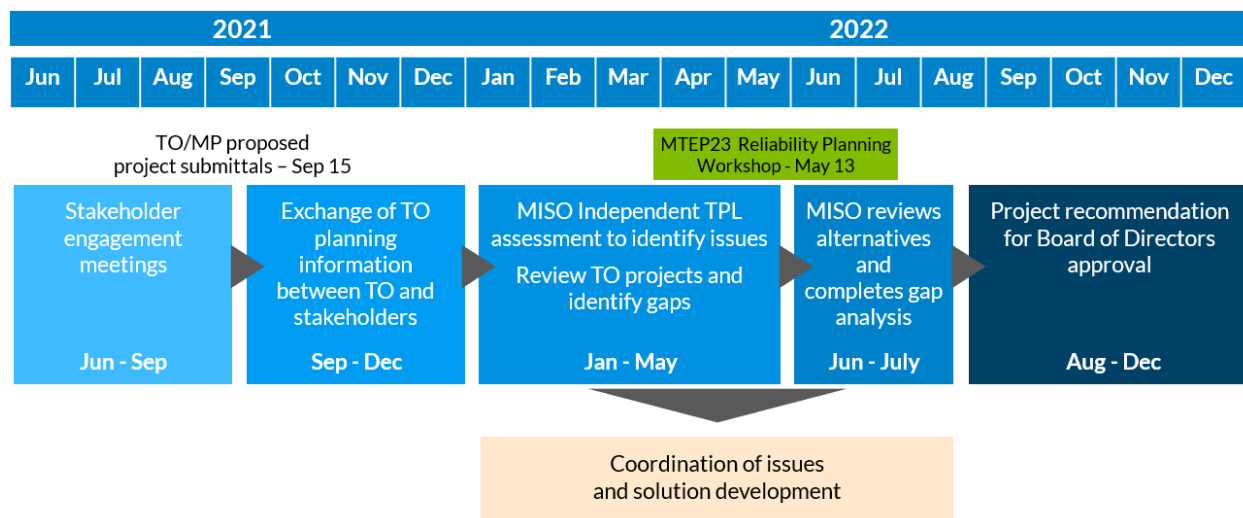


Figure 4.1-1: MTEP22 reliability assessment process

In conjunction with the MTEP planning process, an inclusive, transparent stakeholder process is utilized to facilitate open discussions and allow stakeholders to provide early and meaningful inputs into the development of transmission solutions in each planning cycle. The results of MISO’s independent reliability assessments, along with proposed solution alternatives, are presented to stakeholders through a series of public Sub-regional Planning Meetings (SPM), and additional Technical Study Task Force (TSTF) meetings as needed, for each of the four MISO planning sub-regions: Central, East, South, and West.

After MISO completes its independent review of all proposed projects and associated alternatives and addresses stakeholder feedback received through SPM discussions, MISO staff formally recommends a set of projects to the MISO Board of Directors for review and approval. These projects make up Appendix A of

the MTEP report and represent the preferred solutions to the identified transmission needs of the MISO reliability assessments. Proposed transmission upgrades with sufficient lead times are included in Appendix B for further review in future planning cycles.

The complete results of MTEP22 reliability assessments are detailed in Appendices D3-D10 of the MTEP22 report, which are available on the MISO ShareFile site at <https://misoenergy.sharefile.com/>, subject to Critical Energy Infrastructure Information (CEII) and non-disclosure agreements. These results serve as compliance evidence for a variety of NERC planning standards listed on the MISO public website at: <https://www.misoenergy.org/planning/transmission-planning/reliability-planning>.

MTEP22 project recommendations

As the result of the MTEP22 reliability assessments, 41 baseline reliability projects totaling \$545 million are included in the MTEP22 proposed Appendix A, accounting for 13% of total transmission infrastructure investment in MTEP22. The vast majority of the recommended projects are driven by reliability (either baseline or local reliability), load growth, and age and condition, and are expected to be in service within three years.

Out of the 382 MTEP22 projects submitted in this cycle, MISO Planning Engineers received and evaluated 19 Expedited Project Review requests. These expedited projects were submitted by Transmission Owners who determined that system conditions warrant the urgent development of system enhancements within the current MTEP cycle.

Project justification details of the recommended Appendix A projects are summarized in the following subsections for each of the four MISO planning sub-regions.

Planning Region	BRP	GIP	MPFP	Other	Total
Central	\$131 M	\$270 M		\$1,111 M	\$1,512 M
East	24 M	\$90 M		\$325 M	\$439 M
South	\$72 M	\$117 M		\$817 M	\$1,006 M
West	317 M	\$69 M	\$8 M	\$916 M	\$1,310 M
Total	\$545 M	\$547 M	\$8 M	\$3,169 M	\$4,268 M

Figure 4.1-2: MTEP22 Appendix A new project investment by category and planning region (data as of 9-15-2022)

In the following pages, the region's MTEP22 projects are categorized into three categories, Baseline Reliability, Other, and Generator Interconnection. The definition of each of these categories are detailed below:

Baseline Reliability Projects

According to Attachment FF of the MISO Tariff, "Baseline Reliability Projects are Network Upgrades identified in the base case as required to ensure that the Transmission System is in compliance with applicable national Electric Reliability Organization (ERO) reliability standards and reliability standards adopted by Regional Reliability Organizations and applicable within the Transmission Provider Region."

All costs for Baseline Reliability expansion projects are recovered through Attachment O by the Transmission Owner(s) developing such projects.

Other Projects

The "Other" projects category are projects that do not meet the criteria to be considered as Baseline Reliability Projects (BRP), New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects. Other projects may include projects to satisfy Transmission Owner and/or state and local planning criteria other than NERC or regional reliability standards, interconnection of new Loads, relocate transmission facilities, address aging transmission infrastructure, replace problematic transmission plant, improve operational performance or address other operational issues, address service reliability issues with end-use consumers, improve aesthetics including but not limited to undergrounding overhead transmission facilities, address localized economic issues, and address other miscellaneous localized needs. The tables of project information are broken down by four general categories of project drivers; Local Reliability, Age and Condition, Load Growth, and Other Local Need, but note that these four drivers are not defined in the MISO Tariff.

Generator Interconnection Projects

According to Attachment FF of the MISO Tariff, "Generator Interconnection Projects are New Transmission Access Projects that are associated with interconnection of new, or increase in generating capacity of existing, generation." These represent facilities necessary to physically interconnect the generation resource to the transmission system as well as network upgrades required to facilitate reliable delivery of the output to ultimate load.

The Generator Interconnection Projects (GIPs) noted in the following sections of this chapter have been evaluated through the Generator Interconnection queue and the associated Generator Interconnection Agreements have been signed.

Generation Interconnection Projects are network upgrades associated with interconnection of new, or increase in generating capacity of existing, generation under *Attachments X to the Tariff*. These projects are driven by interconnection study procedures and agreements. The Interconnection Customer is responsible for one-hundred percent (100%) of the costs of network upgrades rated below 345 kV and ninety percent (90%) of the costs of network upgrades rated at 345 kV and above (with the remaining ten percent (10%) being recovered on a system-wide basis.

4.2 Project Justifications – Central Region

Central Region Overview

The MISO Central planning region consists of seventeen Transmission-Ownning members spanning four states: Missouri, Illinois, Indiana, and Kentucky. These Transmission Owners are:

- American Electric Power Service Corporation (AEP)
- Ameren Illinois (AMIL)
- Ameren Missouri (AMMO)
- Big Rivers Electric Corp. (BREC)
- City of Columbia, Mo. (CWLD)
- City of Springfield, Ill. (CWLP)
- Duke Energy Corp. (DEI)
- GridLiance Heartland LLC (GLH)
- Henderson Municipal Power & Light (HMPL)
- Hoosier Energy REC Inc. (HE)
- Indianapolis Power & Light (IPL)
- Northern Indiana Public Service Co. (NIPSCO)
- Pioneer Transmission (PTx)
- Prairie Power Inc. (PPI)
- Republic Transmission (RTx)
- Southern Indiana Gas & Electric (SIGE)
- Southern Illinois Power Cooperative (SIPC)
- Wabash Valley Power Association Inc. (WVPA)

The Bulk Power System (BPS) within these states consists of an extensive 765 kV, 345 kV, 230 kV, 161 kV, and 138 kV networked transmission system. The 345 kV network spans Missouri, Illinois, and Indiana, both north to south and east to west. The 230 kV network spans through Indiana, both north to south and east to west. The 161 kV network spans north to south and east to west in Missouri, Illinois, and Kentucky, and the 138 kV networks span both north and south, and east to west in Illinois, Indiana, and Kentucky. All of Ameren, BREC, CWLD, CWLP, GLH, HMPL, and SIPC belong entirely in the SERC Region. All of DEI, HE, IPL, NIPSCO, PTx, RTx and SIGE belong entirely in the ReliabilityFirst Region. Wabash Valley is split between both ReliabilityFirst and SERC Regions.

Major load pockets in the MISO Central planning region are St. Louis, MO; Peoria, IL; Springfield, IL; Evansville, IN; and Indianapolis, IN (shown in Figure 4.2-1).

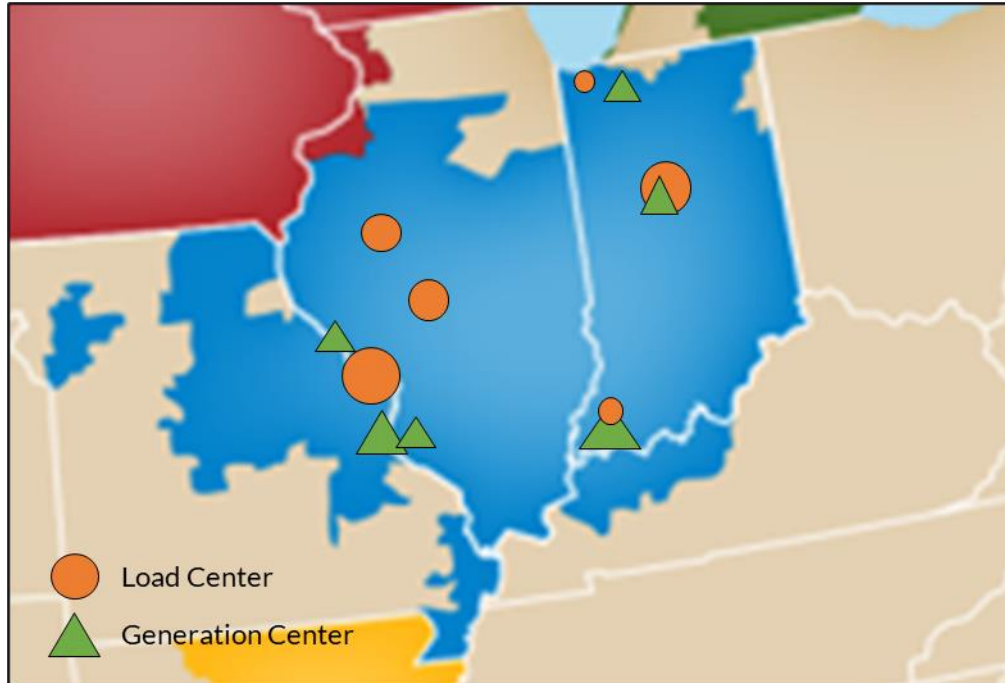


Figure 4.2-1: Generation and load centers in the Central planning region

For MTEP22, MISO Transmission Planning is recommending 126 projects from the Central region for inclusion in Appendix A at an estimated cost of \$1.51 Billion. Of these, nine are Baseline Reliability Projects and 24 are Generator Interconnection Projects. The remaining 93 projects are classified as Other projects because they do not meet the criteria to be considered as Baseline Reliability Projects, New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects.

Of the 126 projects, that are being recommended to be included in MTEP22, 15 have an estimated cost of less than \$1 million, 52 have an estimated cost between \$1 million and \$5 million, and the remaining 59 projects are estimated to cost greater than \$5 million (indicated in Figure 4.2-2 below).

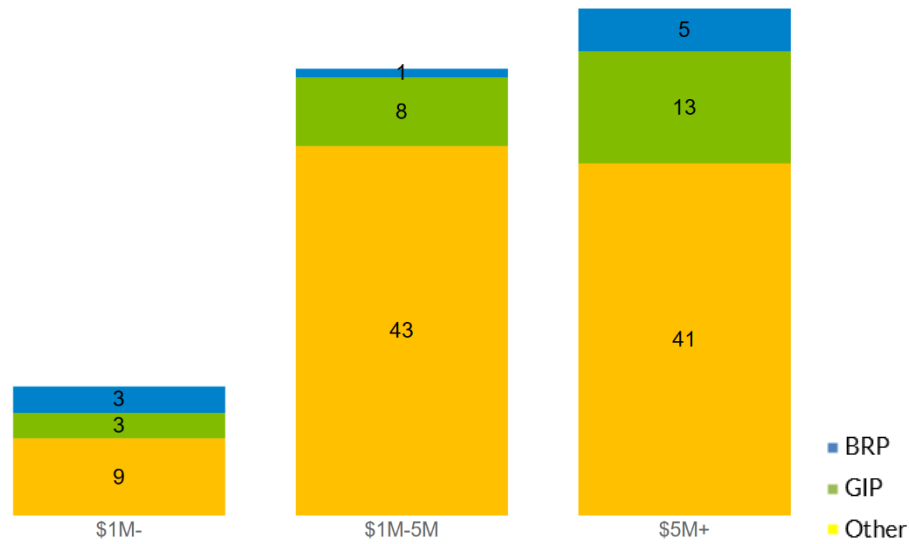


Figure 4.2-2: Project counts by cost category of MISO Central region MTEP22 projects (data as of 9-15-2022)

The majority of the projects in the MISO Central planning region are expected to go in service in the next three years (shown in Figure 4.2-3).

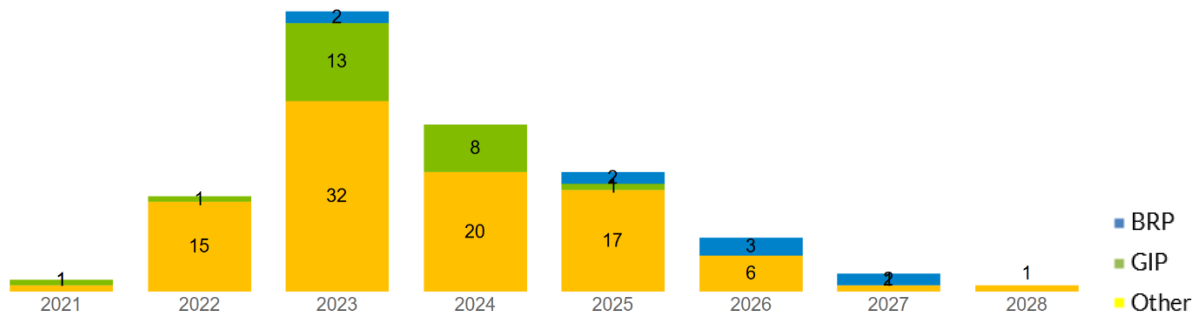


Figure 4.2-3: Central region MTEP22 projects by in-service date (data as of 9-15-2022)

In accordance with Attachment FF of the tariff, in the event a Transmission Owner determines that system conditions warrant the urgent development of system enhancements, an expedited review of the impacts of the project can be requested. MISO shall use a streamlined approval process for reviewing and approving projects proposed by the Transmission Owner(s) and decisions will be provided to the Transmission Owner within 30 Days of the project's submittal to MISO unless a longer review period is mutually agreed upon. During the MTEP22 cycle, MISO received the following projects through the Expedited Project Review (EPR) process:

1. Project ID 21685 New Pratt Paper Load Addition in BREC
2. Project ID 22228 Upgrades Transmission Due to I-69 DOT Project in HMPL
3. Project ID 22766 Replace McCracken-Shell Structure in BREC
4. Project ID 23266 New About Bit 115 MVA Load in HE
5. Project ID 21585 Rebuild Decatur East Main St-Route 51 Decatur 138 kV line in Ameren

Also, in accordance with Attachment FF Section VIII.A.3, the following project(s) are identified as an Immediate Need Reliability Project and excluded from the competitive developer selection process:

- None

The ten highest cost projects in the MISO Central Region represent \$848 million (%) of the \$1.33 billion total recommended projects for the whole MISO region in MTEP22. The locations of these projects are shown in Figure 4.2-4 and the investment is spread across the Central planning region. Projects that are blanket expenditures (relays, physical security, etc.) are excluded from this list.

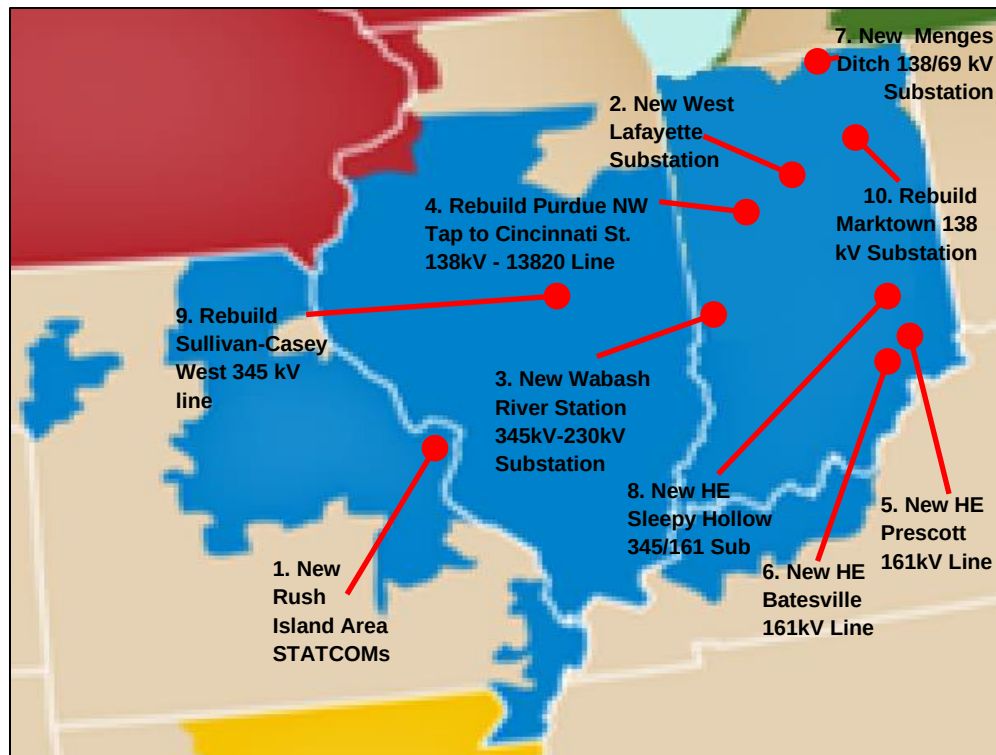


Figure 4.2-4: Central region top ten projects by cost (data as of 9-15-2022)

4.2.1 American Electric Power Service Corporation (AEP)

American Electric Power Service Corporation includes the AEP Transmission business unit which is responsible for the planning and operation of the AEP Transmission system, including assets owned by its affiliates in MISO.

American Electric Power Service Corporation did not submit any new projects for MTEP22.

4.2.2 Ameren Illinois

Ameren Illinois (AMIL) is a regulated electric and gas delivery company based in Collinsville, Ill. Its parent company is Ameren Corp., located in St. Louis, Mo. AMIL serves 1.2 million electric customers and 816,000 natural gas customers in central and southern Illinois. Its transmission system includes approximately 4,500 miles of transmission lines; 46,000 miles of distribution lines; 18,200 miles of natural gas pipe-line transmission and distribution mains; and 12 underground natural gas storage fields. AMIL's total generation is about 12,000 MW, more than half of which comes from Dynegy-owned coal units of 5,698 MW and Prairie State Energy-owned coal units of 1,600 MW.

Ameren Illinois proposed 21 projects at an estimated cost of \$166 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the Ameren area, all of these projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, none are Baseline Reliability projects, 12 projects are Other projects, and nine projects are Generator Interconnection projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21005	New Ina 138 kV line breaker	Add a line breaker on the Benton NW terminal at Ina.	12/1/2023	\$.9
21428	Rebuild Tanner-Xenia 345 kV line	Rebuild the existing Newton-Xenia-1 345 kV line from Xenia-Str. 44. New conductor will be capable of 3000 A and 2 OPGW shield wires will be installed.	12/1/2022	\$9.1
21646	Rebuild Sullivan-Casey West-1 345 kV	Rebuild the Sullivan-Casey West-1 345 kV line with conductors capable of 3000 amps Summer Emergency rating and two OPGWs.	12/1/2023	\$36.1

21970	Reconductor Kinmundy-Tanner 138 kV	Reconductor 27.4 miles Kinmundy-Tanner 138 kV line	6/1/2026	\$10
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Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21385	Rebuild Austin-Taylorville South 138 kV line (1545)	Rebuild Austin-Taylorville South 138 kV line (1545)	12/1/2023	\$8.9
21386	Rebuild Ruby-Sidney 345 kV line (4560)	Rebuild Ruby-Sidney 345 kV line (4560)	12/1/2023	\$2.2
21405	Rebuild Mattoon East-Charleston NE 138 kV line	Rebuild this line due to the condition of the structures, will add OPGW as part of this. This is the Mattoon East to Charleston NE section of the KANW-MTE8-1 line.	6/1/2023	\$3.4
21406	Rebuild Sparta Tap 138 kV radial line	Rebuild Sparta Tap on the Tibbs-Sparta-1723 138 kV line due to issues found during inspections.	12/1/2023	\$2.3
21647	Rebuild Casey West-Kansas 345 kV line (4525)	Rebuild the Casey West-Kansas 345 kV line (4525) with conductors capable of 3000 amps Summer Emergency rating and two OPGWs.	12/1/2024	\$25.2

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21306	New Whitetail 138 kV Substation	Ameren would construct a new 138 kV substation ring bus configuration on the existing Ameren Monument-Macomb West-1646 138 kV line near structure 97. PPI would construct a 138 kV line to connect their Golden 138/69 kV step down station. This is in support of PPI Interconnection.	12/1/2024	\$8
21326	New Duff 138 kV Substation	3-position initial, 4 ultimate ring bus on the Ipava-Meredosia E-1568 138 kV line for PPI Indian Creek 138/69kV substation. This is in support of a PPI Interconnection.	12/1/2023	\$8

21426	Reconductor S. Bloomington-W. Washington 138 kV lines (1334 & 1544)	Replace 3.5 miles and 3.48 miles of 477 KCMIL ACSR/T2 30/7 Hen 138 kV line conductor in SBLM-WWSH-1334 and in SBLM-WWSH-1544 with a new line conductor having a minimum summer emergency rating of 2000 Amps, respectively.	12/1/2022	\$4.2
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Generator Interconnection Projects:

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
20905	New Mulligan Solar (J1102) at Fogarty	Provide interconnection facilities for J1102 70 MW solar farm to connect at Fogarty. Reroute HOWL-FGTY-1417 to position H4 and install metering on position H2 for the customer to connect.	12/1/2021	\$0.5
21645	New Prairie Solar (J1139) Interconnection at Sidney 138kV	One 138 kV terminal in the Sidney Substation. The terminal will consist of all necessary terminal equipment to connect the J1139 lead line to the Sidney Substation 138 kV bus.	6/1/2023	\$.5
21365	New Puro 138 kV Substation POI (J1453)	J1453 Moraine Sands Wind generator 188.3 MW Interconnection: Install Puro ring bus on Havana-Shockey 138 kV line.	12/1/2023	\$7.2
22325	New Mogul 345 kV Ring Bus for J1464	Construct a new 345 kV switching station in Sangamon, Macoupin & Morgan Counties, Illinois to provide a Point of Interconnection for the Generating Facility. The new 345 kV Mogul switching station will have a four-terminal ring-bus design with three terminal positions installed. Split the existing Austin – Turner 345 kV transmission line and terminate the ends at two of the terminal positions.	10/31/2023	\$10.3
22385	New Dion 345 kV Ring Bus for J1115 Greenswitch Wind	The Dion switching station will be a ring bus arrangement with three-line terminal positions and provisions for one additional future terminal position. The existing Latham-Oreana 345 kV transmission line will be cut.	6/1/2024	\$10.5

22405	Add J991 Monarch Solar at Xenia 345 kV	Add 345 kV terminal in the Xenia switching station for the J991 project.	6/1/2023	\$1
22407	Add J974 Fulton Wind at MapleRidge 345 kV	Add position to the existing MapleRidge 345 kV switching station for J974 interconnection facilities.	6/1/2023	\$2.2
22485	Add J1094 Tilden Solar at Prest 138 kV	Expand and convert the Prest switching station 138 kV bus from a 6-position ring to a 4-leg breaker and a half (BAAH) configuration to add J1094 Tilden Solar.	1/1/2024	\$8.5
22566	New Morganfield 138 kV Ring Bus for J1096 Eldorado Solar	Construct a new 138 kV switching station in Saline County, Illinois to provide a Point of Interconnection for the Generating Facility. The new Morganfield 138 kV J1096 Interconnection Switching Station will have a four-terminal ring-bus design with three terminal positions installed. Split the existing Muddy-Norris City North 138 kV transmission line.	1/1/2024	\$6.5

4.2.3 Ameren Missouri

Ameren Missouri (AMMO) is a regulated electric and gas delivery company based in St. Louis, Mo. AMMO provides power to serve 1.2 million electric and 130,000 natural gas customers in central and eastern Missouri. Its service area covers 64 counties and more than 500 communities, including the greater St. Louis area. More than half of the AMMO generation comes from Ameren-owned coal and nuclear generation at more than 6,800 MW. Ameren also owns a good percentage of renewables and hydroelectric generation plants across central and eastern Missouri.

Ameren Missouri proposed 16 projects at an estimated cost of over \$201 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the Ameren Missouri area, all projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, two are Baseline Reliability projects, four are Other projects, and ten are Generator Interconnection projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Baseline Reliability Projects:

P21967: Reconfigure the Dillon 138 kV Capacitor Bank (14 Mvar)

Project Description

This project is expecting to split the existing 28 Mvar capacitance output at Dillon by adding a breaker to the station and operating with two (2) 14 Mvar capacitors. This will reduce the voltage fluctuations caused during a P1-2 contingency event. The total estimated cost of this project is \$0.5 million dollars. The expected in-service date for this project is June 1, 2023.

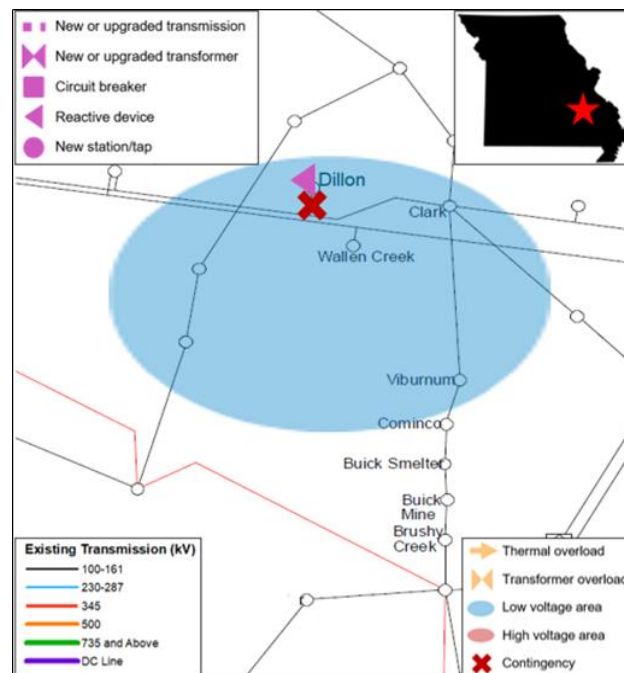


Figure 4.2.3-1: P21967 Geographic transmission map of project area

Project Justification

This project will split the existing 28 Mvar capacitance into two 14 Mvar banks reducing the voltage fluctuations caused during a P1-2 contingency event. The Ameren Local Planning Criteria (LPC) states that voltage fluctuation shall not exceed 3.3% (n-0) or 5% (n-1). In order to achieve this phenomenon, certain steps are required in the powerflow model simulation. 1) turn off capacitor (pre-contingency), 2) simulate contingency plus turn back on capacitor (lock taps, lock auto shunt adjustments). This shows the voltage fluctuation exceeding Ameren's LPC.

Cont. Type	Limiting Element	Voltage w/ cap online	Voltage w/ cap offline	Voltage w/ cap offline + (n-1) event	Voltage Fluctuation
Base case	[AMMO] 138 kV bus	1.02	0.99		2% (limit is 3.3%)
P1-2	[AMMO] 138 kV bus	1.02	0.99	1.10	11% (limit is 5%)
Base case	[AMMO] 138 kV bus	1.02	0.99		2% (limit is 3.3%)
P1-2	[AMMO] 138 kV bus	1.02	1.02	1.06	4% (limit is 5%)

Table 4.2.3-1: P21967 Project contingency drivers

Alternatives Considered

There were no other cheaper or better alternatives for this project proposal.

P21971: Upgrade [AMMO] Callaway 345 kV Substation

Project Description

The project will upgrade [AMMO] Callaway 345 kV Substation with new breakers to prevent Callaway generation from losing synchronism. This will be studied under our Transient Stability dynamics analysis for two faults from a breaker failure on V43 and V71. The total estimated cost of this project is \$3.0 million dollars. The expected in-service date for this project is June 1, 2023.

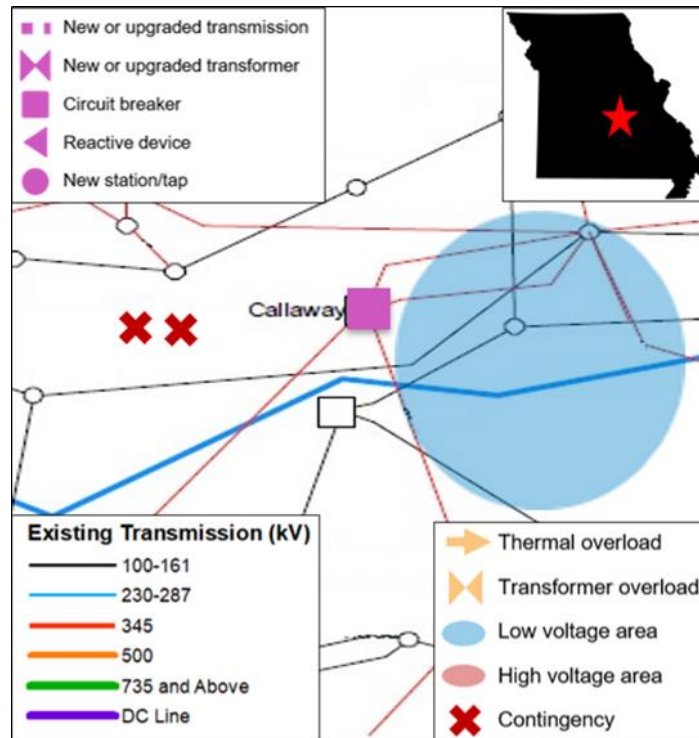


Figure 4.2.3-2: P21971 Geographic transmission map of project area

Project Justification

Several BES 138 kV buses in O'Fallon Missouri observe low voltages below an acceptable operating level in year 2023 for a NERC defined category P6-1-1 contingency event. Building this new capacitor bank will prop up the voltage levels in O'Fallon Missouri for the coincident P6-1-1 events.

Other Projects:

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18063	New Mt. Pleasant 345 kV substation	At the request of Mt. Pleasant Municipal Utilities (MPMU), Ameren Missouri to construct three breaker, 345 kV substation west of Mt. Pleasant, Iowa, along Ameren Missouri's Maywood-Sub T Transmission Line. Ameren Transmission Company of Illinois (ATXI) to install 345/69 kV step-down transformer and construct 69 kV networked loop to interconnect MPMU's distribution substations. MPMU to fund 69 kV facilities.	12/1/2025	\$2.6

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21948	Upgrade New Bourbon bus tie breaker	Replace B40 bus tie breaker.	5/31/2022	\$.4
21948	Rebuild T2B 34.5 kV line	Rebuild 34 kV line that feeds the Mosher B,C,&F plants and the Geneva switching station. This includes most of the switches in the line section.	3/31/2024	\$4.6

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21931	New Crosstown 69/12 kV Substation	Crosstown 69/12kV Substation Build between Park Tap and Altenburg 69kV substation (Citizens project).	6/1/2021	\$.8

Generator Interconnection Projects:

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22786	Upgrade Wildwood 345/138 kV transformer	Upgrade 560 MVA transformer to 700 MVA transformer.	4/1/2024	\$17.5
22807	New Overton 161 kV capacitor (67 Mvar)	Install 67 MVAR Cap bank at Overton 161 kV substation.	4/1/2024	TBD
22808	Upgrade Rush Island 345 kV substation	Upgrade 345 kV bus conductor at Rush Island to 3000 Amps.	4/1/2024	\$120
22365	New Adna 345 kV Ring Bus for J1107 Lutesville Solar	The Adna switching station will be a ring bus arrangement with three-line terminal positions and provisions for one additional future terminal position. The existing Kelso-Lutesville 345 kV transmission line will be cut.	11/1/2023	\$10.8
22406	Add J994 Guthrie Solar at Guthrie 161 kV	Add 161 kV terminal in the Guthrie substation for J994 project.	6/1/2023	\$1.7
22408	Add J987 Warren Solar at Montgomery 161 kV	Add 161 kV terminal in the Montgomery substation for the J987.	6/1/2023	\$2
22465	New Bullion 161 kV Ring Bus for J1087 Kelso Solar	Construct a new 161 kV switching station in Scott County, Missouri to provide a Point of Interconnection for the Generating Facility. The new 161 kV Bullion switching station will have a four-terminal ring-bus design with three terminal positions installed. Split	3/31/2024	\$9.2

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
		the existing Kelso – Miner 161 kV transmission line.		
22527	New Harley 161 kV Ring Bus for J1268 Winfield Solar	New Harley 161 kV Ring Bus for J1268 Winfield Solar.	6/1/2023	\$10.7
22529	New Vanduser 161 kV Ring Bus for J1034 Ringer Solar	Construct a new 161 kV switching station in Stoddard County, Missouri to provide a Point of Interconnection for the Generating Facility. The new Vanduser 161 kV J1034 Interconnection Switching Station will have a four-terminal ring-bus design with three terminal positions installed. Split the existing Morley – Stoddard 161 kV transmission line.	6/1/2024	\$6.7
22546	New Rootbeer 345 kV Ring Bus for J976 Split Rail Solar	Construct a new 345 kV switching station in Warren County, Missouri to provide a Point of Interconnection for the Generating Facility. The new Rootbeer 345 kV J976 Interconnection Switching Station will have a four-terminal ring-bus design with three terminal positions installed. Split the existing Belleau – Montgomery 345 kV transmission line.	6/1/2024	\$10.8

4.2.4 Big Rivers Electric Corporation (BREC)

Big Rivers Electric Corporation (BREC) is a member-owned, not-for-profit, generation and transmission cooperative (G&T) with headquarters in Henderson, KY. Big Rivers provides wholesale electric power and services to three distribution cooperative members across 22 counties in western Kentucky.

Big Rivers Electric Corporation (BREC) proposed three projects at an estimated cost of \$22.2 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the BREC area, all of these projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, all three are Other projects. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22766	Replace Structures on McCracken-Shell 69 kV line (EPR)	Raise or replace existing line structures to provide necessary ground clearance.	10/1/2022	\$1.5

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
20766	New Easton Rd 69 kV line	Add 1 mile tap line to serve new distribution substation.	2/1/2022	\$0.5
21685	New Pratt Paper Load Addition	Conversion of existing switching structure into a switching station and construction of an 8 mile radial transmission line to serve new industrial load.	7/1/2023	\$20.2

4.2.5 City of Columbia, MO (CWLD)

City of Columbia, MO, (CWLD) is a customer-owned utility company located in Columbia, MO. CWLD provides power to serve its nearly 50,000 residents of the City of Columbia with a peak electric load exceeding 250 MW. Its service area covers the city limits of Columbia. Its transmission system consists of 100 miles of both overhead and underground 161 kV and 69 kV network transmission system. The majority of Columbia's electricity comes from electric producers outside of Columbia.

City of Columbia, MO, (CWLD) did not submit any new projects for MTEP22. MISO has not identified any issues in CWLD area.

4.2.6 City of Springfield, IL (CWLP)

City of Springfield, IL, (CWLP) is the municipal electric and water utility for Springfield, IL. CWLP's generation capacity is provided by various fuel mix of generators with a total nameplate capacity of 723 MW. The CWLP electric system's transmission network consists of lines and associated substations operating at voltages of 138 kV and 69 kV. Its 138 kV portion of the transmission network currently includes approximately 63 circuit miles of overhead lines forming a complete loop around the system's service area. The 138 kV transmission lines presently serve nine of the system's substations or switching stations, plus the village of Chatham, IL.

City of Springfield, IL, (CWLP) did not submit any new projects for MTEP22. MISO has not identified any issues in CWLP area.

4.2.7 Duke Energy Corporation (DEI)

Duke Energy Corporation (DEI) is a regulated electric and gas delivery company with operations based in Cincinnati, OH. DEI provides power to serve 840,000 electric customers in Indiana across its 23,000 square miles service area. The DEI generation comes from its own coal and gas generation units totaling more than 6,600 MW. DEI also owns a good percentage of renewables and hydroelectric generation plants across Indiana.

Duke Energy Corporation (DEI) proposed 17 projects at an estimated cost of \$285.4 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the DEI area, all of these projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, six projects are a Baseline Reliability Projects, ten are Other projects and one project is a Generator Interconnection project with a signed Generator Interconnection Agreement. Most of the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Baseline Reliability Projects:

Project 22206: Replacing the [DEI] Dresser 138 kV Breaker

Project Description

Replace five [DEI] Dresser 138 kV over-dutied breakers with higher interruption capacity (63 KIAC rating). The total estimated cost of this project is approximately \$0.56 million dollars. The expected in-service date for this project is March 20, 2026.

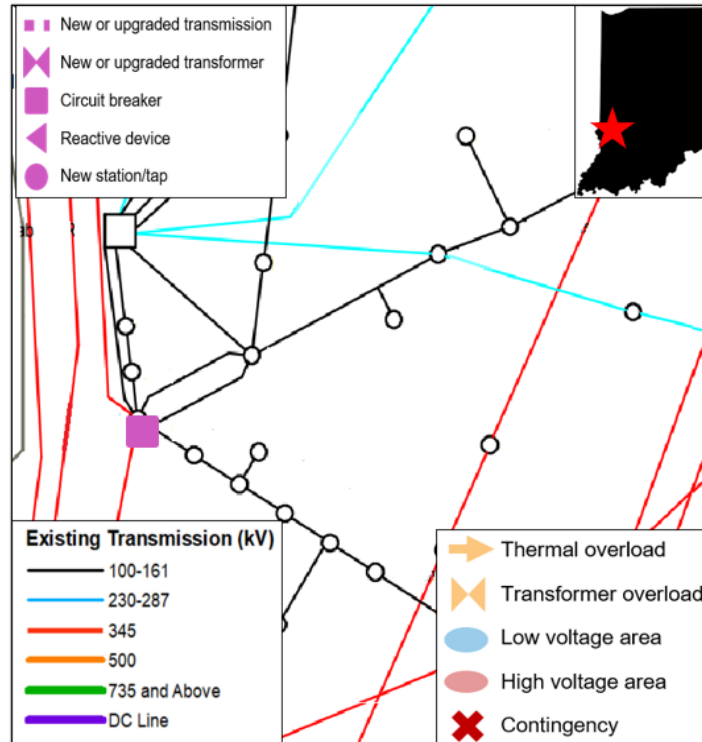


Figure 4.2.7-1: P22206 Geographic transmission map of project area

Project Need

The [DEI] Dresser 138 kV Breaker overload violations are caused by certain fault conditions at Dresser 138 kV substation in MTEP21 short circuit analysis. Replacing these breakers will address the overload violations caused by certain fault conditions.

Alternatives Considered

No other alternatives considered; this breaker replacement project is the best and cheapest option to address this reliability issue.

Project 22208 – Rebuild [DEI] Bedford—[DEI] Mitchell Lehigh 138 kV line

Project Description

The project will Rebuild [DEI] Bedford—[DEI] Mitchell Lehigh 138 kV line. The line rebuild will mitigate an overload on the same line for a P6-1-1 contingency event. The total estimated cost of this project is approximately \$17.96 million dollars. The expected in-service date for this project is June 1, 2027.

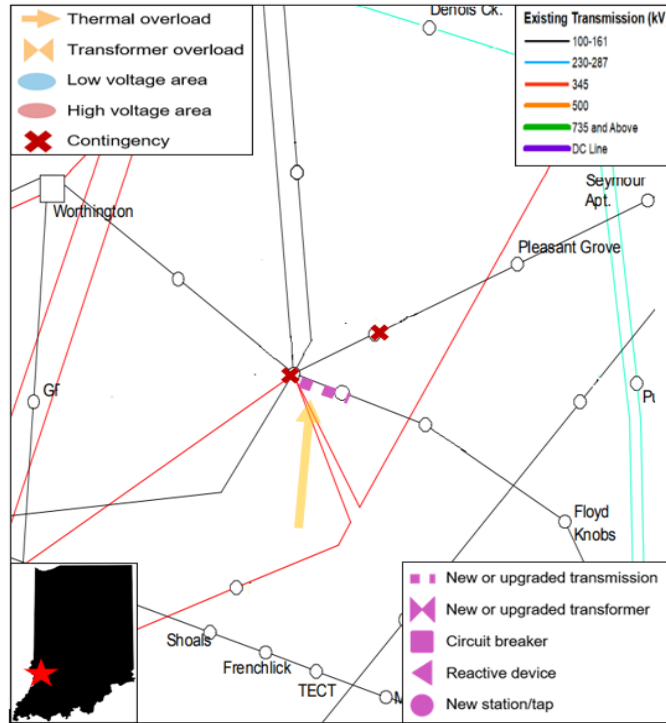


Figure 4.2.7-2: P22208 Geographic transmission map of project area

Project Need

The Rebuild [DEI] Bedford—[DEI] Mitchell Lehigh 138 kV line becomes overloaded to one hundred thirteen (113%) percent in year 2023 for a NERC defined category P6-1-1 contingency event. Upgrading this line will increase the summer emergency rating from 133 MVA to 301 MVA.

Cont.Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
n-0	[DEI] Bedford—[DEI] Mitchell Lehigh 138 kV line	133	69	32
P6-1-1	[DEI] Bedford—[DEI] Mitchell Lehigh 138 kV line	301	106	49

Table 4.2.7-1: P22208 Thermal loading drivers

Alternatives Considered

Generation redispatch or load shed is not feasible for thermal violations due to P6 contingency events if there is no applicable short-time emergency rating. No other alternatives considered; this line rebuild project is the best and cheapest option to address this reliability issue.

Project 22209 – Upgrade [DEI] Columbus—[DEI] Seymour 138 kV line

Project Description

The project will Upgrade [DEI] Columbus—[DEI] Seymour 138 kV line. The line upgrade will mitigate an overload on the same line for a P6-1-1 contingency event. The total estimated cost of this project is approximately \$0.5 million dollars. The expected in-service date for this project is June 1, 2027.

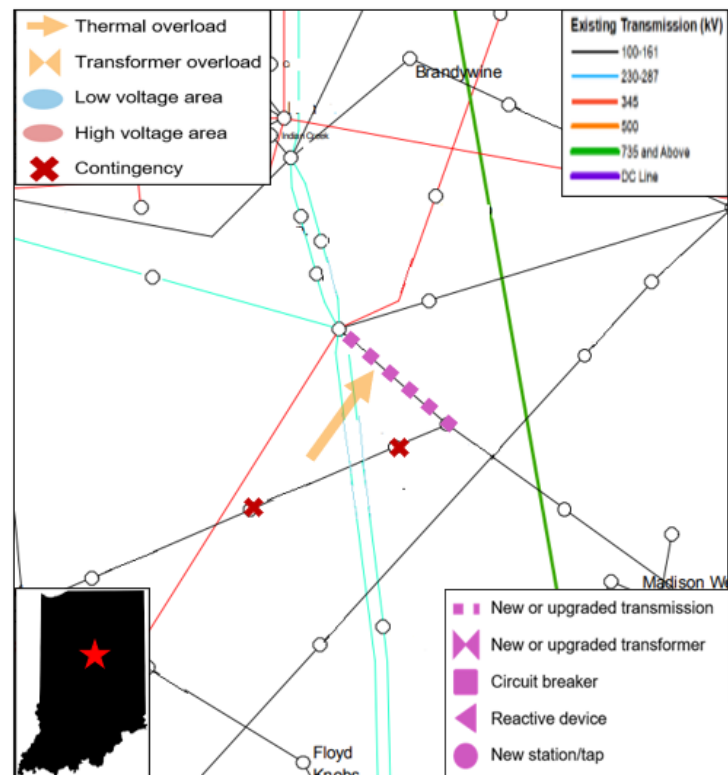


Figure 4.2.7-3: P22209 Geographic transmission map of project area

Project Need

The Upgrade [DEI] Columbus—[DEI] Seymour 138 kV line becomes overloaded 108% percent in year 2023 for a NERC defined category P6-1-1 contingency event. Upgrading this line will increase the summer emergency rating from 151 MVA to 178 MVA.

Cont.Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P0	[DEI] Columbus- [DEI] Seymour 138 kV line	151	20	18
P6-1-1	[DEI] Columbus- [DEI] Seymour 138 kV line	178	110	93

Table 4.2.7-2: P22209 Thermal loading drivers

Alternatives Considered

Generation redispatch or load shed is not feasible for thermal violations due to P6 contingency events if there is no applicable short-time emergency rating. No other alternatives considered; this line rebuild project is the best and cheapest option to address this reliability issue.

Project 22226: New [DEI] Wabash River 345/230 kV Substation

Project Description

The New [DEI] Wabash River 345/230 kV Substation will construct a new 345/230 kV substation with 682 MVA transformer. This project also constructs one Sanford 345 kV three circuit breaker ring bus in the 34513 Cayuga to Sugar Creek and 5.5 miles from Sanford to Wabash River Station 345 kV line. This new substation will mitigate thermal overloads for the loss of P2, P4 and P6 BES events. The total estimated cost of this project is approximately \$50 million dollars. The expected in-service date for this project is October 22, 2026.

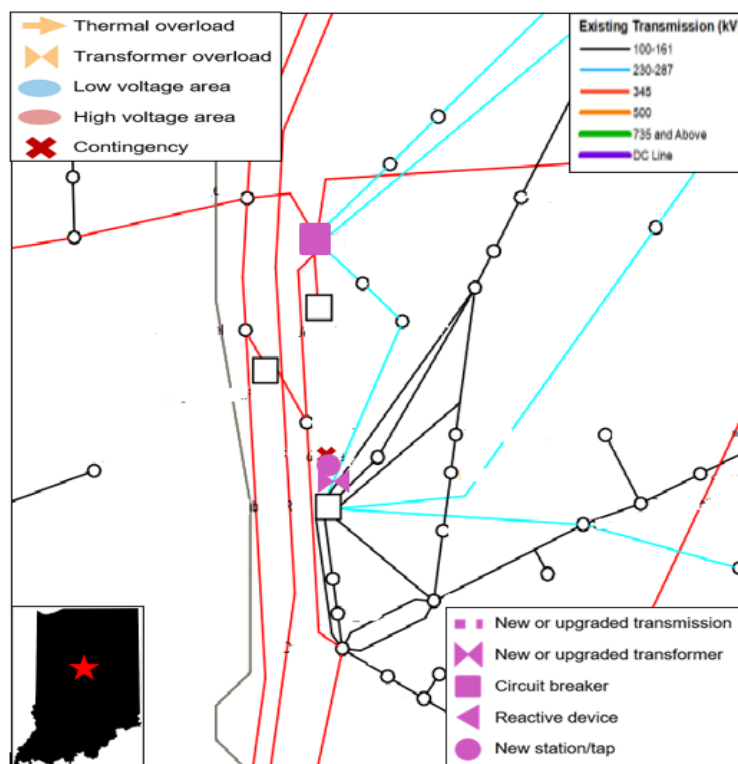


Figure 4.2.7-4: P22226 Geographic transmission map of project area

Project Need

There are a number of BES buses become overloaded beyond one hundred (100%) percent in a year 2023 for multiple TPL contingency events. Adding a New [DEI] Wabash River 345/230 kV substation will reduce the overload on these BES lines during the summer peak season.

Cont. Type	Limiting Element	Rating (MVA)	Pre-project Loading (%)	Post-project Loading (%)
n-0	[DEI] Dresser 345/138 kV transformer 1	550	46	36
n-0	[DEI] Dresser 345/138 kV transformer 2	550	46	36
n-0	[DEI] Dresser 345/138 kV transformer 3	581	45	35
n-0	[DEI] Allendale Jct 2 to [DEI] Dresser 138 kV line	291	65	54
n-0	[DEI] Allendale Jct 2 to [DEI] Margaret 138 kV line	237	69	55
n-0	[DEI] Fruitridge Jct to [DEI] Margaret 138 kV line	243	53	40
n-0	[DEI] Wabash River to Water St. 138 kV line	198	39	18
n-0	[DEI] Dresser to [DEI] Terre Haute S 1st St 138 kV line	291	57	44
n-0	[DEI] Wabash River 230/138/2.4 kV transformer	86	24	10
P6-1-1	[DEI] Dresser 345/138 kV transformer 1	550	110	81
P6-1-1	[DEI] Dresser 345/138 kV transformer 2	550	111	80
P4-3-3	[DEI] Dresser 345/138 kV transformer 3	581	105	76
P6-1-1	[DEI] Allendale Jct 2 to [DEI] Dresser 138 kV line	291	108	79
P6-1-1	[DEI] Allendale Jct 2 to [DEI] Margaret 138 kV line	237	121	85
P6-1-1	[DEI] Fruitridge Jct to [DEI] Margaret 138 kV line	243	104	69
P2-4	[DEI] Wabash River to [DEI] Water St. 138 kV line	198	114	63
P2-4	[DEI] Dresser to [DEI] Terre Haute S 1st St 138 kV line	291	109	75
P2-2	[DEI] Wabash River 230/138/2.4 kV transformer	86	109	37

Table 4.2.7-3: P22226 Thermal loading drivers

Alternatives Considered

No other alternatives were considered. This new substation project is the best and cheapest option to address these reliability issues.

Project 22227 - New [DEI] Staunton 230/138 kV transformer

Project Description

The project will add a New [DEI] Staunton 230/138 kV transformer. The transformer addition will mitigate an overload on [DEI] Staunton 230/138 kV transformers 1 and 2 for P6-1-1 contingency events. The total estimated cost of this project is approximately \$6 million dollars. The expected in-service date for this project is December 31, 2026.

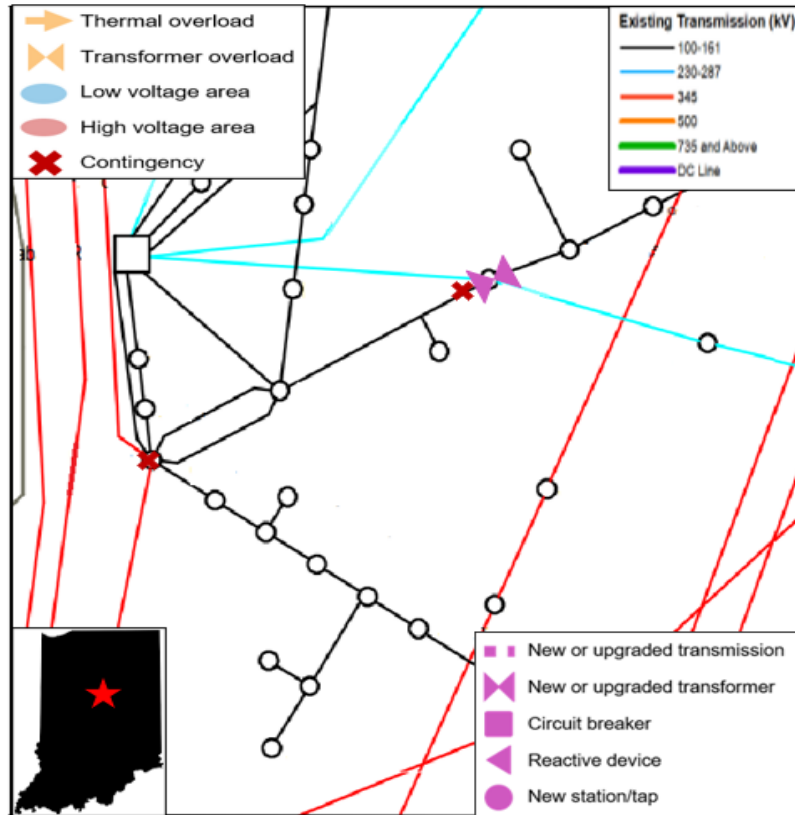


Figure 4.2.7-5: P22227 Geographic transmission map of project area

Project Need

The [DEI] Staunton 230/138 kV transformers are overloaded in a year 2023 Summer peak to 103% percent for a NERC defined category P6-1-1 contingency event. The addition of the third parallel transformer will mitigate this thermal overload on transformer 1 and 2.

Cont. Type	Limiting Element	Rating (MVA)	Pre-project Loading (%)	Post-project Loading (%)
n-0	[DEI] Staunton 230/130 kV transformer 1	181	28	29
n-0	[DEI] Staunton 230/130 kV transformer 2	181	27	29
P6-1-1	[DEI] Staunton 230/130 kV transformer 1	181	112	71
P6-1-1	[DEI] Staunton 230/130 kV transformer 2	181	109	71

Table 4.2.7-5: P22227 Thermal loading drivers

Alternatives Considered

Generation redispatch or load shed is not feasible for thermal violations due to P6 contingency events if there is no applicable short-time emergency rating. No other alternatives considered; the addition of the third transformer project is the best and cheapest option to address this reliability issue.

Project 22265: Rebuild [DEI] Purdue NW Tap—[DEI] Cincinnati St. 138 kV line

Project Description

The project will Rebuild [DEI] Purdue NW Tap—[DEI] Cincinnati St. 138 kV line. The line rebuild will mitigate an overload on the same line for a P6-1-1 contingency event. The total estimated cost of this project is approximately \$45 million dollars. The expected in-service date for this project is December 31, 2025.

This project is the replacement of P20230 in MTEP21 with additional line section and increased costs to the scope per the latest engineering estimate.

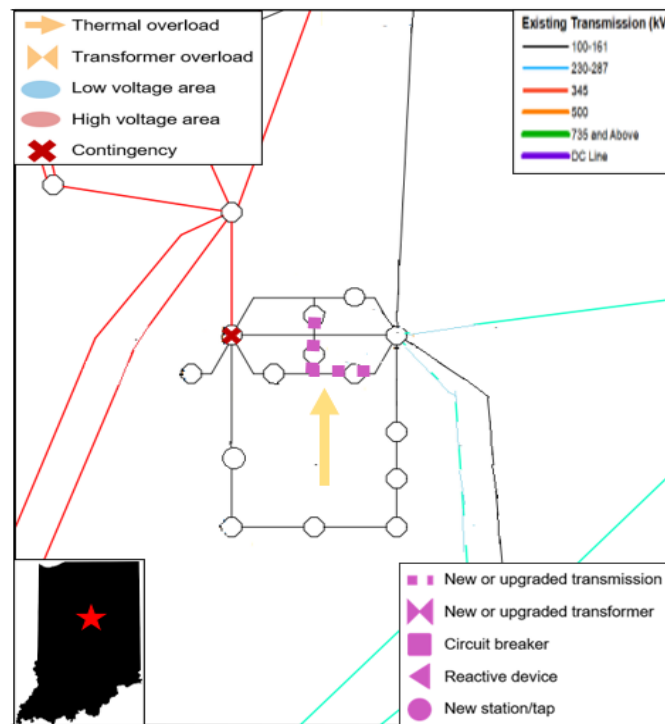


Figure 4.2.7-6: P22265 Geographic transmission map of project area

Project Need

The Rebuild [DEI] Purdue NW Tap—Cincinnati St. 138kV - 13820 Line becomes overloaded to 133% percent in year 2023 for a NERC defined category P6-1-1 contingency event. Rebuilding this line will increase the summer emergency rating of this line from 178 MVA to 301 MVA.

Cont. Type	Limiting Element	SE Rating (MVA)	Pre-project Loading (%)	Post-project Loading (%)
n-0	[DEI] Lafayette Cincinnati St - [DEI] Purdue 138 kV line	178	34	18
n-0	[DEI] Purdue NW Jct Tap 2 - [DEI] West Lafayette Airport	243	48	45
P6-1-1	[DEI] Lafayette Cincinnati St - [DEI] Purdue 138 kV line	301	107	57

Cont. Type	Limiting Element	SE Rating (MVA)	Pre-project Loading (%)	Post-project Loading (%)
P6-1-1	[DEI] Purdue NW Jct Tap 2 – [DEI] West Lafayette Airport	243	100	93

Table 4.2.7-6: P22265 Thermal loading drivers

Alternatives Considered

Generation redispatch or load shed is not feasible for thermal violations due to P6 contingency events if there is no applicable short-time emergency rating. No other alternatives considered; the addition of the third transformer project is the best and cheapest option to address this reliability issue.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
2125	Upgrade Centerton 138/69kV Bk 1	Upgrade/replace existing 75MVA 138/69kV bank with 150MVA bank.	9/11/2024	\$4
22210	Rebuild Collamer Jct to N. Manchester 69kV (6923) Line	Collamer Jct to N. Manchester 6923 (Pt2) Rebuild with 477ACSR@120C.	5/19/2023	\$7.3
22211	Rebuild and Reconfigure Huntington 138kV Sub	Huntington 138kV Sub rebuild and reconfigure: establish 4-position 138kV ring bus; replace (2)138/69/12 xfmr with (2)50MVA 138/69 xfmr and (1)22.4MVA 138/12 xfmr; establish 4-position 69kV ring bus.	7/19/2024	\$28.4
22213	Replace Dresser 69kV - 6901 Tower	Dresser 69kV - 6901 Tower Replacements - Replace 1.25 miles of steel lattice towers - six total (802-1012 to 802-1007, including the Wabash River crossing).	7/28/2023	\$4.8
22245	Improve 23008 Tipton W-Cammack Rd Line	Noblesville to Tipton West 23008 ckt: reinsulate and implement anti-gallop line conductor modifications.	11/30/2026	\$4.7

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22230	Upgrade Gibson 34509 Line Terminal	Gibson 345 Switchyard 34509 Line Terminal Upgrades: M210313 - (for Ameren Albion ring bus project) WT / Relays replacement; M210371 - replace (2)345kV CBs, free-standing CTs, and disconnect switches.	12/3/2023	\$3

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22207	New WVPA Hendricks Belle Union 138/12kV Distribution Sub	New 138/12 kV - 22.4MVA Belle Union substation for Hendricks Power to be constructed by WVPA. Substation 138kV - 13867 source to be looped through with SCADA controlled switches.	12/31/2023	\$1.4
22212	Rebuild Morgantown to Samaria Jct 69kV Line	Rebuild 6949 Line from Morgantown to Samaria Jct; 3.63 miles; 954ACSR@120C.	5/1/2023	\$2.9
22246	New IMPA Frankfort 300W 69/12kV Distribution Substation	IMPA Frankfort 300W 69/12kV Distribution Substation.	6/1/2023	\$5
22425	New West Lafayette Substation	West Lafayette New Customer: construct 345kV and 138kV facilities to serve new proposed industrial load of approximately 353MVA.	1/1/2025	\$100

Generator Interconnection Projects:

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22185	New J1063 Hardy Hills Solar	J1063 - Hardy Hills Solar.	8/1/2023	\$3.6

4.2.8 GridLiance Heartland LLC (GLH)

GridLiance Heartland LLC (GLH) is an electric utility that owns and operates transmission assets in Illinois and Kentucky. The assets include 50 miles of 161 kV transmission lines and a 161 kV switching station. The transmission assets are interconnected to the Joppa Generating Station in Illinois.

GridLiance Heartland LLC (GLH) did not submit any new projects for MTEP22. MISO has not identified any issues in GLH area.

4.2.9 Henderson Municipal Power & Light (HMPL)

Henderson Municipal Power & Light (HMPL) is a municipally owned generation, transmission, and distribution electric utility providing services to the City of Henderson, KY. Henderson owns a transmission network consisting of 56 miles of 161 kV and 69 kV facilities. HMPL is regulated by the City of Henderson, KY Utility Commission.

Henderson Municipal Power & Light (HMPL) proposed one project at an estimated cost of \$3.8 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the HMPL area, this project was recommended by MISO to be included in MTEP22 Appendix A as an Other Project. The project's expected in-service date and estimated cost are provided as of September 15, 2022.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22228	Upgrades HMPL Transmission Due to I-69 DOT Project	Rerouting of HMPL 69 kV transmission lines (HMPL Sub 6 to HMPL Sub 1, HMPL Sub 6 to (BREC) HMPL6 Tap, HMPL Sub 6 to HMPL Sub 3, and HMPL Sub 3 to (BREC) Wolf Hills Tap) due to I-69 KY DOT project.	6/30/2022	\$3.8

4.2.10 Hoosier Energy REC, Inc. (HE)

Hoosier Energy REC Inc. (HE) is a generation and transmission electric cooperative providing wholesale power and services to 18 electric cooperative members in central and southern Indiana and southeastern Illinois. HE's generation includes coal, natural gas and renewable energy resources and delivers power through a nearly 1,700-mile transmission network consisting of 345 kV, 161 kV, 138 kV, 69 kV, and 34.5 kV voltage levels.

Hoosier Energy REC, Inc. (HE) proposed five projects at an estimated cost of \$155.2 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the HE area, these projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, all five are Other Projects. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Other Projects:

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22646	New Decatur-Napoleon 161 KV Station	New 161 kV transmission station between Decatur and Napoleon.	10/1/2024	\$26
22648	New HE Prescott 161 kV Line	New 161 kV line from DEI Prescott to new 161 kV station.	6/1/2026	\$45
22666	New HE Batesville 161 kV Line	New 161 kV line from DEI Batesville to new 161 kV station (22646).	6/1/2027	\$45
22628	New HE Sleepy Hollow 345/161 Sub	New 345/161 kV transmission station and 161 kV transmission line to new station (22646).	6/1/2028	\$38
23266	New [HE] AboutBit 115 MVA load interconnection	Extend a new 69 kV transmission line from an available bay position at Merom transmission station to the new customer. The 69 kV line will split at about 0.3 miles from Merom station, extending to two separate substations. Total estimated load is 115 MVA.	1/1/2023	\$1.2

4.2.11 Indianapolis Power & Light Company (IPL)

Indianapolis Power & Light Company (IPL) is an investor-owned utility. IPL provides retail electric service to approximately 500,000 customers in Indianapolis, IN, and the surrounding communities. The IPL transmission system consists of approximately 458 circuit miles of lines at 345 kV, 408 circuit miles of 138 kV transmission and associated substations. IPL owns and operates four generating stations with a total generating capacity of 3,555 MW.

Indianapolis Power & Light Company (IPL) proposed three projects at an estimated cost of \$12 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the IPL area, these projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, all three are Other projects. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21836	Replace Sunnyside 345 kV and 138 kV Breakers	Replacement of three (3) 345 kV breaker #21, #22 & #23 and two (2) 138 kV breakers #1, & #3.	12/31/2024	\$5.1
21838	Replace Southwest 138 kV Breakers	Replacement of five (5) 138 kV breakers #4, #5, #3, #26, #27.	12/31/2024	\$4.5
21839	Replace West 138 kV Breakers	Replacement of three (3) 138 kV breakers #1, #4, #7.	12/31/2024	\$2.4

4.2.12 Northern Indiana Public Service Company (NIPSCO)

Northern Indiana Public Service Company (NIPSCO) is a vertically-integrated gas and electric company providing electric generation, transmission, and distribution services as well as natural gas distribution service across the northern third of Indiana. NIPSCO is headquartered in Merrillville, IN and is a subsidiary of NiSource Inc. (NiSource). The NIPSCO transmission system consists of 21 circuit miles of 765 kV lines, 453 circuit miles of 345 kV lines, and 810 circuit miles of 138 kV lines.

Northern Indiana Public Service Company (NIPSCO) proposed 17 projects at an estimated cost of \$502.8 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the NIPSCO area, these 17 projects have been recommended by MISO to be included in MTEP22 Appendix A at an estimated cost of \$502.8 million. Of these projects, zero are Baseline Reliability Projects, fifteen are Other projects, and two projects are Generator Interconnection projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21167	New Menges Ditch 138/69 kV Substation	New 138/69 kV Substation source from AEP's E. Elkhart substation.	12/31/2025	\$42.3
21168	New Hiple to Northport 138 kV Line	New Hiple to Northport 138 kV line.	12/31/2024	\$27.5

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21867	Rebuild Marktown 138 kV Substation	Rebuild Marktown 138 kV Substation.	12/31/2024	\$30.1
21870	Rebuild S. Chalmers - Oakdale 69 kV Circuit 6972	Rebuild circuit 6972, South Chalmers to Oakdale substations.	12/31/2022	\$4.9
21875	Rebuild Lagrange 69 kV line-Circuit 6980	Rebuild 3.4 miles of circuit 6980.	12/31/2023	\$3
21876	Upgrade Transmission Substation - 2022 Program	Transmission Age & Condition Projects.	12/31/2022	\$26.2
21878	Upgrade transmission lines - 2022 program	Rebuilds/upgrades 69 kV and up line based on age and condition, including structures, conductor, switches and communication upgrades (fiber).	12/31/2022	\$36.8
21879	Upgrade Transmission Substation - 2023 Program	Upgrades/replacements 69 kV and up Substation based on age and condition, including but not limited to switches, breakers, transformers, and protection and communication equipment.	12/31/2023	\$42.3
21880	Upgrade transmission lines - 2023 program	Rebuilds/upgrades 69 kV and up line based on age and condition, including structures, conductor, switches and communication upgrades (fiber).	12/31/2023	\$19.3
21881	Upgrade Transmission Substation - 2024 Program	Upgrades/replacements 69 kV and up Substation based on age and condition, including but not limited to switches, breakers, transformers, and protection and communication equipment.	12/31/2024	\$32.7
21882	Upgrade transmission lines - 2024 program	Rebuilds/upgrades 69 kV and up line based on age and condition, including structures, conductor, switches and communication upgrades (fiber).	12/31/2024	\$31.3
21883	Upgrade Transmission Substation - 2025 Program	Upgrades/replacements 69 kV and up Substation based on age and condition, including but not limited to switches, breakers, transformers, and protection and communication equipment.	12/31/2025	\$26.9
21884	Upgrade transmission lines - 2025 program	Rebuilds/upgrades 69 kV and up line based on age and condition, including structures, conductor, switches and communication upgrades (fiber).	12/31/2025	\$30.2
21885	Upgrade Transmission Substation - 2026 Program	Upgrades/replacements 69 kV and up Substation based on age and condition, including but not limited to switches, breakers, transformers, protection and communication equipment.	12/31/2026	\$54.9

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21886	Upgrade transmission lines - 2026 program	Rebuilds/upgrades 69 kV and up lines based on age and condition, including structures, conductor, switches and communication upgrades (fiber).	12/31/2026	\$68.3

Generator Interconnection Projects:

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22630	Reynolds to Burr Oak 345 kV line (J1067)	Network upgrades necessary to connect J1067 to interconnect at NIPSCO's Reynolds to Burr Oak 345 kV line.	10/20/2023	\$23.6
22649	Reynolds 345 kV substation upgrades (J968 & J1069)	New 345 kV breaker in Bay 6 at Reynolds substation to accommodate new lead line into northern position, plus all associated equipment .	4/30/2023	\$2.2

4.2.13 Pioneer Transmission, LLC

Pioneer is a joint venture between Duke Energy and American Electric Power formed to build, own and operate the Reynolds to Greentown 765 kV transmission line.

Pioneer Transmission, LLC did not submit any new projects for MTEP22. MISO has not identified any issues in the Pioneer Transmission area.

4.2.14 Prairie Power, Inc. (PPI)

Prairie Power, Inc. (PPI) is a member-owned, not-for-profit electric generation and transmission cooperative. PPI produces and supplies wholesale electricity to ten electric distribution cooperatives in central Illinois. PPI's distribution cooperatives provide retail electric service to approximately 78,000 members within its local service territories. PPI owns and operates approximately 590 miles of transmission lines at 138 kV, 69 kV and 34.5 kV voltage levels. PPI generates 141 MW of oil and gas-fired peaking units and 79 distribution and transmission substations to serve its members.

PPI proposed 24 projects at an estimated cost of \$85 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the PPI area, these 24 projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, one project is a Baseline Reliability Project, and 23 projects are Other projects, and there are zero Generator Interconnection Projects with signed a Generator Interconnection Agreement. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Baseline Reliability Projects:

P21794: New Forest City 138/69 kV Substation

Project Description

The new [PPI] **New Forest City 138/69 kV Substation** will connect to the Ameren [AMIL] Havana—[AMIL] Cincinnati 138 kV line in order to provide adequate backup voltage support for the loss of the [AMIL] Havana—[AMIL] Cane 138 kV line plus the [AMIL] Quiver—[AMIL] Mason City 138 kV line. The total estimated cost of this project is \$7.4 million dollars. The expected in-service date for this project is December 31, 2025.

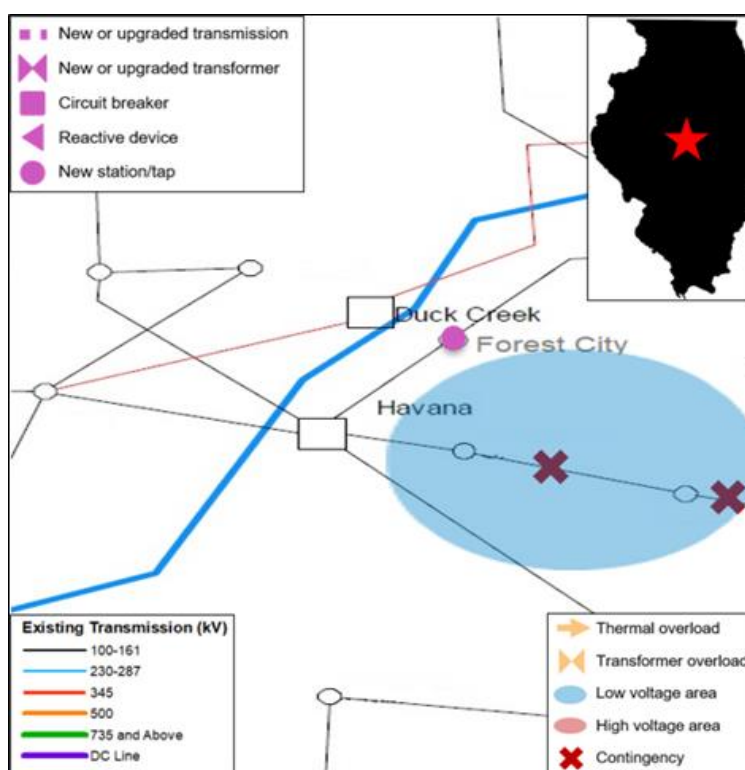


Figure 4.2.14-1: P21794 Geographic transmission map of project area

Project Justification

The addition of the new [PPI] New Forest City 138/69 kV Substation improves the voltages in and around Grand Island during the 2027 SPK season for a NERC defined category P6-1-1 contingency event. See Table P21794-1 and Figures One-Line P21794-1 through One-Line P21794-4.

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
Base case	[AMIL] Cane 138 kV bus	0.95	1.02	1.02
P6-1-1	[AMIL] Cane 138 kV bus	0.93	0.44	.094
Base case	[AMIL] Mason City 138 kV bus	0.95	1.02	1.02
P6-1-1	[AMIL] Mason City 138 kV bus	0.93	0.44	0.94
Base case	[PPI] Grand Island 138 kV bus	0.95	1.02	1.02
P6-1-1	[PPI] Grand Island 138 kV bus	0.93	0.44	0.94

Table 4.2.14-1: P21794-1 Project Contingency Drivers

Alternatives Considered

This was the best solution to address the need for adding a back-up voltage support station.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
19945	New Topeka-Bishop 69 kV Line	New Topeka--Bishop 69 kV line (~8-mile) w/ 556 MCM in MEC.	12/31/2025	\$5.1
19950	New St. David-Wee-Ma-Tuk-Canton 69 kV line	New St. David-Wee-Ma-Tuk-Canton 69 kV line (10.4-mile) in Spoon River Electric Cooperative.	12/31/2023	\$5.6
21746	New Bluff Hall-E. Hannibal 69 kV line	New Bluff Hall-E. Hannibal 69 kV line (~9-mile) 69 kV Line.	12/31/2022	\$4.2
21749	New Trailblazer 69 kV Substation	New Trailblazer 69 kV breaker substation near Marblehead, IL.	12/31/2022	\$2.9
21750	New Rock Creek 69 kV Substation	New Rock Creek 69 kV breaker station near Ursa substation in Adams Electric Cooperative. Breaker station will have terminals looking towards Elvaston, Golden, Payson and Ursa.	12/31/2025	\$2.9
21751	New Panther 138/69 kV Substation	New Panther 138/69 kV substation near Pana, IL in Shelby Electric Cooperative.	12/31/2025	\$8.4

21752	New Panther-Pana Tap 69 kV line	New Panther-Pana Tap 69 kV line (~2.5-mile) in Shelby Electric Cooperative.	12/31/2025	\$2.3
21753	New Pana Tap-Lakewood 69 kV line	New Pana Tap-Lakewood 69 kV line (~12-mile) in Shelby Electric Cooperative.	12/31/2025	\$7.7
21754	New Herrick-ISHI 69 kV line	New Herrick-ISHI 69 kV line (~9.7-mile) in Shelby Electric Cooperative. This is part of the line from Panther to ISHI.	12/31/2025	\$5.5
21760	New Rushville 69 kV Substation	New 69 kV Rushville breaker substation in Adams Electric Cooperative. Terminals will look to Golden and Liberty.	12/31/2024	\$2.9
21791	Rebuild E. Lanesville-Lanesville 69 kV line	Rebuild E. Lanesville-Lanesville 69 kV line (~0.9-mile) in Menard Electric Cooperative. Replace 336.4 MCM ACSR line built in 1981 with 556.5 MCM ACSR conductor.	12/31/2024	\$4.5
21793	New Forest City-Bishop 69 kV line	New Forest City-Bishop 69 kV line (~2-mile) in Menard Electric Cooperative.	12/31/2025	\$1.4
21795	New Middletown 69 kV Substation	New 69 kV breaker substation near Middletown, IL in Menard Electric Cooperative. Substation will have terminals looking to Fancy Creek, Grand Island, Forest City, and Middletown Distribution substation.	12/31/2025	\$2.1

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
19952	Rebuild Walkerville-Eldred 69 kV line	Rebuild Walkerville-Eldred 69 kV line (7.3-mile) in Illinois Electric Cooperative.	12/31/2023	\$3.9
21755	Rebuild Hadley-E. Hannibal Tap 69 kV line	Rebuild Hadley-E. Hannibal Tap 69 kV line (0.54-mile) in Illinois Electric Cooperative.	12/31/2025	\$0.27
21756	Rebuild Hadley Breaker-Hadley 69 kV line	Rebuild Hadley Breaker-Hadley 69 kV line (1.61-mile) in Illinois Electric Cooperative	12/31/2025	\$0.8
21757	Rebuild E. Hannibal Tap-Liberty 69 kV line	Rebuild E. Hannibal Tap-Liberty 69 kV line (6.35-mile) between Illinois Electric Cooperative and Adams Electric Cooperative.	12/31/2023	\$3.2
21758	Rebuild Kampsville-Kampsville River Crossing 69 kV line	Rebuild Kampsville-Kampsville River Crossing 69 kV line (0.7-mile) in Illinois Electric Cooperative.	12/31/2025	\$0.35

21788	Rebuild Andrew-Barclay 69 kV line	Rebuild Andrew-Barclay 69 kV line (~7.5-mile) in Menard Electric Cooperative. Replace 4/0 ACSR line built in 1964 with 556.5 MCM ACSR conductor.	12/31/2024	\$3.8
21789	Rebuild Lanesville - Barclay 69 kV Line	Rebuild ~11.5-mile Lanesville to Barclay 69 kV line in Menard Electric Cooperative. Replace 4/0 ACSR line built in 1964 with 556.5 MCM ACSR conductor.	12/31/2025	\$5.7
21792	Rebuild Fancy Creek-Andrew 69 kV line	Rebuild Fancy Creek-Andrew 69 kV line (~4.5-mile) in Menard Electric Cooperative. Replace 336.4 MCM ACSR line built in 1969 with 556.5 MCM ACSR conductor.	12/31/2026	\$2.3

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21747	New Barry 69 kV Substation	New Barry 69 kV Transmission Substation near Barry, IL. Terminals to Bluff Hall, Liberty, Pittsfield, and Barry Distribution substation.	12/31/2022	\$1.6
21748	Rebuild E. Hannibal Tap-Barry 69 kV line	Rebuild E. Hannibal Tap-Barry 69 kV line to double circuit. Split the Pittsfield to Liberty 69 kV line into a Pittsfield to Barry circuit and a Liberty to Barry circuit.	12/31/2022	\$3.8

4.2.15 Republic Transmission, LLC (RTx)

Republic Transmission, LLC (Republic Transmission) is an electric transmission utility company, with the mission to ensure reliable and cost-effective electric transmission in the Central region of the MISO transmission system. Republic Transmission is a subsidiary of LS Power, a power generation, transmission, and investment group.

Republic Transmission owns and operates a 345 kV transmission line connecting the Vectren Duff substation to the Big Rivers 345 kV high voltage electrical system ultimately terminating at the Big Rivers Coleman Substation.

Republic Transmission did not submit any new projects for MTEP22. MISO has not identified any issues in the area.

4.2.16 Southern Indiana Gas & Electric Company (SIGE)

Vectren Energy Delivery (doing business as Southern Indiana Gas and Electric (SIGE)) is a subsidiary of CenterPoint Energy, headquartered in Houston, Texas. Vectren provides electricity to 144,000 customers in southwestern Indiana. Vectren owns and operates 63.8 miles of 345 kV transmission lines, 375.4 miles of 138 kV transmission lines, 562.9 miles of 69 kV transmission lines. Vectren (SIGE) serves a peak load of approximately 1.1 GW and owns and operates 1 GW of coal generation, 150 MW of natural gas generation, 4 MW solar generation, and 1 MW of battery storage.

Southern Indiana Gas & Electric Company (SIGE) proposed nine projects at an estimated cost of \$41.2 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the SIGE area, all nine of these projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, zero are Baseline Reliability Projects, seven projects are Other Projects, and two projects are Generator Interconnection projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22004	Replace Southwind 69/12 kV T2 transformer	Replace 69/12 kV distribution transformer T2 at the Southwind substation.	6/1/2023	\$2.0
22005	Replace Duff 345/138 kV substation 345 kV breakers	Replace two 345 kV breakers (#s 3200 and 3300) at the Duff 345/138 kV substation.	6/1/2023	\$1.5

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22006	New Anderson Road 69/12 kV Substation	Build new 69/12 kV distribution substation.	6/1/2023	\$8.5
22007	Rebuild Warrick North 138 kV switching station	Expand Warrick North 138 kV switching station from ring bus to breaker and a half to allow for new terminal for large industrial customer.	7/1/2023	\$3.5
22008	New Kaiser Aluminum 138 kV Sub	Build new 138 kV substation for new large industrial customer.	7/1/2023	\$12.0
22009	New Midwest Fertilizer 138 kV Sub	Build new 138 kV substation on existing 138 kV line for large industrial customer.	12/1/2023	\$8.0
22011	New Warrick N. - Kaiser 138 kV Line	New 138 kV line from existing Warrick North 138 kV switching station to new Kaiser 138 kV substation to serve large industrial customer.	7/1/2023	\$2.3

Generator Interconnection Projects:

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22002	Francisco Gen Interconnection (J1074)	Francisco Substation modifications needed for the interconnection of Queue	12/15/2022	\$1.8
22010	Scott Gen Interconnection (J856)	New 138 kV terminal at existing Scott 138/69 kV substation to facilitate	10/1/2023	\$1.5

4.2.17 Southern Illinois Power Cooperative (SIPC)

Southern Illinois Power Cooperative (SIPC) is a generation and transmission cooperative providing wholesale electric power to seven-member distribution cooperatives and two wholesale customers in southern Illinois. Its member cooperatives provide electricity to more than 100,000 end-use customers. SIPC generates around 600 MW from coal fired and natural-gas fired generation plants and owns more than 900 miles of 161 kV, 138 kV and 69 kV transmission lines.

Southern Illinois Power Cooperative did not submit any new projects for MTEP22. MISO has not identified any issues in the SIPC area.

4.2.18 Wabash Valley Power Association, Inc. (WVPA)

Wabash Valley Power Association, Inc. (WVPA) is a generation and transmission cooperative providing wholesale electric power to 23-member distribution cooperatives in Illinois, Indiana, and Missouri. Its member cooperatives provide electricity to more than 321,000 end-use customers. WVPA generates 33% of its energy from coal and 11% from natural gas fired generation plants; WVPA owns 744 miles of transmission lines ranging from 345 kV to 34.5 kV along with 94 substations.

WVPA proposed 13 projects at an estimated cost of \$43.28 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the WVPA area, all 13 projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, all 13 are categorized as Other Projects. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Other Projects:

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21948	Upgrade New Bourbon bus tie breaker (B40)	Replace B40 bus tie breaker.	12/31/2022	\$0.4
21949	Rebuild T2B 34.5 kV line	Rebuild 34 kV line that feeds the Mosher B,C,&F plants and the Geneva switching station. This includes most of the switches in the line section.	3/31/2024	\$4.6
21959	Upgrade Moody 69 kV sub	Purchase the Moody Substation from NIPSCO and replace the transformer (14.4MVA from Fair Oaks), regulators, fuses, and jumpers. Add oil containment around a new foundation.	9/1/2022	\$1.5
21965	Upgrade Otter sub equipment	Otter substation in Steuben has an inoperable switch and transrupter that need replaced. Replacing switch with a load break manual switch and replacing the transrupter with a S&C Circuit Switcher.	7/31/2023	\$0.9

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21950	New Timmons 69/12 kV substation	Build the new Timmons 69/12 kV 14.4 MVA sub.	12/31/2023	\$5.3
21954	New Rosebud 69/12kV Sub	Building new Rosebud 69/12 kV substation to feed load normally fed from existing Parr substation.	11/29/2022	\$4.9
21955	Upgrade Fair Oaks 2nd bank 138/69/12 kV transformer	Add second 138/69/12 kV 22.4 MVA transformer to Fair Oaks substation to handle new load in area.	11/17/2024	\$4.2
21956	New Yellow River 69 kV Sub	Build a new 69 kV/12.5 kV 14.4 MVA substation near the Knox Industrial Park. Install 13 Mvar cap bank. Build 0.5 miles of 69 kV to new sub.	12/31/2023	\$5.3
21957	Upgrade Oregon 69 kV Sub	Replace existing 7 MVA substation transformer at the KVREMC owned Oregon Substation to a 14 MVA. Add a backup transmission line for Oregon Substation and tap the nearby 69105 (0.25mi). New ATO scheme to be installed on KVL property on transmission line poles and to be owned by WVPA.	12/31/2023	\$1.3
21961	New Marco 69/12 kV substation	Build new Marco substation near the existing NIPS Argos substation. 69/12kV 14.4 MVA Xfmr, design for 4 ckts used initially. Will replace Marshall MP. Add external control Hut for Fiber connections	12/30/2022	\$4.9

		(Rochester Fiber). Build short tap from Nipsco 6968 to new sub.		
21962	New Berthaville 69/12 kV sub	New Berthaville substation needed south of Goodland for new dairy farm and added load near Newton office. 14.4 MVA transformer, 4 distr. exits, 69kV with manual switch to Nipsco 6964 line.	9/30/2022	\$4.2
21964	Upgrade Colfax 138/69/12 kV substation	Add 2nd transformer to Colfax substation. 138/69/12 kV 22.4 MVA transformer. Substation work to make space for new transformer.	11/30/2023	\$5

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21931	New Crosstown 69/12 kV Substation	Crosstown 69/12kV Substation Build between Park Tap and Altenburg 69 kV substation (Citizens project).	6/1/2021	\$0.8

4.3 Project Justifications – East Region

East Region Overview

The MISO East Planning Region consists of six Transmission-Ownning members within Michigan. These Transmission Owners are:

- ITC Transmission (ITCT)
- Michigan Electric Transmission Co. (METC)
- Wolverine Power Supply Cooperative Inc. (WPSC)
- Michigan Public Power Agency (MPPA)
- Michigan South Central Power Agency (MSCPA)
- Lansing Board of Water & Light (LBWL)

The region contains 9,830 circuit miles of transmission lines ranging from 120 kV to 345 kV. It also contains 1216 circuit miles of 69 kV sub-transmission system. The MISO East Region is interconnected with non-MISO systems: Hydro One Networks Inc. and American Electric Power to the east.

The 2023 Summer Peak planning model indicates the region contains more than 30.3 GW of generation. Installed generation capacity in the region consists mostly of coal, gas, and wind. Figure 4.3-1 shows the major load centers and generation pockets within the East Region. The load centers are typically found around larger cities in the region, i.e., Detroit, Lansing, and Grand Rapids. According to the 2023 Summer Peak planning model, the region's load exceeds 20.56 GW.

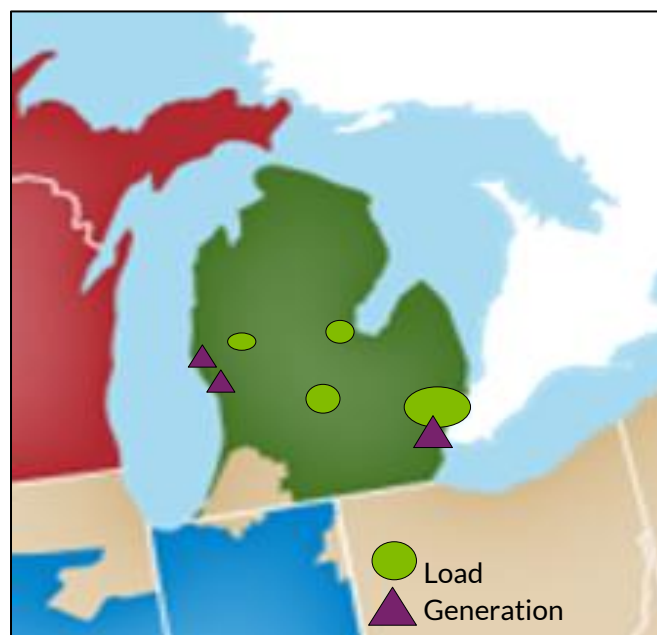


Figure 4.3-1: Generation and load centers in the East planning region

For MTEP22 as of September 15, 2022, MISO Transmission Planning is reflecting 46 projects from the East region for inclusion in Appendix A at an estimated cost of \$439 million. Of these, six are Baseline Reliability Projects and seven are Generator Interconnection Projects. The remaining 33 projects are classified as Other projects because they do not meet the criteria to be considered as Baseline Reliability Projects, New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects.

Of the 46 projects, that are being recommended to be included in MTEP22, four have an estimated cost of less than \$1 million, 24 have an estimated cost between \$1 million and \$5 million, and the remaining 18 projects are estimated to cost greater than \$5 million (shown in Figure 4.3-2).

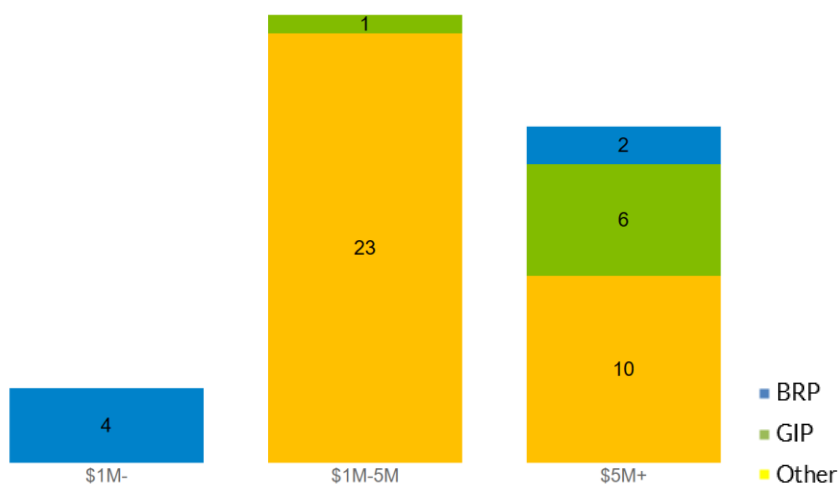


Figure 4.3-2: Project cost category by project type for MTEP22 East projects (as of 9-15-2022)

The projects in the MISO East planning region are expected to go into service in the next five years as shown in Figure 4.3-3. There are 0 GIP projects that are being approved in MTEP22 with in-service dates in 2022.

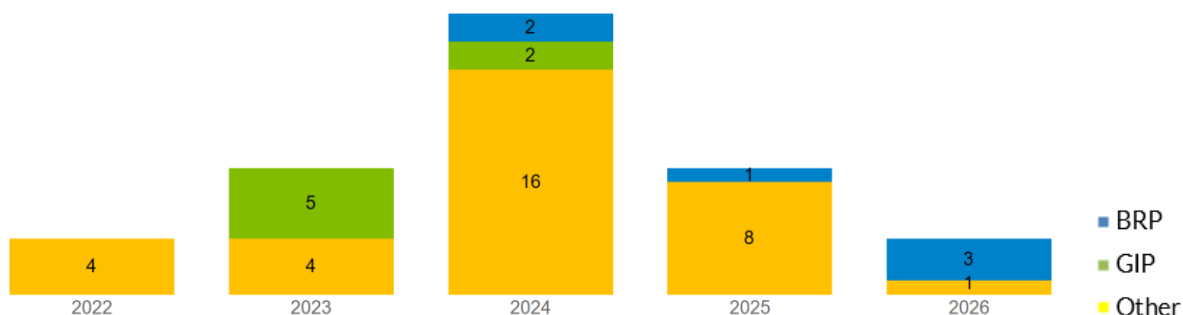


Figure 4.3-3: Project in-service date by project type for MTEP22 East projects (as of 9-15-2022)

In accordance with Attachment FF of the MISO tariff, if a Transmission Owner determines system conditions warrant the urgent development of system enhancements, MISO will perform an expedited review of the impacts of the project. MISO shall use a streamlined approval process for reviewing and approving such projects proposed by the Transmission Owner(s) so that decisions will be provided to the Transmission Owner within 30 Days of the project's submittal to MISO, unless a longer review period is mutually agreed upon. During the MTEP22 cycle, MISO received the following project through the Expedited Project Review (EPR) process:

1. Project ID 23246, Riverbend Load Interconnection

Also, in accordance with Attachment FF Section VIII.A.3, the following project was identified as an Immediate Need Reliability Project and is excluded from the competitive developer selection process:

- None

The ten highest cost MISO East projects represent \$222 million (46%) of the \$487million total recommended projects for the region in MTEP22. The locations of these projects are shown in Figure 4.3-4 with the investment spread across the East Planning region. Projects that are blanket expenditures (such as relays, physical security, etc.) are excluded from this list.



Figure 4.3-4: East region top ten projects, by cost (data as of 9-15-2022)

4.3.1 ITC Transmission (ITCT)

ITCT proposed 17 projects at an estimated cost of \$92 million targeting MTEP22 Appendix A. Of these projects, seven are Baseline Reliability Projects, nine are Other projects and one is a Generator Interconnection project with a signed Generator Interconnection Agreement. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022. After MISO's independent reliability analysis only four of the Baseline Reliability Projects were approved for inclusion in MTEP22 Appendix A, therefore the remaining three have been moved to appendix B.

Baseline Reliability Projects:

ITCT proposed seven Baseline Reliability Projects targeting Appendix A in MTEP22. MISO's independent reliability analysis recommends one Baseline Reliability Project to be included in MTEP22 Appendix A. MISO identified open issues in the ITCT area as listed in Appendix D3 (CEII) and will be coordinating with ITCT to develop a solution in MTEP22.

Project 17999 – Re-conductor Corktown – Waterman 120kV UG

Project Description

Re-conductor approximately 3.7 miles of the Corktown – Waterman 120 kV underground cable utilizing 5000 Kcmil underground cable.

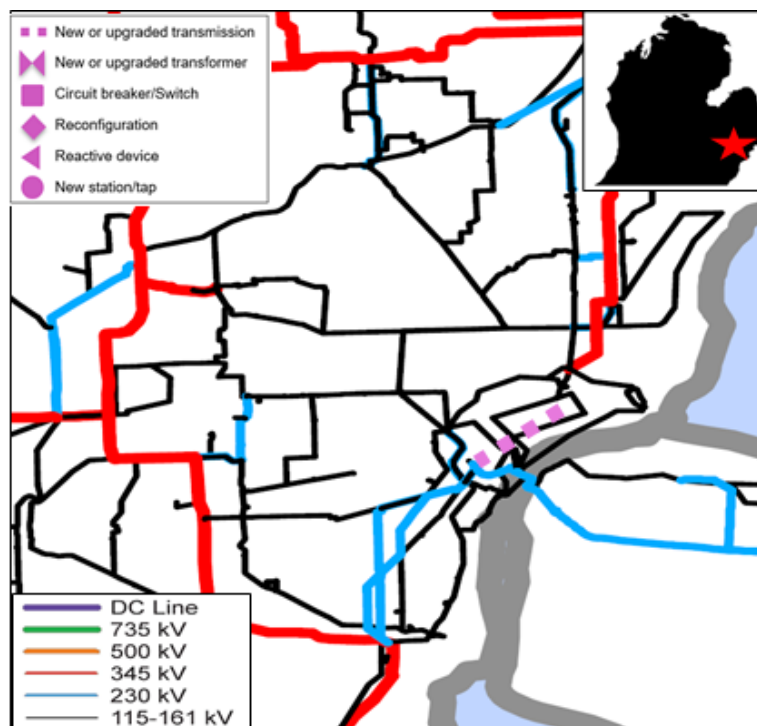


Figure 4.3.1-1: P17999 Geographic transmission map of project area

Project Need

The Corktown – Waterman 120 kV underground cable is projected to marginally overload.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
N1	264984 19WTRMN4 120 265667 19CORKTWN2 120 1	241	115	<99
P6	264984 19WTRMN4 120 265667 19CORKTWN2 120 1	241	155	<99

Table 4.3.1-1: P17999 Thermal loading drivers

Alternatives Considered

No alternatives were considered.

Project 21745 – Cato – Waterman 120kV Re-conductor

Project Description

The Cato – Waterman 120 kV underground cable is projected to overload for P1-1 and P6 contingencies during peak load. The limiting elements on this circuit are the cable conductor.

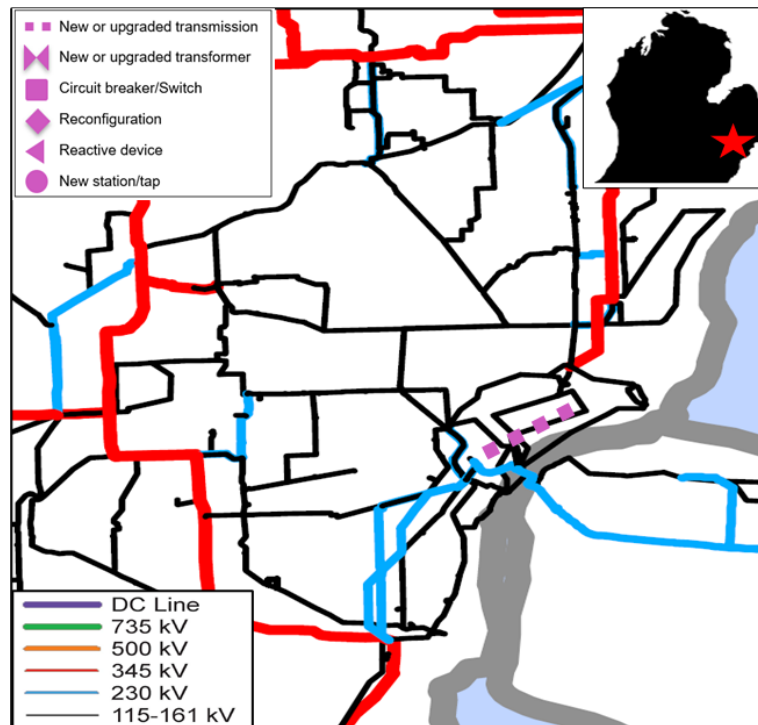


Figure 4.3.1-2: P21745 Geographic transmission map of project area

Project Need

The proposed solution is to re-conductor approximately 4.43 miles utilizing 1600MM2 CU XLPE underground cable.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
N1	264541 19CATO1 120 264985 19WTRMN5 120 1	241	110	<99
P6	264541 19CATO1 120 264985 19WTRMN5 120 1	241	155	<99

Table 4.3.1-2: P21745 Thermal loading drivers

Alternatives Considered

No alternatives were considered.

Project 15875 – Kentucky – Luzon 120kV Sag Remediation

Project Description

Completely remove the sag limit on the Kentucky – Luzon 120 kV line.

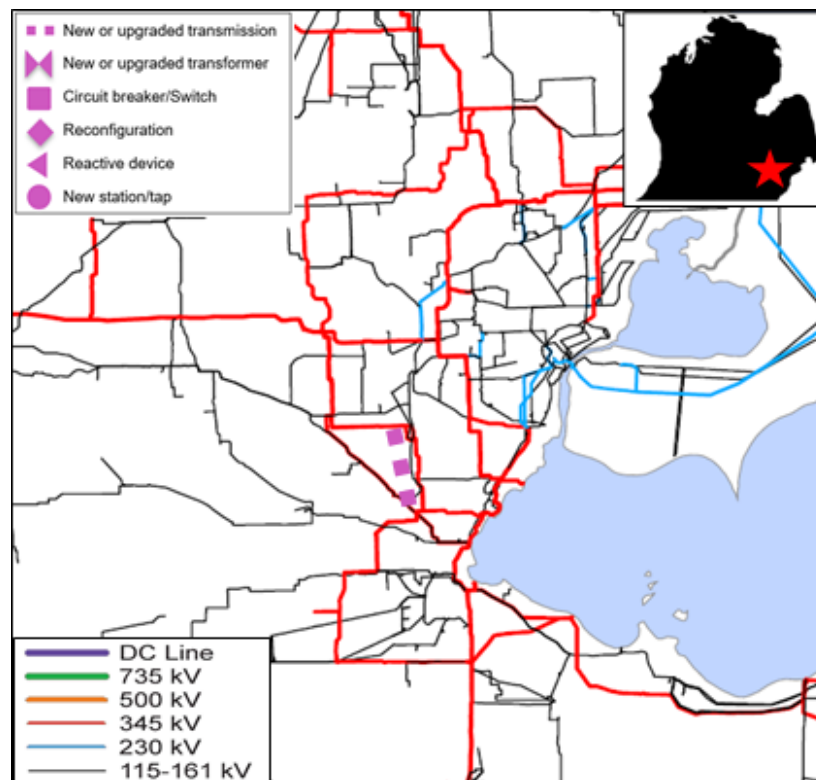


Figure 4.3.1-3: P15875 Geographic transmission map of project area

Project Need

The Kentucky – Luzon 120 kV line is projected to be overloaded for several shutdown-plus-contingencies that involve the loss of the Brownstown – Swan Creek 120 kV line. The identified overloaded equipment on this circuit is the sag limit. The thermal violations are found in peak and off-peak conditions.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
N1	264595 19LUZON1 120 264718 19KNTKY1 120 1	130	103	<99
P6	264595 19LUZON1 120 264718 19KNTKY1 120 1	130	114	<99

Table 4.3.1-3: P15875 Thermal loading drivers

Alternatives Considered

No alternatives were considered

Project 21726 – Warren 120kV Pos JJ Station Equipment Upgrades

Project Description

Replace station equipment at Warren to meet or exceed 528 MVA (2540 Amps)

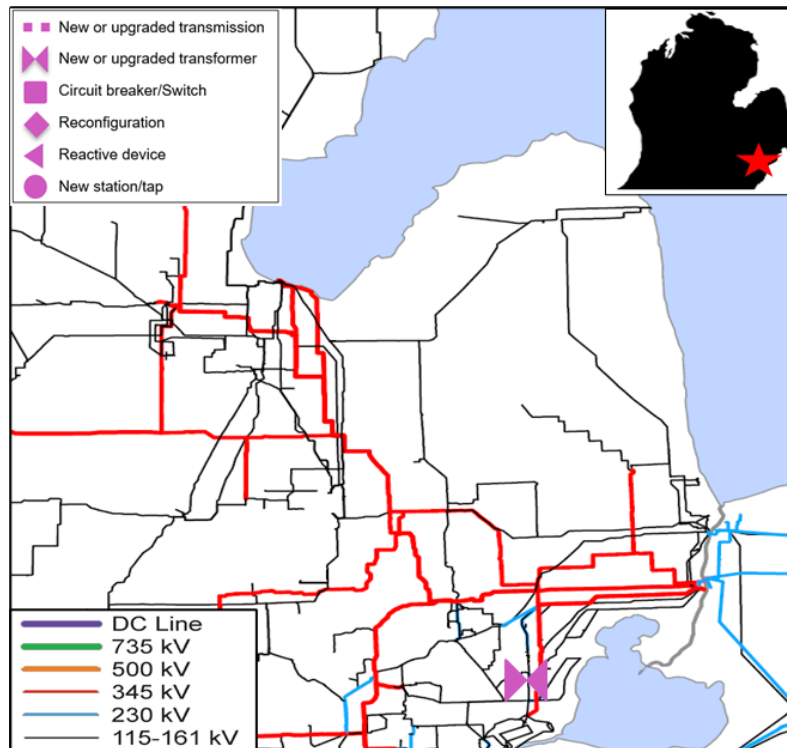


Figure 4.3.1-5: P21726 Geographic transmission map of project area

Project Need

The station equipment at Warren Pos JJ is projected to be overloaded for P2-1 contingency during peak load condition.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
N1	265648 19WAREN5 120 265649 19WAREN6 120 Z1	317	102	<99
P6	265648 19WAREN5 120 265649 19WAREN6 120 Z1	317	<99	<99

Table 4.3.1-5: P21726 Thermal loading drivers

Alternatives Considered

No alternatives were considered.

Other Projects:

Projects Driven by Other Local Needs

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
20525	ITCT Line Relocation Blanket	Relocate portions of transmission lines located in public right of way at the government entity's mandate.	12/31/2022	\$2

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21958	Augusta 2 120kV PTS Additions	Install 2 new 120 kV, 1-way, full-load-break 2,000 A GOAB switches near the Augusta 2 Tap.	12/31/2024	\$1.3
22013	ITCT Pole Top Switch Additions/Replacement Program 2023	Install or replace an existing pole top switch with a new 120 kV, 1-way full-load-break pole top switch at tap points.	12/31/2023	\$2.4
20182	Northeast 120kV 55 MVAR Reactor	Install a 120 kV, 55 MVAR shunt reactor along with a 3000 A synchronizing breaker and disconnect switches at Northeast 120 kV bus.	12/31/2024	\$1.3
20183	Stephens 345kV 55 MVAR Reactor	Install a 345 kV, 55 MVAR shunt reactor along with a 3000 A synchronizing breaker	12/31/2024	\$3.5
20185	Warren 120kV 55 MVAR Reactor	Install a 120 kV, 55 MVAR shunt reactor along with a 3000 A synchronizing breaker and disconnect switches at Warren 120kV bus.	12/31/2024	\$1.7
20186	Mack 120kV 55 MVAR Reactor	Install a 120 kV, 55 MVAR shunt reactor along with a 3000 A synchronizing breaker and disconnect switches at Mack 120kV bus.	12/31/2024	\$1.7

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22017	2024 ITCT Transmission Asset replacement Program	Replace aging and outdated equipment on a cycle that will ensure each piece of equipment is replaced near its expected end of life. Modern equipment can improve reliability, use state of the art technology, and typically will allow for longer maintenance intervals. New equipment is also commonly equipped with better monitoring and alarming functionality giving improved remote supervision. All of this will help to reduce overall maintenance costs.	12/31/2024	\$37

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22016	2025 ITCT Customer Interconnections	ITCT will individually evaluate each request to ensure it does not adversely impact reliability. Projects that result in a system solution where the cost estimate is less than \$1 Million that have an in-service date within the year 2025 will be associated with this project. When Customer Interconnection projects are evaluated to amount to a cost greater than \$1 Million a separate project will be submitted for approval.	12/31/2025	\$2

Generator Interconnection Projects:

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21166	Maple - J1062 Generator Interconnection	Build a new 345kV 3 breaker substation cut-in from the Lemoyne - Majestic 345kV circuit. Remove terminal equipment at Majestic 345kV substation.	06/01/2023	\$16.4

4.3.2 Michigan Electric Transmission Co. (METC)

METC proposed 32 projects targeting MTEP22 Appendix A. Of these projects, nine are Baseline Reliability Projects, 17 are Other projects, and six are Generator Interconnection projects with signed Generator Interconnection Agreements. Two other projects were initially proposed for appendix B. After MISO's independent reliability analyses only two of the nine Baseline Reliability Projects were justified to be included in MTEP22, therefore the remaining seven will be moved to Appendix B. One EPR was added to appendix A bringing the total appendix A projects to 26 for METC. All the projects' expected in-service dates and estimated costs totaling approximately \$305 million are provided as of September 15, 2022.

Baseline Reliability Projects:

Project 21485 – Lawndale – Murphy 138kV Sag Remediation

Project Description

The proposed solution is to remove sag limits on the Lawndale to Murphy 138 kV line to conductor limit.

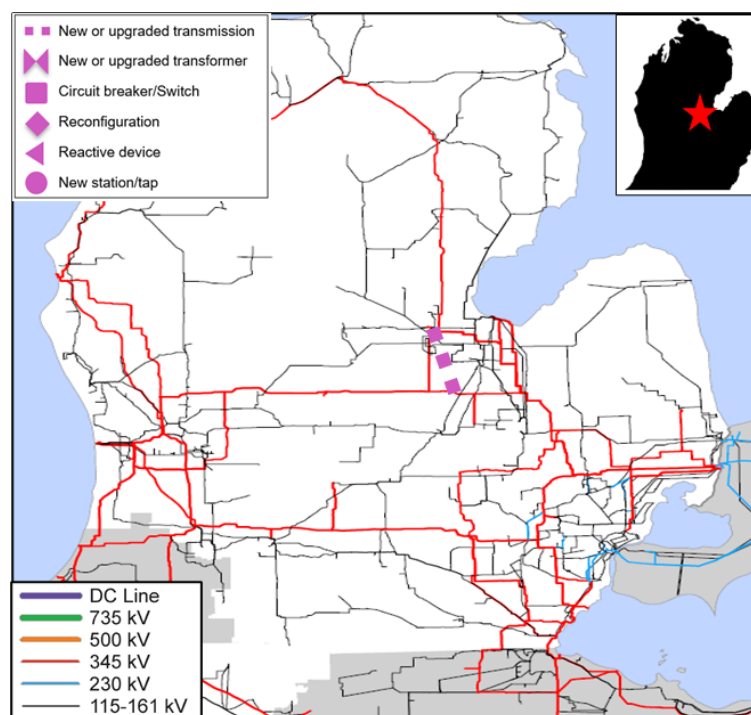


Figure 4.3.2-1: P21485 Geographic transmission map of project area

Project Need

Lawndale to Murphy 138 kV line is projected to be overloaded for P1, P2, P3, P6, and P7 contingencies in peak and off-peak scenarios. This consequently causes the Lawndale to Murphy 138kV line section to exceed its summer emergency limit. The overloaded equipment identified on this circuit is the overhead conductor, 795 ACSR 26/7.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
N1	256193 18LAWNDL 138 256499 18MURPHY 138 1	229	104	<99
P6	256193 18LAWNDL 138 256499 18MURPHY 138 1	229	112	<99

Table 4.3.2-1: P21485 Thermal loading drivers

Alternatives Considered

No alternatives were considered.

Project 21890 – Gallagher - Ogemaw 138kV Sag Remediation

Project Description

Remediate sag on the 336 ACSR conductor on the Gallagher to Ogemaw 138 kV circuit up to at least 60 MVA summer emergency rating.

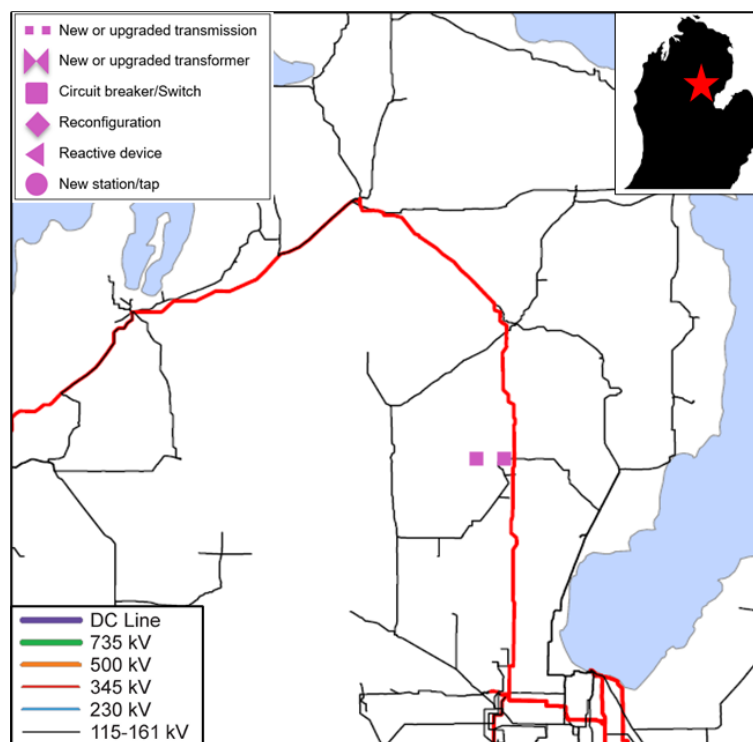


Figure 4.3.2-2: P21890 Geographic transmission map of project area

Project Need

The Gallagher - Ogemaw 138 kV circuit is projected to be overloaded for P2, P5, and P6 contingencies during peak load conditions. The identified overloaded equipment on this circuit is the sag limit on the Gallagher - Ogemaw 138 kV sections.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
N1	256148 18GALAGR 138 256777 OGEMAW J 138 1	47.8	115	<99
P6	256148 18GALAGR 138 256777 OGEMAW J 138 1	47.8	114	<99

Table 4.3.2-2: P21890 Thermal loading drivers

Note: Project causes new thermal overload up to 120% in other Michigan models for the same facility, METC agreed to update the scope of this project to elevate the sag to 85 MVA instead of 60 MVA as originally proposed, and this will resolve these addition issues.

Alternatives Considered

No alternatives were considered.

The remaining four Baseline Reliability Projects submitted will be moved to Appendix B:

Project 21972 – Kenowa – Nelson Road 345kV Sag Remediation

Project Description

Remediate the sag limit up to a summer emergency rating of 1350 MVA.

Project 21986 – Vergennes – Muskrat (J1043) 345 kV Sag Remediation

Project Description

Remediate the sag limit up to the conductor's limit.

Project 22015 – Blackstone – Leoni #1 and #2 138kV Rebuild

Project Description

The proposed solution is to rebuild approximately 4.2 miles of the Blackstone – Leoni #1 and #2 138 kV lines using 954 ACSR on double-circuit structures constructed at 138 kV. Install OPGW along the entire length from Blackstone to Leoni.

Project 22018 – Blackstone - Leoni #3 138 kV Rebuild

Project Description

The proposed solution is to rebuild approximately 4.2 miles of the Blackstone – Leoni #3 138 kV line using 954 ACSR on double circuit structures constructed at 138 kV Standards, with future double circuit on the northeast side of the towers. Install OPGW along the entire length from Blackstone to Leoni.

Other Projects:

Projects Driven by Other Local Needs

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22586	Buskirk - New CE Interconnection	METC to construct a new 138 kV switching station, Buskirk, with three 138 kV breakers in a ring-bus configuration, expandable for a fourth line exit, and cut the Buck Creek-Hazelwood 138 kV line in.	12/30/2024	\$7.8
20965	METC Line Relocation Blanket	Relocation portions of transmission lines located in public right of way at the government entity's mandate.	12/31/2022	\$2

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
15910	Marquette-Bingham 138kV Rebuild	Rebuild 18.76 miles of the Marquette – Bingham circuit using 954 ACSR conductor with double-circuit steel structures.	12/31/2025	\$27.6
21945	Parr Road-Whiting 138kV Rebuild	Rebuild approximately 12 miles from Parr Rd - Tangent (J875 Generator Interconnection) and approximately 28.6 miles from Raisin Tap - Whiting on 138kV future double circuit steel poles with 1431 ACSR conductor with OPGW (coordinating with J875 Generator Interconnection project).	5/15/2025	\$53.5
21947	Roedel Road 138 kV PTS Replacements	The proposed solution is to replace 3185 and 3198 pole top switches with 2 new 138kV, 1-way, full-load-break 2,000 A GOAB switch near the Roedel Road Tap.	12/31/2024	\$1.2
21952	Dutton 138 kV PTS Replacements	Replace 3004, 3017 and 3493 pole top switches with 3 new 138kV, 1-way, full-load-break 2,000 A GOAB switch near the Dutton Tap.	12/31/2024	\$1.7
21966	Gaines-East Paris-Plaster Creek Cut In	Cut East Paris – Gaines 138kV at STR #122T46 (Meadowbrooke Jct.) and construct 0.09 mile using 1431 ACSR single circuit structure into Gaines Pos. BM9. Convey ~ 5.6 miles from CE the existing circuit from STR #122T46 to Plaster Creek and station equipment at Meadowbrooke to become METC's assets. Install (3) 138 kV, 3000 A continuous, 50 kA min. interruptible breakers at Gaines. Close N.O 5353 switch at Meadowbrooke.	06/01/2026	\$3.9

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21983	Bard Road-Salt River 138kV	The proposed solution is to construct a new 138 kV circuit from Bard Road to Salt River, using 954 ACSR conductor. This new circuit will be strung on the vacant side of the existing structures that were already pre-built to future double-circuit standard all along the path. The work at Bard Road 138kV station involves a new 138kV breaker at position 37W8, re-terminate Gallagher-Bard Road 138kV to position WM37, and terminate the new Bard Road – Salt River 138kV line to where the Gallagher connection is now (WM39). The work at Salt River 138kV station requires installation of 4 new 138kV breakers to complete the third row and swapping terminations to avoid the two Bullock connections terminated in the same row.	06/01/2025	\$15.5
21984	METC Pole Top Switch Additions/Replacement Program 2023	Install or replace an existing pole top switch with a new 138 kV, 1-way full-load-break pole top switch at tap points.	12/31/2023	\$2.4
21990	Cut losco-Karn into Twinning 138kV	Cut the losco – Karn 138 kV circuit into the Twining 138 kV substation. This will require expanding the Twining 138 kV substation and constructing approximately 0.15 miles of new right of way to bring the line into the substation.	06/01/2026	\$4.1
21993	METC Pole top switch additions/replacement program 2024	Install or replace an existing pole top switch with a new 138 kV, 1-way full-load-break pole top switch at tap points.	12/31/2024	\$2.4

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22000	2024 METC Asset Replacement Program	Replace aging and outdated equipment on a cycle that will ensure each piece of equipment is replaced near its expected end of life. Modern equipment can improve reliability, use state of the art technology, and typically will allow for longer maintenance intervals. New equipment is also commonly equipped with better monitoring and alarming functionality giving improved remote	12/31/2024	\$39.4

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
		supervision. All of this will help to reduce overall maintenance costs.		
22020	Leoni-Parr Rd 138V Rebuild	Rebuild approximately 19 miles from Leoni to Parr Rd utilizing 1431 ACSR on standard 138 kV double circuit steel poles with OPGW.	12/31/2025	\$32.6
22021	Chopin-Dupont – White Lake 138kV Rebuild	Rebuild approximately 21 miles of the Chopin – Dupont – White Lake 138 kV Circuit from around structure 003F410 to Dupont utilizing 954 ACSR conductor on standard 138 kV double circuit structures with OPGW and relocate Cobb - White Lake and Dupont White Lake lines to the other side of the new 138 kV standard structures utilizing 954 ACSR conductor.	12/31/2025	\$39

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21505	Houghton Lake Interconnection	Create a three-breaker ring bus switching station. Install three 138kV breakers, bus work, and associated disconnects. Loop in the METC Bard Rd - Higgins 138kV circuit to the new switching station. Connect CE Houghton Lake Substation to new switching station.	06/01/2025	\$6.16
21706	Mill Creek Interconnection	Install tap pole and two 1-way switches to facilitate the Mill Creek tap on the Mcgulin-Riggsville line.	06/30/2023	\$1.83
21995	2025 METC Customer Interconnections	METC will individually evaluate each request to ensure it does not adversely impact reliability. Projects that result in a system solution where the cost estimate is less than \$1 Million that have an in service date within the year 2025 will be associated with this project. When Customer Interconnection projects are evaluated to amount in a cost greater than \$1 Million a separate project will be submitted for approval.	12/31/2025	\$2
22001	Westwood 138kV Interconnection	Install a 138 kV breaker and associated equipment at Plum station to accommodate Wolverine's interconnection request.	12/31/2024	\$3.7
22627	Day – New WPSC Interconnection	Tap the Deja Jct. – Derby Jct. section of Eureka – Vestaburg 138kV line,	6/1/2024	\$1.1

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
		approximately 2.5 miles West of Deja Tap. Transmission tap will be ~ 4 miles long. Install tap pole at the tap location, install two new 138 kV, 1-way, full-load-break 2000 A.		

Generator Interconnection Projects:

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
20906	Styx-J984 Generator Interconnection	Build a new 345kV three breaker ring bus station and cut-in the Murphy - Nelson Road 345kV line. Remote relay upgrades at Murphy 345kV and Nelson Road 345kV substations, as well as jumper work at Slate 345kV substation.	1/31/2023	\$15.3
21145	J996 Generator Interconnection	Build a new 138 kV, 3 breaker ring switching station cut in from the METC Beecher - Morocco 138 kV circuit.	3/31/2023	\$6.6
21146	J1172 Generator Interconnection	Build a new 138 kV, 3 breaker ring switching station cut in from the METC Dort - Garfield 138 kV circuit.	11/15/2023	\$10.9
21185	Muskrat - J1043 Generator Interconnection	Build a new three breaker ring bus station and cut-in the Vergennes - Nelson Road 345kV line. Install remote relay upgrades at Vergennes 345kV and Nelson Road 345kV substations.	05/31/2024	\$9.6
21445	Donaldson Creek - White Lake 138 kV Rebuild (J1071)	Rebuild approximately 18.43 miles of the 336 ACSR conductor utilizing 954 ACSR on standard 138 kV double circuit structures with OPGW. Upgrade 336 ACSR Line Drop and 336 ACSR Bus at White Lake Position 188, as well as the 336 ACSR line drop at Donaldson Creek position BM11 to meet or exceed the summer emergency rating of the new conductor.	06/01/2024	\$30.1
21446	J1071 Generator Interconnection	Install 138kV breaker at position 13W8. Install 138kV disconnect switches at positions 13W8 & 13M6.	12/31/2023	\$1.2

4.3.4 Wolverine Power Supply Cooperative Inc. (WPSC)

Wolverine Power Supply Cooperative Inc. proposed five projects in the Other Projects type category at an estimated cost of \$40.5 million, targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the Wolverine Power Supply Cooperative Inc. area, all of the projects are recommended by MISO to be included in MTEP22 Appendix A. The projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Other Projects:

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
15652	Rebuild Atlanta to Gaylord 138kV	Rebuild 22 miles to Wolverine's 138 kV standard	12/31/2024	\$16.5
21785	Grawn Station Protection System update	Replace obsolete electromechanical relays that require replacement by modern digital relays.	12/31/2024	\$1.5
21787	Westwood 138kV Interconnection	The Westwood Transmission station was fully rebuilt in 2009 and can accommodate the addition of a new 138kV/69kV power transformer. This project includes adding the power transformer, bus equipment, and protection systems, to accommodate the addition of a 138kV line terminal.	12/31/2025	\$18

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21425	Mills Creek Interconnection	Add .9 Miles Transmission and 138kV BUS to accommodate load interconnection.	6/29/2023	\$4.5
21665	Day Interconnection	New ROW and line for interconnection.	12/31/2022	\$3.5

4.3.5 Michigan Public Power Agency (MPPA)

Michigan Public Power Agency proposed one new project in the Other Projects type category at an estimated cost of \$1.5 million, targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the Michigan Public Power Agency area, this project is recommended by MISO to be included in MTEP22 Appendix A. The projects' expected in-service date and estimated cost is provided as of September 15, 2022.

Other Projects:

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21761	Zeeland Riley Transformer Addition	Install a second 138/69 kV Transformer at Riley Substation.	5/26/2022	\$1.5

4.3.6 Lansing Board of Water and Light (LBWL)

Lansing Board of Water and Light did not submit any new projects for MTEP22. MISO has not identified any open issues in LBWL area.

4.3.7 Michigan South Central Power Agency (MSCPA)

Michigan South Central Power Agency did not submit any new projects for MTEP22. MISO has not identified any open issues in MSCPA area.

4.4 Project Justifications – South Region

South Region Overview

The MISO South Planning Region consists of eleven Transmission-Owning members spanning four states, Arkansas, Louisiana, Mississippi, and parts of Texas. These Transmission Owners are:

- Arkansas Electric Cooperative Corporation (AECC)
- City of Alexandria (AXLA)
- CLECO Power LLC (CLEC)
- Cooperative Energy (SMEPA)
- East Texas Electric Cooperative (ETEC)
- Entergy Arkansas LLC (EAL)
- Entergy Louisiana LLC (ELL)
- Entergy Mississippi LLC (EML)
- Entergy New Orleans LLC (ENO)
- Entergy Texas Incorporated (ETI)
- Lafayette Utilities Systems (LAFA)

The region contains approximately 16,500 circuit miles of transmission lines ranging from 115 kV to 500 kV. There is also a significant 69 kV sub-transmission network interspersed across the footprint.

In the 2023 Summer Peak planning model, the region contains more than 38.1 GW of generation. The MISO South generation profile consists of mostly combine cycle, nuclear, gas, and coal fuel types, serving major load centers such as Little Rock, New Orleans, etc. Approximately 53 percent (20.2 GW) of the South region's generation capacity is made up of Combine Cycle (CC) units. Major generation centers are in central Arkansas, lower Louisiana, and western Mississippi (Figure 4.4-1).

Major load centers are typically found around larger cities in the region such as Little Rock, Jonesboro, and Pine Bluff in Arkansas; Monroe, Alexandria, Lake Charles, Lafayette, New Orleans, and Baton Rouge in Louisiana; Jackson, Hattiesburg, Natchez, Vicksburg, and Greenville in Mississippi. Texas major load centers in the Western load pocket include Bryan and the Woodlands area. The major load center in the WOTAB load pocket portion of Texas is in South Beaumont and the Port Arthur Area (Figure 4.4-1). According to the 2023 Summer Peak planning model, the regional load is over 36.1 GW.

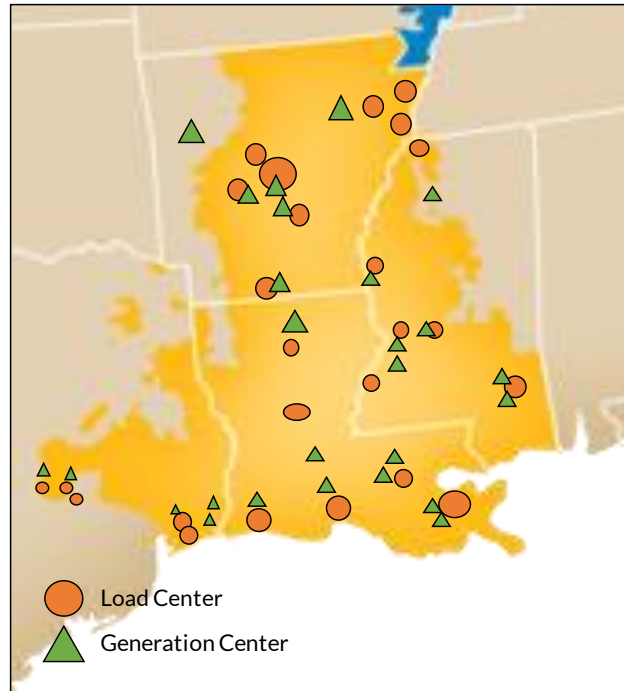


Figure 4.4-1: Generation and load centers in the South planning region

For MTEP22, MISO Transmission Planning is recommending 44 projects from the South region for inclusion in Appendix A at an estimated cost of \$1 billion. Of these, four are Baseline Reliability Projects, and 14 are Generator Interconnection Projects. The remaining 26 projects are classified as Other projects because they do not meet the criteria to be considered as Baseline Reliability Projects, New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects.

Of the 44 projects, that are being recommended to be included in MTEP22, three have an estimated cost of less than \$1 million, seven have an estimated cost between \$1 million and \$5 million, and the remaining 27 projects have an estimated cost greater than \$5 million (Figure 4.4-2).

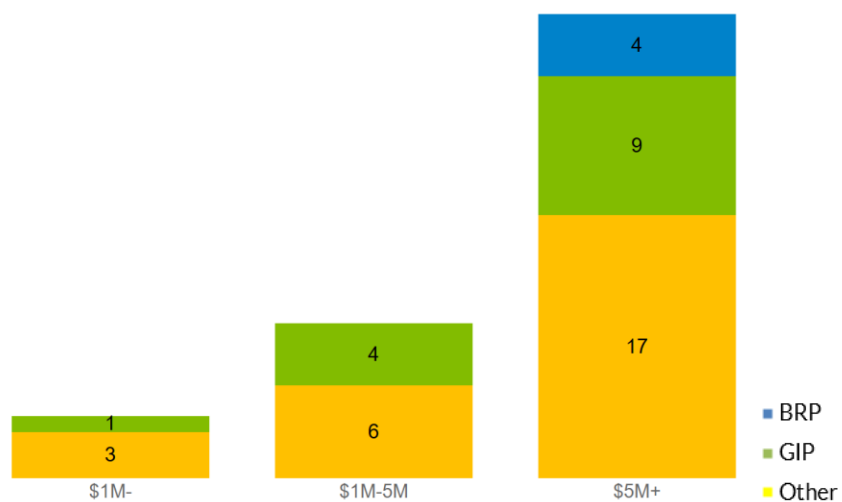


Figure 4.4-2: Cost category by project types (project data as of 9-15-2022)

The majority of the projects in the MISO South planning region are expected to go in service in the next three years (figure 4.4-3).

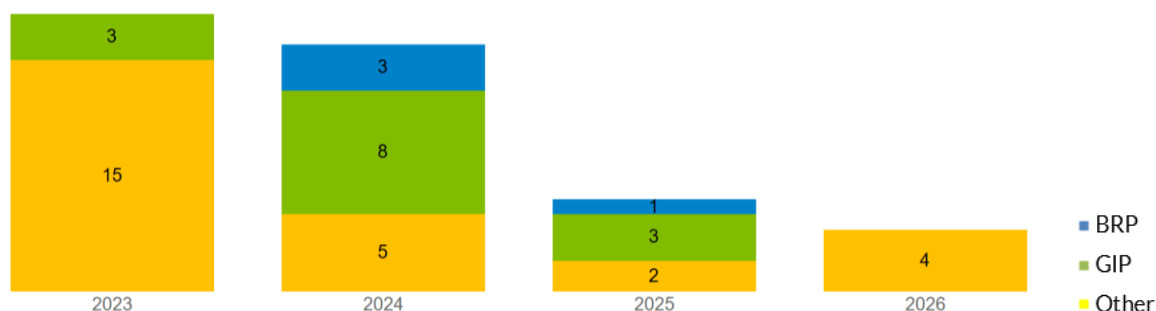


Figure 4.4- 3: Project in-service date by project type (project data as of 9-15-2022)

In accordance with Attachment FF of the tariff, in the event a Transmission Owner determines that system conditions warrant the urgent development of system enhancements an expedited review of the impacts of the project can be requested. MISO shall use a streamlined approval process for reviewing and approving projects proposed by the Transmission Owner(s) and decisions will be provided to the Transmission Owner within 30 Days of the project's submittal to MISO unless a longer review period is mutually agreed upon. During the MTEP22 cycle, MISO received seven projects in the South planning region through the Expedited Project Review (EPR) process:

1. Project ID 18228 Golden Meadow to Clovelly 115 kV: Rebuild line
2. Project ID 21085 Sugar House 230 kV: New Station
3. Project ID 23307 Flare Hill 230 kV New Customer Substation
4. Project ID 23209 Legend 230 kV New Customer Connection
5. Project ID 23327 Cleco Marthaville Substation
6. Project ID 23328 Cleco Vernon Substation
7. Project ID 22205 Cleco Cole Substation

Also, in accordance with Attachment FF Section VIII.A.3, none of the projects were identified as an Immediate Need Reliability Project and excluded from the competitive developer selection process. The ten most expensive projects account for \$650 million (65%) of the \$993 million total recommended projects for the South in MTEP22. The locations of the ten most expensive projects are shown in Figure 4.4-4 and it is seen that they are spread across the southern part of the South planning region. Projects that are blanket expenditures (relays, physical security, etc.) are excluded from this list.

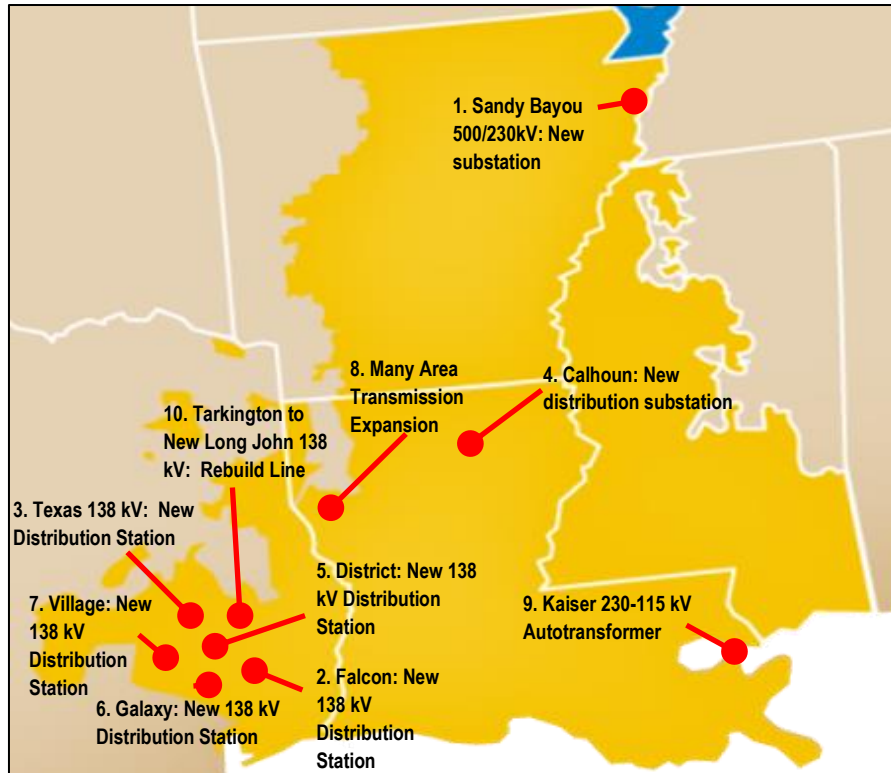


Figure 4.4-4: South region top ten projects by cost (data as of 9-15-2022)

4.4.1 Arkansas Electric Cooperative Corporation (AECC)

Arkansas Electric Cooperative Corporation proposed one project at an estimated cost of \$3 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the AECC area, this project is recommended by MISO to be included in the MTEP 22 Appendix A as an Other project category at an estimated cost of \$3 million. The project's expected in-service date and estimated cost is provided as of September 15, 2022

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21565	Ward Breaker Additions	Install five (5) breakers, one of which will be a transfer breaker, at AECC's Ward 115kV Switching Station.	12/1/2023	\$3

4.4.2 City of Alexandria (AXLA)

City of Alexandria did not submit any new projects for MTEP22. MISO has not identified any open issues in the City of Alexandria area.

4.4.3 CLECO Power LLC (CLEC)

CLECO Power LLC proposed four projects at an estimated cost of \$86.2 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the CLECO Power LLC area, all four projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, one is a Baseline Reliability Project, and three are Other projects. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Baseline Reliability Projects:

Project 20163 - Many Area Transmission Expansion

Project Description

Tap the existing Fisher to Mansfield Compressor 230 kV line 8.3 miles from Fisher. From the Tap point, build two new 2.8 mile 230 kV lines to a new substation (Belmont) tapped into the Pelican to Loring and Loring to Many 138 kV lines. At Belmont substation add a 230/138 kV 300 MVA Auto. The total estimated cost of this project is \$38.1 million dollars. The expected in-service date for this project is June 1, 2025.

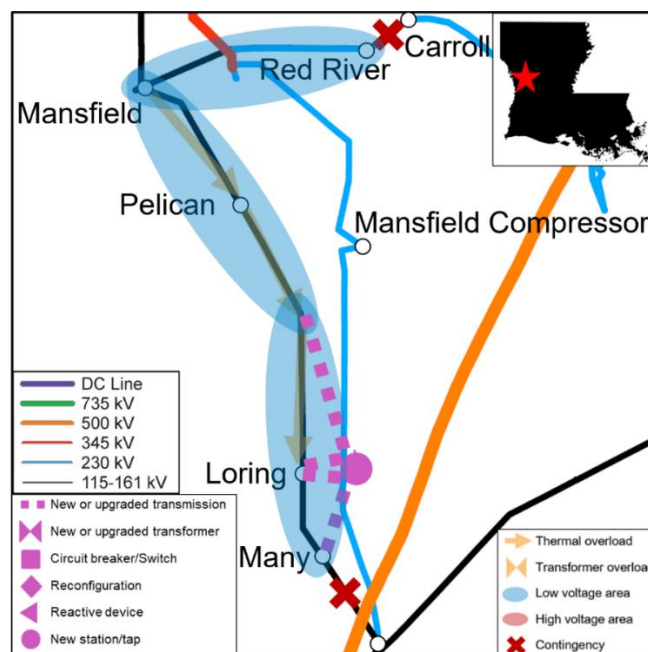


Figure 4.4.3-1: P20163 Geographic transmission map of project area

Project Need

In 2025, a P6 contingency on Many to Fisher 138 kV line followed by a contingency on Carroll to Red River 138 kV line causes an overload greater than 125 % on South Shreveport to Wallace lake 138 kV line. Voltage violations also exist for this contingency. Load shed is the only mitigation for this P6 contingency. With projected industrial load growth in this area this project has been modified to account for the additional load.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	South Shreveport – Wallace Lake 138 kV	188	133	6
P6	Wallace Lake – International Paper 138 kV	188	111	6
P6	Mansfield – Pelican 138 kV	143	112	22
P6	Pelican– Loring 138 kV	143	97	N/A

Table 4.4.3-1: P20163 Thermal loading drivers

Cont. Type	Limiting Element	Rating (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P6	Many 138 kV	0.9	0.48	1.035
P6	Loring 138 kV	0.9	0.48	1.032
P6	Pelican 138 kV	0.9	0.56	1.016
P6	Mansfield 138 kV	0.9	0.60	1.006
P6	International Paper 138 kV	0.9	0.80	0.991

Table 4.4.3-2: P20163 Voltage Performance drivers

Alternatives Considered

No alternatives were considered for this project.

Other Projects:

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22205	Cole Substation	Tap the existing Cooper to Penton 230 kV line 2.4 miles from Cooper and build a new substation (Cole).	3/27/2023	\$18.8
23327	Marthaville Substation	New load substation tapping the Fisher to Mansfield Compressor 230 kV line	4/23/2023	\$15.7
23328	Vernon Substation	Adding new load substation by tapping the Toledo to Leesville 138 kV line	4/23/2023	\$13.7

4.4.4 Cooperative Energy (SMEPA)

Cooperative Energy proposed one project targeting MTEP22 Appendix A categorized as Other. After MISO's independent reliability analysis of the Cooperative Energy area, this project was recommended by MISO to be included in MTEP22 Appendix A. The expected in-service date and estimated cost for this project are provided as of September 15, 2022.

Other Projects:

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22345	East Winstonville Delivery Point	Tap the Energy owned Delta SES - Shelby 115kV line with switching station and build a 115kV line to the new member East Winstonville delivery point.	7/1/2023	\$1.9

4.4.5 East Texas Electric Cooperative (ETEC)

East Texas Electric Cooperative Arkansas Electric did not submit any new projects for MTEP22. MISO has not identified any open issues in the ETEC area.

4.4.6 Entergy Arkansas LLC (EAL)

Entergy Arkansas proposed six projects at an estimated cost of \$179.3 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the EAL area, all six of these projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, three are Other projects, and three are Generator Interconnection Projects with a signed Generator Interconnection Agreement. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21806	Junction City 115kV: Upgrade CT's	This project will upgrade the existing 600 amp CT's at Junction City to >1000 amps.	12/30/2023	\$1.4

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21805	2023 EAL Asset Renewal Program	The AM renewal asset program is to address aging and failing transmission assets across EAL's territory.	12/31/2023	\$28.5

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22530	Sandy Bayou 500/230kV: New substation	Construct a new 500/230kV substation to serve a new industrial customer. The substation will consist of a 4 breaker 500kV ring bus, two 500/230kV autotransformer banks (7 single-phase 250 MVA units that includes one spare), a two bay breaker and a half 230kV bus with two 230kV sources to the customer. This new substation will be cut into the existing Driver - Shelby 500kV transmission line.	3/1/2024	\$122

Generator Interconnection Projects:

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
23154	KEO 500kV: Expansion for J1258 & J1415	The KEO 500kV substation will be expanded to interconnect the J1258/J1415 gen-tie 500kV line. The Network Upgrades include a new 500 kV gas circuit breaker, two new 500 kV disconnect switches, relay panels, revising the station RTU configuration, wiring additions to the DFR, and telecommunication additions.	10/15/2024	\$8.9
23267	Fountain Hill 115kV: J1281 Cut In Switching Station	Entergy will provide oversight for the construction of the transmission owners interconnection facilities and stand-alone network upgrades associated with	10/31/2024	\$3.8

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
		the interconnection of J1281 on the Crossett - Monticello East 115kV line. The Crossett North 115kV substation will have new line protection installed. The Monticello East 115kV substation will have new line protection installed with new CCVT's and a new breaker.		
23326	Sunset 230kV: J1215 New Switching Station Cut-In	A new 230kV three breaker ring bus substation will be cut-in to the Woodward - Stuttgart Ricuskey 230kV transmission line. Remote end upgrades will be performed at Stuttgart Ricuskey and Woodward substations. Approximately 2.6 miles of OPGW fiber will be installed between Stuttgart Ricuskey and the new J1215 stations. 230kV Circuit breakers B8414 and B8406 will be replaced at Stuttgart Ricuskey.	2/2/2025	\$14.7

4.4.7 Entergy Louisiana LLC (ELL)

Entergy Louisiana proposed 13 projects at an estimated cost of \$210 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the ELL area, all of these projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, two are Baseline Reliability Projects, five are Other projects, and six are Generator Interconnection projects with signed Generator Interconnection Agreements. The projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Baseline Reliability Projects:

Project 21811 - Richard 500-138 kV AT1 Relay Improvement SPOF

Project Description

Ensure the Richard 500-138 kV autotransformer #1 has redundant high-speed protection with independent CT and PT windings. The total estimated cost of this project is \$10.3 million dollars.

The expected in-service date for this project is December 31, 2024.

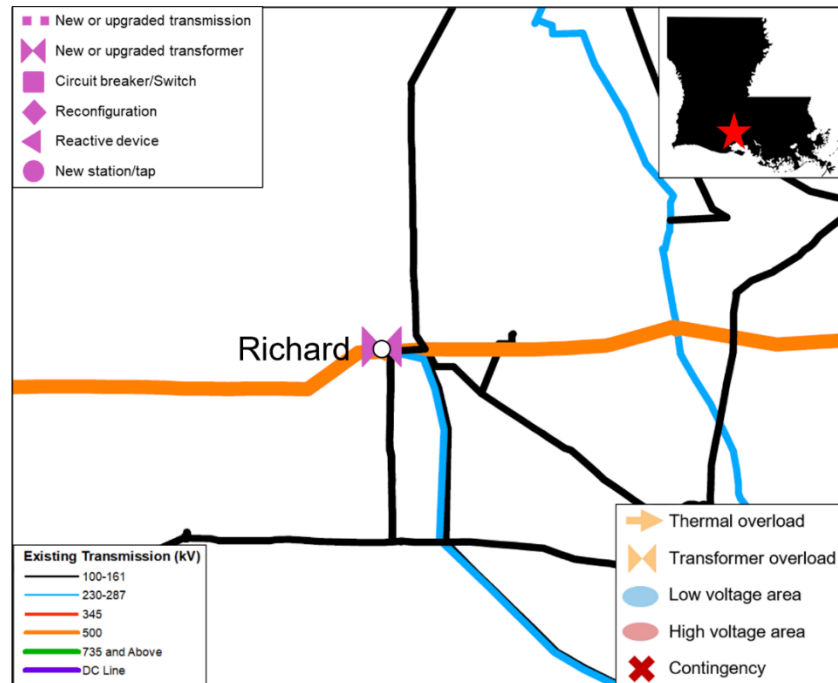


Figure 4.4.7-1: P21811 Geographic transmission map of project area

Project Need

NERC TPL-001-4 Reliability Standards and Entergy's Planning Guidelines. P5 contingency SLG fault on the Richard 500 kV bus with non-redundant relay failure for the Richard 500/138 kV auto #1 protection has the potential to cause 600MW of generation to trip.

Alternatives Considered

No alternatives were considered for this project.

Project 21832 - Dow Meter to Tiger 230 kV Re-Termination project

Project Description

Re-terminate the Addis to Tiger and Dow Meter to Addis 230 kV lines to create the Dow Meter to Tiger 230 kV line, which will bypass the Addis 230 kV station. The total estimated cost of this project is \$5.7 million dollars. The expected in-service date for this project is June 1, 2024.

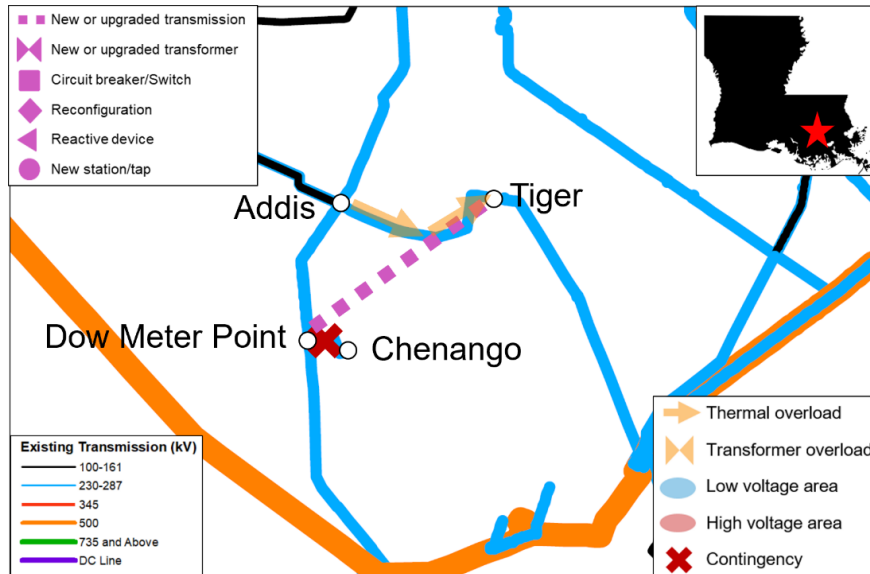


Figure 4.4.7-3: P21832 Geographic transmission map of project area

Project Need

The 2018 MISO Deliverability study shows an overload on the Addis to Tiger 230 kV line for the P1.2 loss of the Dow Meter to Chenango 230 kV or the Richardson to Iberville 230 kV line in 2023. This project replaces project 15566, which was shown to be ineffective in mitigating deliverability constraints in the area.

Alternatives Considered

No alternatives were considered for this project.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21816	Drusilla to Jefferson 69 kV: Upgrade Switches	Upgrade switches 9716 and 9717 at Jefferson 69 kV station to 1200 Amps. Upgrade switch 4474 at East 69 kV station to 1200 Amps. Upgrade any remaining terminal equipment to achieve a rating of 1200 Amps on the Drusilla to Jefferson to East 69 kV path.	6/1/2023	\$2.7
21835	Kaiser 230-115 kV Autotransformer	Install 300 MVA, 230/115 kV Autotransformer to tie Kaiser 230 kV to Conquest 115 kV. Kaiser and Conquest stations are adjacent to each other. Remove Behrman to Pt Nickel 115 kV line from the model. This facility includes an under the Mississippi River section that has failed. The Kaiser Auto alleviates	12/1/2024	\$26.3

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
		some of the potential radial concerns during outages with this line removed.		

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21809	2023 ELL Asset Renewal Program	This project is for the ELL Asset Renewal Program that will be executed in 2023.	12/1/2023	\$55.1
21822	Hartburg to Rhodes 500 kV River Crossing Tower Replacement	Replace the existing structure with a new structure that will be installed further to the east to avoid future washing out. Access to the structure is especially difficult and will require extensive matting and/or the use of specialized marsh equipment.	8/31/2023	\$5.8

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
13910	Calhoun: New distribution substation	Construct a 230 kV distribution substation named Calhoun in western Ouachita parish near the town of Calhoun and construct a 230 kV transmission line from the new substation that terminates into the Cadeville substation.	6/1/2026	\$56.8

Generator Interconnection Projects:

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
23147	J1219/J1257 115 kV station(Hickory)	Construct a 115 kV substation between the existing Amite and Chicken Farm 115 kV stations.	9/24/2024	\$14.4
23308	J1246 Bayou Labutte 500 kV Interconnection	The Bayou Labutte 500 kV substation will be expanded to interconnect the J1246 gen-tie 500 kV line. The Network Upgrades include relocation of the Willow Glen 500 kV line position, new 500 kV gas circuit breaker, two new 500 kV disconnect switches, along with other additional station equipment and upgrades at Bayou Labutte. Also Included in the network upgrades for J1246 are the replacement of two 230 kV breakers at the Point Pleasant 230	4/9/2025	\$14.3

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
		kV station to 63 kA IPO breakers along with relay upgrades for the associated equipment.		
23330	Rilla 115 kV: Expand Station (J1239)	The Rilla 115 kV station will be expanded to interconnect J1239.	10/15/2024	\$7.2
23386	MPFCA J1281, J1294, J1458 Adams Creek 230 kV and Bogalusa 500-230 kV Upgrades	MPFCA J1281, J1294, J1458 Adams Creek 230 kV and Bogalusa 500-230 kV Upgrades. Replace breakers S4402/S4405 and switches at Adams Creek 230 kV with 4000 Amp equipment. Upgrade bus at Bogalusa 500-230 kV and install slipover current transformers on 230 kV bushings of existing 500-230 kV autotransformers. These upgrades will increase the rating of the circuit between the Adams Creek 230 kV station and the Bogalusa 500-230 kV station to 1281 MVA.	4/4/2024	\$3.9
23421	J1368/J1372 Interconnection: Ventress 230 kV Station	Construct a new Ventress 230 kV substation between the existing Fancy Point and Waterloo 230 kV stations.	4/21/2023	\$5.4

4.4.8 Entergy Mississippi LLC (EML)

Entergy Mississippi proposed eight projects at an estimated cost of \$93.6 million targeting MTEP22 Appendix A. Of these projects, three are Other projects and five are Generator Interconnection projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 30, 2022.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
20063	Choctaw - TVA French Camp 500 kV: Decommission Line	Decommission and demolish the existing Choctaw-TVA French Camp 500 kV line that was recently acquired by Entergy in November of 2019 (~0.75mi).	12/15/2024	\$0.5
20064	Choctaw - Wolf Creek 500 kV: Build New Line	Utilize ~0.1mi of existing Choctaw-French Camp 500 kV line to tie into Wolf Creek (Choctaw - French Camp	12/15/2024	\$7.1

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
		500 kV decommissioned with a separate project). Convert Wolf Creek 500 kV switching yard to 4 breaker ring bus to accommodate 2nd line.		

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21815	2023 EML Asset Renewal Program	The AM renewal asset program is to address aging and failing transmission assets across EML's territory.	12/31/2023	\$56

Generator Interconnection Projects:

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
23268	Franklin - Ray Braswell 500 kV Upgrade - MPFCA	Multi-Party Facilities Construction Agreement for J1215, J1281, J1294, J1425, J1434, J1440, and J1458 to upgrade the Franklin 500 kV Substation: Replace four (4) 500 kV switches with motor operators, and one (1) 500 kV ground switch.	12/19/2023	\$1.7
23387	Gerald Andrus - Tinnin 230 kV Cut-in and Line Upgrade (J1458)	Generator Interconnection on the Gerald Andrus - Tinnin 230 kV line and Network Upgrade rebuild of the Greenville - SE Greenville 115 kV line.	3/13/2024	\$10.4
23482	Isola - Belzoni 115 kV Upgrade - MPFCA	MPFCA. Belzoni 115 kV Substation: Replace line terminal bus and insulators for the line terminal to Isola 115 kV to a new through-path rating of 106 MVA.	11/10/2023	\$0.4
23518	Tunica - Lula 115 kV Cut-in (J1440)	Generator Interconnection J1440 new Dundee 115 kV station cutting in on the Tunica - Lula 115 kV transmission line. Conditional on Ray Braswell - Franklin 500 kV MPFCA	5/23/2024	\$13.6
23519	Indianola - Cleveland 115 kV Cut-in (J1294)	Generator Interconnection for J1294 GI on the Indianola - Cleveland 115 kV transmission line. This generation project is conditional on: Ray Braswell - Franklin 500 kV Terminal Equipment Upgrades (DPP 2019 Shared Network Upgrade -	5/23/2024	\$3.9

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
		MPFCA executed) Isola – Belzoni 115 kV Upgrade (DPP 2019 Shared Network Upgrade – MPFCA executed) Bogalusa – Adams Creek 230 kV Upgrade (DPP 2019 Shared Network Upgrade – MPFCA executed)		

4.4.9 Entergy New Orleans LLC (ENO)

Entergy New Orleans (ENO) proposed one project at an estimated cost of \$1.9 million targeting MTEP22 Appendix A. After MISO’s independent reliability analysis of the ENO area, this project is recommended by MISO to be included in MTEP22 Appendix A categorized as an Other project at an estimated cost of \$1.9 million. The project’s expected in-service dates and estimated costs are provided as of September 15, 2022.

Other Projects:

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21808	2023 ENOL Asset Renewal Program	This project is for the ENOL Asset Renewal Program that will be executed in 2023.	12/31/2023	\$1.9

4.4.10 Entergy Texas, Inc. (ETI)

Entergy Texas Incorporated proposed 11 projects at an estimated cost of \$433 million targeting MTEP22 Appendix A. After MISO’s independent reliability analysis of the ETI area, these projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, one is a Baseline Reliability Project, nine are Other projects, and one is a Generator Interconnection project with a signed Generator Interconnection Agreement. All the projects’ expected in-service dates and estimated costs are provided as of September 15, 2022.

Baseline Reliability Projects:

Project 18262 – Tarkington to New Long John 138 kV: Rebuild line

Project Description

The rebuild of the Tarkington to New Long John (L870) 138 kV line will address an overload on the same line during a P5 contingency event. The total estimated cost of this project is \$18.23 million. The expected in-service date for this project is June 1, 2024.

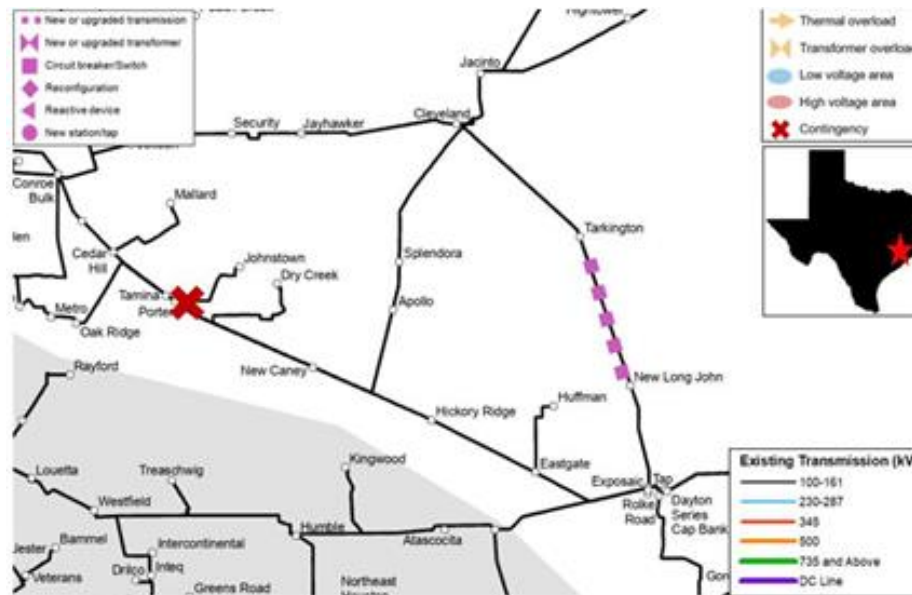


Figure 4.4.10-1: P18262 Geographic transmission map of project area

Project Need

NERC TPL-001-4 Reliability Standards and Entergy's Planning Guidelines. The P5 contingency loss of Porter 138 kV bus differential relaying plus a fault causes thermal overloads on the line in multiple sensitivity cases. Tarkington to New Long John 138 kV line also overloads for the loss of Porter to New Caney 138 kV line.

Alternatives Considered

No alternatives considered.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21841	Porter to Oak Ridge 138 kV Station Equipment Upgrade	Upgrade Line Bay Bus at Porter 138 kV to achieve a through path rating of 1766 A on the Porter to Oak Ridge 138 kV line.	12/1/2023	\$0.46

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21810	This project is for the ETI Asset Renewal Program that will be executed in 2023	Address aging and failing equipment such as structures, breakers, relay panels, RTU's, arresters, circuit switchers, etc.	12/31/2023	\$31.6
21843	Neches River Crossing Structure Mitigation	Mitigation includes: Replace coffers, de-water and excavate soils. Perform structure foundation inspection(s), fortify/repair where possible. Restore soils around structure foundations, fill coffers to grade.	6/1/2025	\$28

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
15649	Texas 138 kV: New Distribution Station	Cut the Texas substation in and out of the Lewis Creek to Goree 138 kV line.	12/1/2025	\$69.4
21847	Galaxy: New 138 kV Distribution Station	Cut the Galaxy substation in and out of the Porter to Apollo 138 kV line.	6/1/2026	\$65.9
21850	District: New 138 kV Distribution Station	Cut the District substation in and out of the Crystal to Security 138 kV line.	12/1/2026	\$85.2
21851	Falcon: New 138 kV Distribution Station	Cut the Falcon substation in and out of the Luce Bayou to Tarkington 138 kV line.	6/1/2027	\$113.8
23209	Legend: New Customer Connection	Entergy is proposing to construct new 230 kV transmission facilities at the existing Legend 230 kV substation which will serve a new industrial customer in Port Arthur, TX. The proposed new facilities include installing two additional 230 kV breakers into the Legend substation and extending a 230 kV radial transmission line towards the customer's site.	1/31/2023	\$4.25
23307	Flare Hill: 230 kV New Customer Substation	Entergy has been notified that a customer is proposing to construct a new 230 kV substation called Flare Hill, which will serve an industrial customer's new load in Port Arthur, TX. The substation will be cut in on the existing Port Acres to Southside circuit 1.	4/30/2023	\$0.98

Generator Interconnection Projects:

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22125	R1003 Generator Interconnection at Sabine 230/138 kV Station (OCAPS)	This project is to perform the required upgrades to accommodate the interconnection at the existing Sabine 230/138 kV station identified in the GIA for R1003(OCAPS).	3/1/2025	\$14.76

4.4.11 Lafayette Utilities System (LAFA)

Lafayette Utilities System, LAFA did not submit any new projects for MTEP22. MISO has not identified any open issues in the LAFA area.

4.5 Project Justifications – West Region

West Region Overview

The MISO West Planning Region consists of 20 Transmission-Owning members spanning eight states in the upper Midwest. It includes Iowa, Minnesota, Wisconsin, and parts of North Dakota, South Dakota, Montana, Michigan, and Illinois. These Transmission Owners are:

- American Transmission Company (ATC)
- Cedar Falls Utilities (CFU)
- Central Minnesota Municipal Power Agency (CMMPA)
- City of Ames, IA (COA)
- Dairyland Power Cooperative (DPC)
- Great River Energy (GRE)
- ITC Midwest (ITCM)
- MidAmerican Energy Company (MEC)
- Minnesota Municipal Power Agency (MMPA)
- Minnesota Power (MP)
- Missouri River Energy Services (MRES)
- Montana-Dakota Utilities Co. (MDU)
- Muscatine Power and Water (MPW)
- Northwestern Wisconsin Electric (NWECE)
- Otter Tail Power Company (OTP)
- Rochester Public Utilities (RPU)
- Southern Minnesota Municipal Power Agency (SMMPA)
- Wilmar Municipal Utilities (WMU)
- WPPI Energy (WPPI)
- Xcel Energy (Northern States Power, XEL/NSP)

The West planning region contains approximately 27,300 miles of transmission ranging from 57 kV to 500 kV. In the 2023 Summer Peak planning model, the region contains more than 67.2 GW of generation. Installed generation capacity in the region consists mostly of coal, gas and wind. Approximately 33.4 percent (22.5 GW) of the West region's generation capacity is made up of wind units. Major generation centers are located in central North Dakota; the Twin Cities in Minnesota; and the Quad Cities in Iowa and Illinois, with wind generation located in the eastern Dakotas and western Iowa and Minnesota (Figure 4.5-1).

Major load centers are typically found around larger cities in the region: Minneapolis/Saint Paul, Milwaukee, and Des Moines. According to the 2023 Summer Peak planning model, the regional load exceeds 42.2 GW. Power generally flows from generation-rich areas in the western portion of the region through Minnesota, Iowa, and Wisconsin, toward large load centers in the east. This is especially prevalent in times of high wind output.

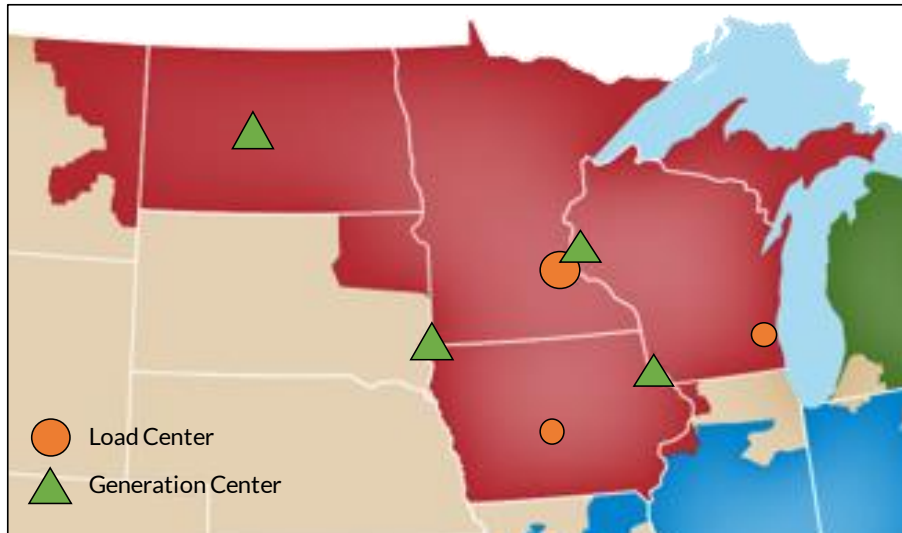


Figure 4.5-1: Generation and load centers in the West Planning Region

For MTEP22, MISO Transmission Planning is recommending 166 projects from the West region for inclusion in Appendix A at an estimated cost of \$1.3 billion. Of these, 22 are Baseline Reliability Projects, 24 are Generator Interconnection Projects, 2 are Market Participant Funded Projects, and the remaining 118 projects are classified as Other because they do not meet the criteria to be considered as Baseline Reliability Projects, New Transmission Access Projects, Market Efficiency Projects, or Multi-Value Projects.

Of the 166 projects, that are being recommended to be included in MTEP22, 34 have an estimated cost of less than \$1 million, 58 have an estimated cost between \$1 million and \$5 million, and the remaining 75 projects are estimated to cost greater than \$5 million (Figure 4.5-2).

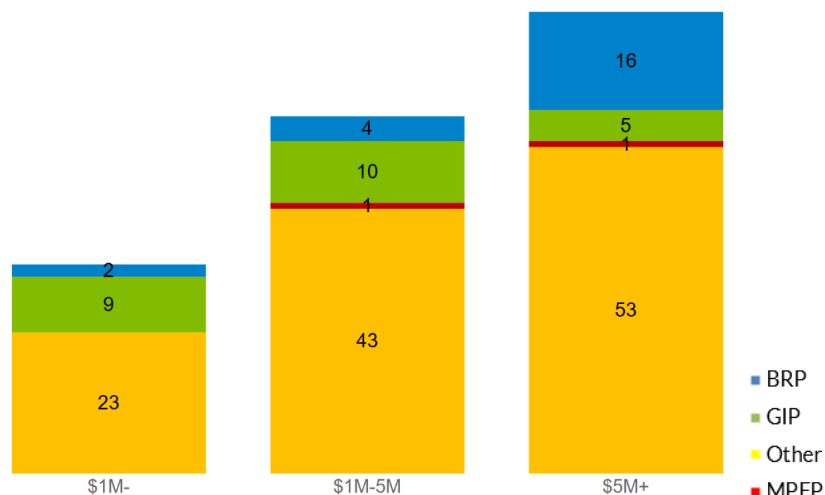


Figure 4.5-2: Project cost category by project type (data as of September 15, 2022)

The majority of the projects in the MISO West planning region are expected to go in service in the next three years (Figure 4.5-3). There were a handful of projects that are being approved in 2022 that either already went into service in 2022 or are expecting to go into service in 2022. Some of those were transmission network upgrades (GIP) correlated with generation projects that were justified by a separate Generator Interconnection process. Others fit into a bucket of timing or constructability considerations. Examples of these are the retirement of catastrophic failure devices, load additions which required short time frames to come online, or those that take advantage of outages already scheduled and planned the upgrade accordingly to meet the compliance.

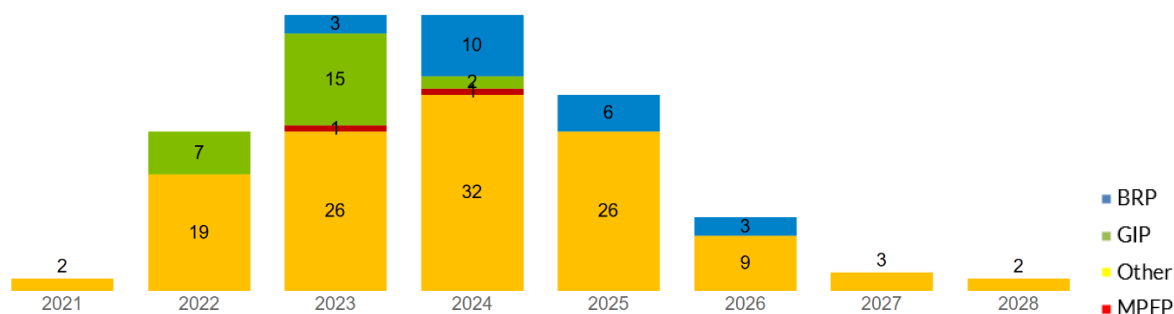


Figure 4.5-3: Project in-service date by project type (data as of September 15, 2022)

In accordance with Attachment FF of the tariff, in the event that a Transmission Owner determines that system conditions warrant the urgent development of system enhancements that would be jeopardized unless MISO performs an expedited review of the impacts of the project, MISO shall use a streamlined approval process for reviewing and approving projects proposed by the Transmission Owner(s) so that decisions will be provided to the Transmission Owner within 30 Days of the project's submittal to MISO unless a longer review period is mutually agreed upon. During the MTEP22 cycle, MISO received the following projects through the Expedited Project Review (EPR) process:

1. Project ID 22606, OTP Jamestown, ND Load Addition
2. Project ID 22885, MP Brainerd Crypto Load Addition
3. Project ID 22766, Replace McCracken- Shell Structure
4. Project ID 23346, MRES Hutchinson Substation
5. Project ID 23452, Brookings – Lyon, Hampton – Helena 2nd 345kV Circuits
6. Project ID 23633, MDU Ellendale Load Addition

Also, in accordance with Attachment FF Section VIII.A.3, the following project was identified as an Immediate Need Reliability Project and is excluded from the competitive developer selection process:

- None

The ten most expensive projects represent \$387 million of the \$1.3 billion total recommended projects for the West in MTEP22. The locations of these projects can be seen in Figure 4.5-4 and the investment is spread across the West Planning region. Projects that are blanket expenditures (relays, physical security, etc.) are excluded from this list.

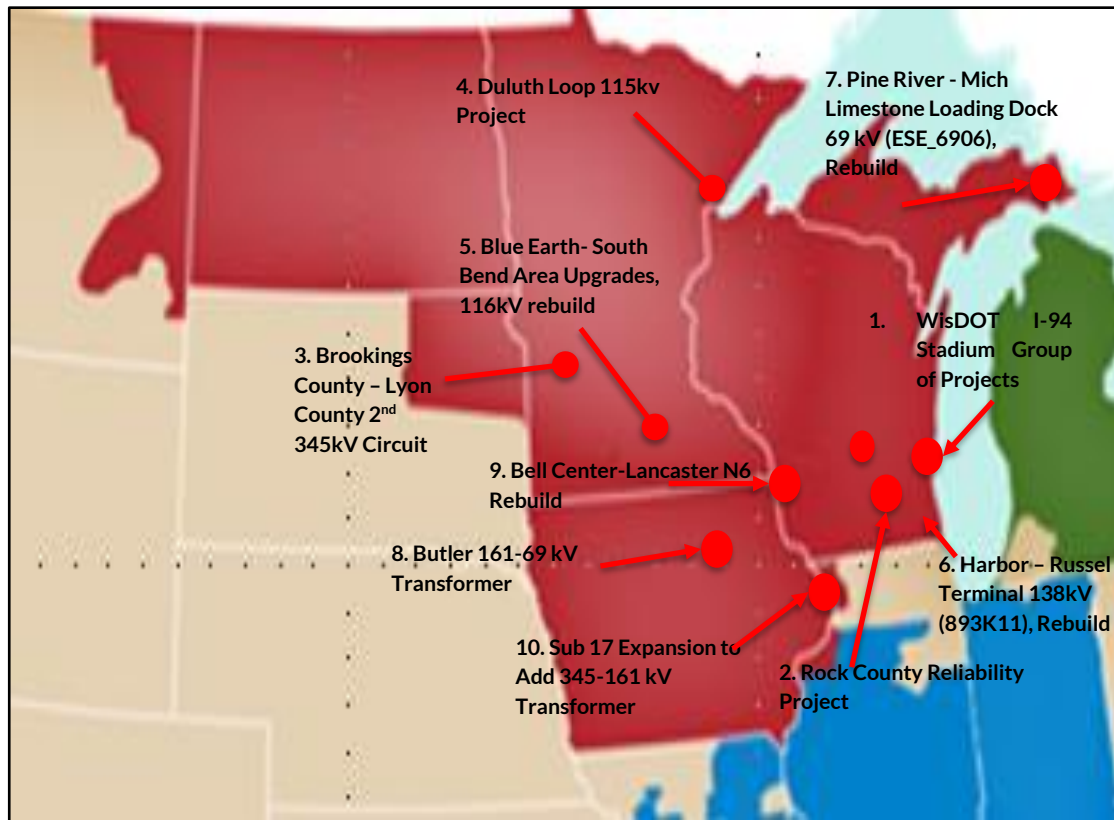


Figure 4.5-4: West region top ten projects by cost (data as of September 15, 2022)

4.5.1 American Transmission Company (ATC)

American Transmission Company proposed 50 projects at an estimated cost of \$539 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the ATC area, all 50 of these projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, five are Baseline Reliability Project, 31 are Other projects, and 14 are Generator Interconnection projects with signed Generator Interconnection Agreements. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Baseline Reliability Projects:

Project 19966 Munising Area Reactive Power Project

Project Description

Install one +/- 10 MVA, 69-kV STATCOM (Static Synchronous Compensator) at Munising substation. The total estimated cost of this project is \$10 million dollars. The expected in service

date for this project is December 31, 2025.

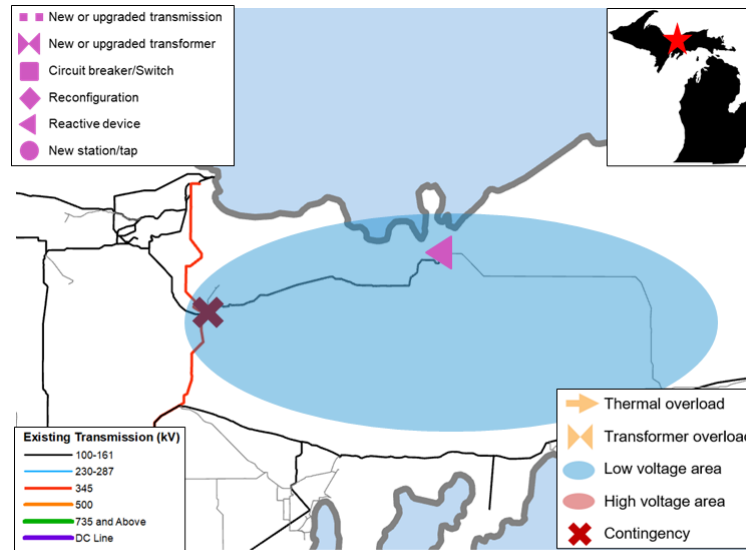


Figure 4.5.1-1: P19966 Geographic transmission map of project area

Project Need

NERC Category P2 Contingency causes voltage collapse in the Munising Area.

Cont. Type	Limiting Element	Rating (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P22	MUNISING 138	0.9	0.80	0.99
P22	SHINGLETON	0.9	0.82	1.00
P22	TIMBER 6	0.9	0.82	1.00
P22	TMBR PDS TAP	0.9	0.82	1.00
P22	MUNISING 69	0.9	0.82	1.01
P22	ALGER	0.9	0.82	1.01
P22	AD HIAW	0.9	0.83	1.00
P22	AD HIAW TAP	0.9	0.83	1.00
P22	FOREST L	0.9	0.84	0.99
P22	CHATHAM	0.9	0.85	0.99

Table 4.5.1-1: P19966 Voltage Performance drivers

Alternatives Considered

No alternatives were considered for this project.

Project 21899 Port Washington – Saukville 138 kV (X-132/X-133) partial rebuild

Project Description

Full rebuild of 3.5 miles of X-132/X-133 and reconductor only for the remaining 1.4 miles due to newer steel poles using ATC standard T2 Dove ACSR 556 conductor. Replace legacy lattice towers, install new OPGW and static line. The total estimated cost of this project is \$7 million dollars. The

expected in service date for this project is June 30, 2025.

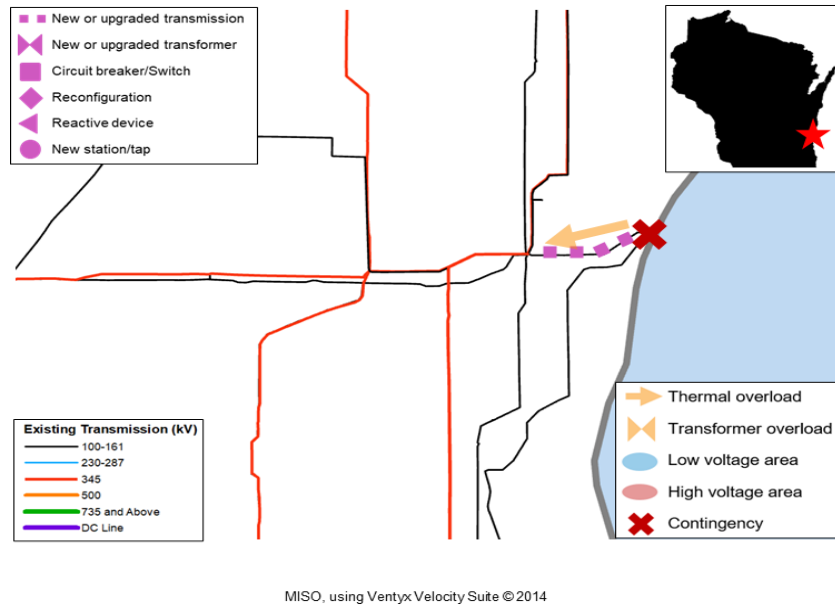


Figure 4.5.1-2: P21899 Geographic transmission map of project area

Project Need

NERC Category P7 and P6 contingencies causes line loading in excess of emergency ratings

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P71	PORT WSH 138 - SAUKVL6 138 Ckt 1	402	103.39	94.56
P71	PORT WSH 138 - SAUKVL1 138 Ckt 1	402	102.49	94.65

Table 4.5.1-2: P21899 Thermal Loading drivers

Alternatives Considered

No alternatives were considered for this project.

Project 21286 Rocky Run SS Power Transformer Replacement

Project Description

Replace T1, a 1968 vintage 345/115 kV, 150 MVA power transformer with a 345/115 kV, 500 MVA system spare. The total estimated cost of this project is \$5.4 million dollars. The expected in service date for this project is December 31, 2023.

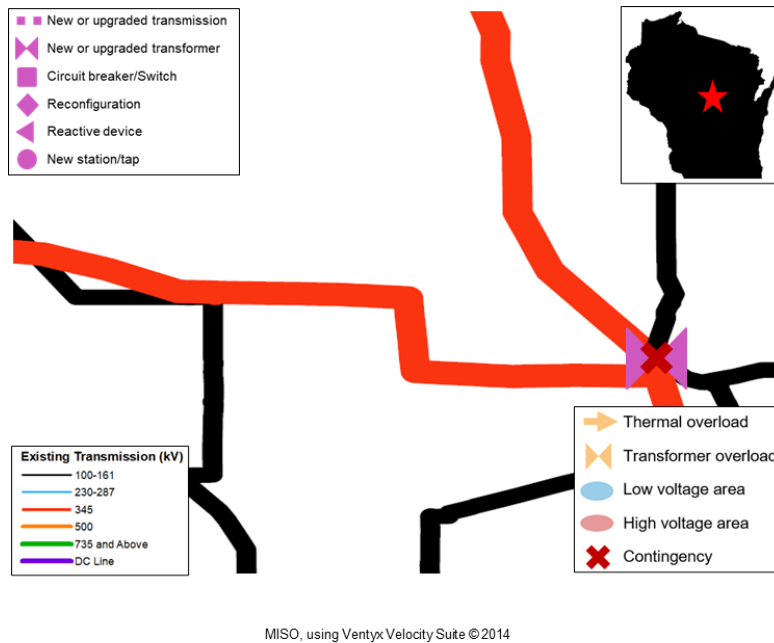


Figure 4.5.1-3: P21286 Geographic transmission map of project area

Project Need

T1 loading exceeds the emergency rating for the NERC Category P2.3 contingency of Rocky Run 115 kV breaker 1153.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P23	'ROCKY RN B1' 345 - 'ROCKY RN' 115 Ckt 1	223	103.8	47

Table 4.5.1-3: P21286 Thermal Loading drivers

Alternatives Considered

No alternatives were considered for this project.

Project 20805 Rock County Reliability Project

Project Description

To address the limited connections, ATC is proposing to develop 345-138 kV connections at Kittyhawk. Project scope includes adding two 500-MVA, 345-138 kV transformers at the Kitty Hawk substation, adding a 138 kV line between the Kittyhawk and Town Line Road substations, looping into Kittyhawk existing Paddock-Town Line and Town Line-Bass Creek 138 kV lines, uprating Rock River-Marine 138 kV line, and increasing ground grid fault capability at several nearby substations. The total estimated cost of this project is \$49.7 million dollars. The expected in service date for this project is December 31, 2024.

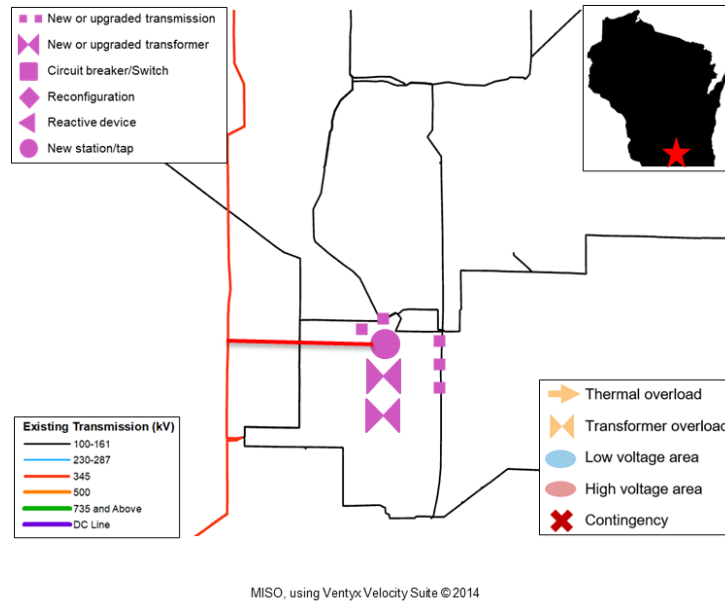


Figure 4.5.1-4: P20805 Geographic transmission map of project area

Project Need

Several factors drive the need for this project resulting in a variety of TPL-001 Planning Event limitations. While many might be addressed by re-dispatching 345 and 138 kV generation, the future availability of the 138 kV generation is very uncertain. Also, new 345 and 138 generation additions to parts of the system that aggravate the limitations will make it increasingly difficult to find enough generation to redispatch. In addition, there is long-lead time equipment at the Paddock substation that further aggravates the limitations. In 2020 actual system events highlighted this need and make it a near and long term need. All of this points to the need for an additional 345-138 kV source for the Janesville area.

Contingency types of P2, P3, and P6 were the main drivers. These contingencies were reviewed and tested on ATC's sensitivity models developed from the MTEP20 series of models. Overloads on the 138 kV system as well as overloads on the Paddock 345/138 kV transformer were prevalent. As noted above, redispatch was an option when allowed by TPL-001 to address some of the issues, but a larger area solution was preferred as there is only one 345 kV source in the area.

The proposed project addresses these needs and creates resiliency in the system given the uncertainty of future generation flexibility.

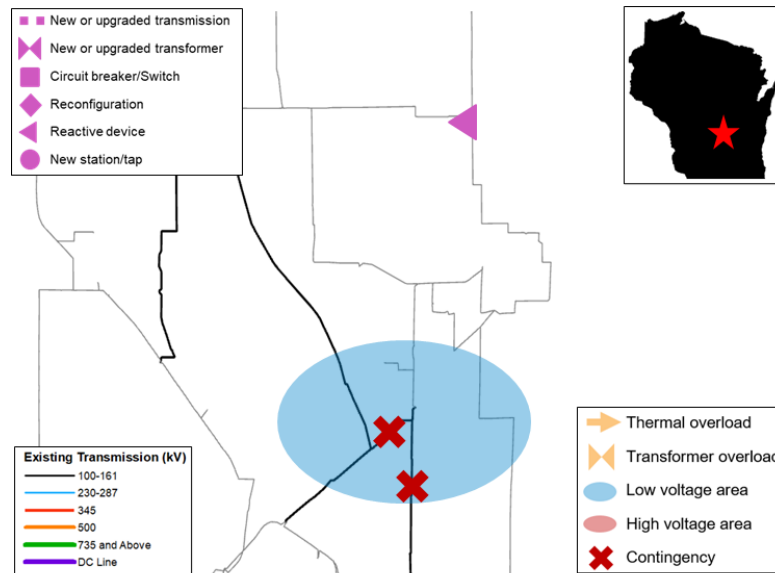
Alternatives Considered

No alternatives were considered for this project.

Project 22986 Vinburn Substation 69 kV Capacitor Addition

Project Description

Install a 69 kV capacitor bank configuration and associated automatic control features at Vinburn SS. The total estimated cost of this project is \$3.6 million dollars. The expected in service date for this project is December 31, 2024.



MISO, using Ventyx Velocity Suite © 2014

Figure 4.5.1-5: P22986 Geographic transmission map of project area

Project Need

NERC Category P6 contingencies causes low voltages on the 138 kV and 69 kV system

Cont. Type	Limiting Element	Rating (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P6	RNR_138	.9	0.85	0.95
P6	SPR_138	.9	0.85	0.95

Table 4.5.1-5: P22986 Voltage Performance drivers

Alternatives Considered

No alternatives were considered for this project.

Other Projects:

Projects Driven by Other Local Needs

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
16769	Physical Security Program 2023	Physical Security projects typically have shorter project life cycles. The Physical Security Program is guided by the NERC Standard CIP-014 and is designed to harden substation infrastructure. Specific project locations and scopes of work are not made public. Initial capital spend for this program expected in 2023.	1/1/2025	\$1.8
16767	Communications Reliability Program 2023	Communications network system upgrades typically require limited infrastructure modifications and additions with shorter project life cycles. Substation Communications Reliability Upgrade projects are typically (but not limited to): - OPGW additions, replacements, relocations, or removals. - Project drivers typically are communications support for SCADA, Relay protection, Security systems, Small communications network upgrades, and Telecom industry market transitions. These projects have limited scope and cost. Initial capital spend for this program expected in 2023.	6/30/2026	\$19.6
23531	Cottonwood - Summit Underground Fiber	This project will establish a fiber link between the Cottonwood to Summit substations, which will enhance communications resiliency by interconnecting into ATC's larger substation fiber network infrastructure at Summit.	12/31/2024	\$3.1

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
16768	Line Clearance Mitigation Program 2023	Line Clearance Mitigation projects have shorter project life cycles. Projects are driven by the ongoing assessment and analysis of field line clearances using Light Detection and Ranging (LiDAR) technology. Clearance mitigation work immediately follows data assessments and issue identification to maintain safety and system reliability. Initial capital spend for this program expected in 2023.	6/30/2026	\$3.9

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
16766	Small Capital Project and Asset Renewal Program 2023	Asset replacements and upgrades typically require limited infrastructure modifications and additions with shorter project life cycles. Transmission Line Projects typically (but not limited to): Structure, Cross-arm, Insulator, Surge-arrester, and Pole hardware replacements. Substation Projects typically (but not limited to): Relays, Circuit breakers, Switches, Instrument Transformers (CTs & PTs etc.), Batteries, RTUs, and IT/OT/Communications hardware replacements. Project drivers may be Asset Renewal or small system limit upgrades. These projects have limited scope and cost. Initial capital spend for this program expected in 2023.	6/30/2026	\$34.9
18046	Darlington – Rock Branch, (Y-109), 69kV, Rebuild & OPGW	T-Line Scope: Rebuild approximately 14 miles of the Y-109 line from Darlington - Rock Branch. Install new conductor IT/Communications: Install fiber and terminate at substations. Substation Scope: Fiber terminations at remote ends. Asset Renewal of substation equipment: Rock Branch-Replace disconnect switches, jumpers, strain bus and (1) breaker. Darlington- Replace jumpers and strain bus section.	12/31/2025	\$16.7

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18047	South Beaver Dam - Hubbard, (Y-134), 69kV Rebuild & OPGW	T-Line Scope to replace: South Beaver Dam SS to Str 230 (~13.2 miles) Str 230 to Hubbard (~3.3miles) Install new conductor and install new OPGW fiber. Substation Scope: South Beaver Dam- replace jumpers and 600A switches, fiber terminations Horicon- replace existing 336 jumpers, fiber terminations. Hubbard- Fiber terminations, no other work planned.	12/31/2026	\$20.1
18049	Portage St - 3 Mile 69 kV (6901/ESE_6902), Re-insulate & Underground Fiber	9 mile SW STA – Portage (6901/6902), 69 kV, Re-insulate	12/1/2025	\$5.4
19525	Saratoga SS - Control House and Breaker Asset Renewal	Replace control building and contents. Replace 69 kV oil breakers. Replace 2 CCVT's. Install normally-open GOAB line switch outside the fence to enable the connection of radial lines Y-301 and Y-302. Retire inoperable 922-A and 923-A breaker bypass switches which will no longer be required with the installation of the new GOAB. Reconductor 138 kV Bus 2 to match the higher capacity X-43 line rating. Install Dual batteries to meet future TPL-002-2 standards.	12/1/2025	\$8.2
19962	Nelson Dewey - Bloomington (Y-184), 69 kV, rebuild & OPGW	Rebuild and install OPGW on approximately 15 miles on Y-184 from Nelson Dewey substation to Bloomington substation. Like for like wood pole replacement with woodpecker mesh where necessary.	12/31/2025	\$16.3
21345	National - Presque Isle SW YD 138 kV (446), Re-insulate & OPGW	Reinsulate ~50% of structures on the line with like for like porcelain bells. Replacement of structures (~15% based on ground inspection) for woodpecker damage. Install OPGW.	12/31/2025	\$8.3
21346	Ontonagon - Victoria Hy 69 kV (Ontonagon69 69 kV), Re-insulate	Reinsulate ~65% of structures on the line with like for like porcelain bells. Replacement of structures (~5% based on ground inspection) for woodpecker damage.	12/31/2024	\$4.7
21347	Huiskamp - Ruskin 69 kV (6937), Partial Rebuild	Replace 23 legacy steel lattice towers with 23 new structures.	6/30/2025	\$7.9
21348	Tower Dr - Pulliam 69 kV (P-16/R-70), Partial Rebuild	Reroute approximately .65 miles of double circuit 69 kV transmission line on steel lattice towers due to condition and performance from ice damage including	4/1/2024	\$4.1

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
		removal of 10 legacy steel lattice towers with replacement with 7 double circuit steel poles and approximately .65 miles of double circuit conductor and replace with T2 4/0 conductor.		
21627	North Lake SS, Power Transformer Asset Renewal	Replace TG2, a 1971 vintage 138/69 kV 47 MVA power transformer with a new 138/69 kV 60 MVA replacement.	12/31/2025	\$4.2
21628	Rockdale and Christiana SS - Control House and Relay Asset Renewal	Replace Rockdale 138 kV control building and contents, reusing three bus differential panels installed in 2011 as appropriate. Replace X-65 and X-66 line relay panels at Christiana SS and coordinate with Rockdale work. Install Dual batteries to meet future TPL-002-2 standards in the Rockdale 138 kV building.	12/31/2025	\$5.8
21629	Oak Creek, Racine SS and Remote Ends, Control Buildings and Relaying Asset Renewal	Replace control building and contents at Racine and Oak Creek 138 kV. Replace 9 138 kV oil breakers at Racine, Dewey and Lincoln Replace 21 138 kV Model ACBL, ATA-2A and MK-40 at Dewey and Racine. Install approximately 35 relay panels. Provide room for WEC distribution equipment at Racine, demo existing building.	12/31/2025	\$20.9
21915	Fitzgerald SS – Transformer and Breaker Asset Renewal	Replace T1, a 1972 vintage 345/138 kV, 336 MVA power transformer with a new 345/138 kV 500 MVA standard three phase transformer. Replace three 138 kV oil circuit breakers with SF6 gas, eliminating all oil breakers at this station. Replace one 138 kV 1600 A disconnect switch and associated jumpers. Remove bus arresters off bus 4 and install line arresters.	12/31/2025	\$7.8
21916	Browntown - South Monroe 69 kV (Y-155), Rebuild	Rebuild approximately 8.3 miles on Y-155 from Browntown substation to South Monroe substation.	12/31/2026	\$10.3
21917	West Marinette SS – Transformer and Breaker Asset Renewal	West Marinette SS – Transformer and Breaker Asset Renewal.	12/31/2025	\$8.9
21918	Pine River - Mich Limestone Loading Dock 69 kV (ESE_6906), Rebuild	T-Line rebuild of approximately 25.5 miles with steel poles and reconductor. Construction-Utilize two winter cycles for constructability.	12/31/2027	\$31.1
21923	Everett GIS Asset Renewal	Rebuild the GIS substation in the same location and re-use existing HPFF (High-Pressure-Fluid-filled) terminations.	12/31/2025	\$16.9

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21926	Norwich - Harbor 138 kV, (893K11), underground line replacement	Norwich - Harbor 138 kV, (893K11), underground line replacement.	12/31/2025	\$37.0
23008	WisDOT I-94 Stadium Group of Projects	WisDOT Road Move: ATC's 138 kV lines and the Park Hill substation have been identified as impacted by the roadway construction in all WisDOT alternatives and will require relocation between the end of Calvary Cemetery and the 28th Street Substation to maintain the transmission network and distribution interconnections. Final transmission line routing and substation site selection to be determined through project investigations and coordination. Asset Renewal Rebuild of Structures 4145-4149: Install structures, OPGW/static, and new conductors. Communications: Install fiber communications between 96th Street Substation to structure 4145 and from the new substation location for the WisDOT road move to Valley Substation.	12/31/2026	\$63.7

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
16770	Load Interconnection Program 2023	Load Interconnection Project life cycles are customer need driven and typically have shorter project life cycles. These Load Interconnections typically require limited infrastructure modifications and additions. These projects have limited scope and cost. Initial capital spend for this program expected in 2023.	6/30/2026	\$18.0
21925	Plymouth #5 - New DIC	Plymouth #5 - New DIC	9/1/2025	\$21.5
22987	Northview SS, DIC, New Substation	ATC to install a new 138 kV circuit on new ROW (~0.6 miles) to the new substation. ATC to install two manual load break switches and one manual disconnect switch at the new tap point on line X-107 and 10. New structures and 8 foundations are planned.	6/1/2023	\$3.2
22988	Gaston Road SS, DIC, Additional Transformer	New DIC, additional transformer.	12/31/2024	\$3.0
22989	Columbus SS, DIC, Transformer Replacement	Columbus SS, new DIC, transformer replacement.	6/1/2024	\$8.7
23007	Cambria SS, DIC, New Substation	Distribution Interconnection, new substation.	12/31/2024	\$3.0
23520	Johnstown SS, DIC, New Substation	DIC, New Substation	5/1/2024	\$3.0

Generator Interconnection Projects:

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21648	J1000 Comet SS Network Upgrades and Interconnection Facilities	Interconnection Customer shall install a Solar Generating Facility modeled at 50 MW in Grant County, Wisconsin.	3/27/2023	\$6.6
21812	J1002 Wautoma SS Network Upgrades and Interconnection Facilities	Interconnection Customer shall install a 99 MW solar facility connecting into the Wautoma SS in Waushara County.	1/26/2023	\$3.2
21813	J1003 North Beaver Dam SS Network Upgrades and Interconnection Facilities	Interconnection Customer shall install a 50MW solar farm interconnecting at North Beaver Dam SS in Dodge County.	6/6/2023	\$1.0
21814	J986 J1002 MPFCA	Each individual Interconnection Customer is responsible for a proportionate percentage share of the Common Use Upgrades (CUU) estimated cost based on its proportionate impact on the constrained facilities causing the need for the CUU.	12/31/2023	\$7.5

		J986 is seeking interconnection service for 149.76 MW. J1002 is seeking interconnection service for 99 MW.		
22045	J855 J864 J870 J871 MPFCA	Installation of ground conductor, ground rods and possibly ground wells along the perimeter of the Spring Green substation.	12/31/2022	\$0.2
22046	J1188 Sheepskin SS Network Upgrades and Interconnection Facilities	Installation of a 50 MW solar farm located in Rock County, Wisconsin.	5/11/2022	\$1.1
22047	J1154 Crawfish River Solar Network Upgrades	Installation of a 75 MW solar farm located in Jefferson County, Wisconsin.	5/15/2023	\$2.3
22048	J986 J1002 J1003 MPFCA	Ground grid mitigation at Roeder substation in Green Lake County, Wisconsin.	12/31/2023	\$0.2
22049	J1101 Kewaunee SS Network Upgrades and Interconnection Facilities	Installation of a 20 MW battery storage project at the J886 Generating Facility.	8/15/2023	\$0.15
22085	J1002 J1003 MPFCA	Shared work for ground grid upgrades at North Beaver Dam SS, cost shared jointly between J1002 and J1003. J1002 - 5.54% J1003 - 94.46%	12/31/2023	\$0.55
22105	J1042 North Lake Geneva SS Network Upgrades and Interconnection Facilities	Interconnection Customer shall install a 180 MW solar farm interconnecting at North Lake Geneva SS in Walworth County.	9/15/2023	\$1.4
22145	J1153 Holland SS Network Upgrades and Interconnection Facilities	Interconnection Customer shall install a 150 MW solar farm interconnecting at Holland SS in Sheboygan County.	7/20/2022	\$8.1
22146	J1171 Butternut SS Network Upgrades and Interconnection Facilities	Interconnection Customer shall install a 100 MW solar farm interconnecting at Butternut SS in Dodge County.	1/26/2023	\$3.9
22225	J1042 J1188 MPFCA	Each individual Interconnection Customer is responsible for a proportionate percentage share of the Common Use Upgrades (CUU) estimated cost based on its proportionate impact on the constrained facilities causing the need for the CUU. J1042 is seeking interconnection service for 180 MW. J1188 is seeking interconnection service for 50 MW.	11/1/2023	\$4.3

4.5.2 Cedar Falls Utilities (CFU)

Cedar Falls Utilities did not submit any new projects for MTEP22. MISO has not identified any issues in CFU area.

4.5.3 Central Minnesota Municipal Power Agency (CMMPA)

CMMPA proposed three projects at an estimated cost of \$6.85 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the CMMPA area, these projects are recommended by MISO to be included in MTEP22 Appendix A. These projects are classified as Other projects. The projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21975	Blue Earth Light and Water- New Northeast 69/12.47 kV Sub	Build a new North East 69/12.47.kV substation and self-supporting steel pole with 69.kV 3-way switch.	1/16/2023	\$4.1

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21973	Blue Earth Light and Water- Rice Street Transformer Replacement	Blue Earth Light and Water has a Rice Street sub which carries a 10/14 MVA 69/12.47 kV Transformer. The transformer is showing signs of age as it is 28 years old and needs to be replaced because of potential load growth.	5/2/2022	\$1.3
21974	Blue Earth Light and Water- West Sub Transformer Replacement	Blue Earth Light and Water West Sub Transformer replacement.	5/2/2022	\$1.5

4.5.4 City of Ames, IA (COA)

City of Ames, IA did not submit any new projects for MTEP22. MISO has not identified any issues in COA area.

4.5.5 Dairyland Power Cooperative (DPC)

Dairyland Power Cooperative proposed one Other – Age and Condition project with an estimated cost of \$23.9 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the DPC area, this project was recommended by MISO to be included in MTEP22 Appendix A. Expected in-service date and estimated cost are provided as of September 15, 2022.

Other Projects:

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22531	Bell Center-Lancaster N6 Rebuild	This project is a rebuild of DPC's existing 69 kV line (N6) connecting the Bell Center and Lancaster transmission substations. The recommended plan is to rebuild 40.51 miles of new 69 kV transmission line on existing right-of-way with 477 ACSR conductor. The recommended plan will address the age and condition issues, constructing the line with new poles and new 477 ACSR conductor.	10/15/2025	\$23.9

4.5.6 Great River Energy (GRE)

Great River Energy proposed nineteen projects at an estimated cost of \$79.5 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the GRE area, all of these projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, twelve are Other projects, and seven are Baseline Reliability projects. Great River Energy did not propose any Generator Interconnection projects. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Baseline Reliability Projects:

Project #20707: Granite Falls - Willmar (WB) Line Upgrade

Project Description

Upgrade the Granite Falls - Willmar (WB) line to 212 F by replacing 10 structures. The total estimated cost of this project is \$0.6 million dollars. The expected in-service date for this project is June 1, 2025.

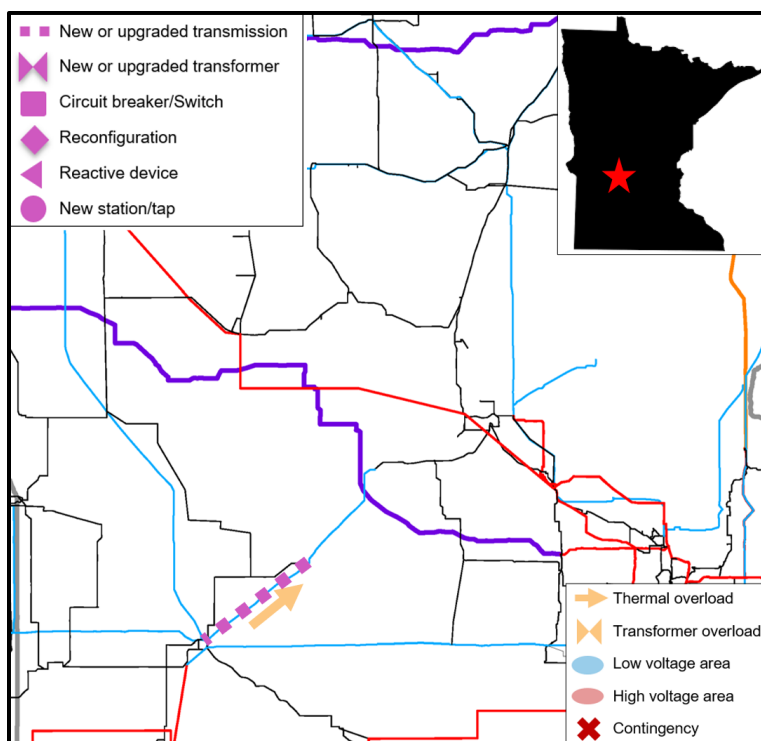


Figure 4.5.6-1: P20707 Geographic transmission map of project area

Project Need

Thermal violations caused by NERC Category P6 events in both 2026 Shoulder High Wind (wind dispatched at 83% of nameplate) and 2026 Spring Light Load (wind dispatched at 70% of nameplate) Sensitivity models.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	GRANITF4 - WMU-WILLMAR4 230 kV Ckt 1	264.9	119.6	78
P6	GRANITF4 - WMU-WILLMAR4 230 kV Ckt 1	264.9	122.3	84

Table 4.5.6-1: P20707 Thermal loading drivers

Alternatives Considered

No alternatives were considered for this project.

Project #22027: Birch Lake - Pleasant Lake - Wabedo 69 kV Line Rebuild

Project Description

Rebuild the Birch Lake - Pleasant Lake - Wabedo 69 kV line. The total estimated cost of this project is \$14.42 million dollars. The expected in-service date for this project is June 1, 2024.

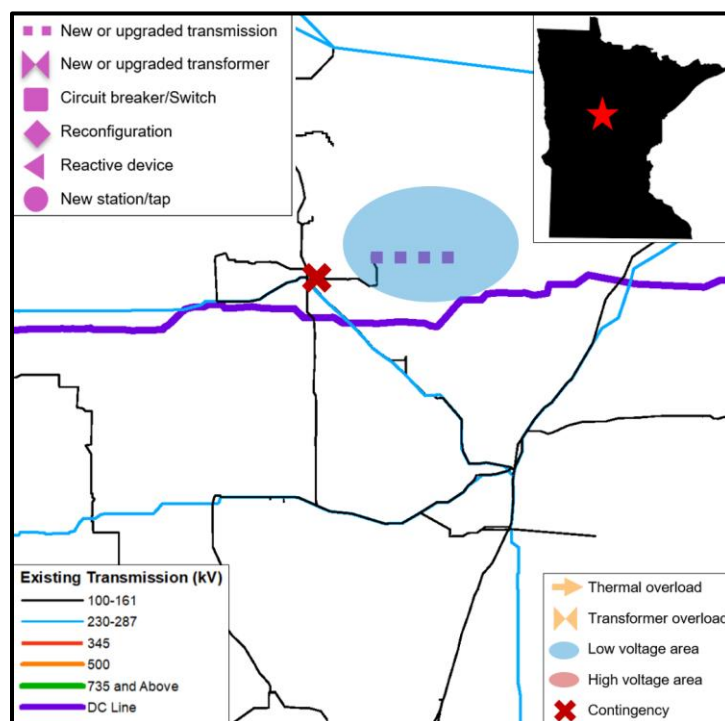


Figure 4.5.6-2: P22027 Geographic transmission map of project area

Project Need

Age and condition. For the loss of the Badoura - Tripp Lake 115 kV line, the voltages in the Blind Lake area go into violation. The existing line is 1/0 and 2/0 ACSR and is planned to be rebuilt with 795 ACSR to help the voltage profile of the Blind Lake area loads.

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P6	GRE-BIRCHLK8 69 kV	0.90	0.863	0.909
P6	GRE-PLSNTLK8 69 kV	0.92	0.911	0.935

Table 4.5.6-2: P22027 Voltage performance drivers

Alternatives Considered

No alternatives were considered for this project.

Project #22028: Denham - Harry Maser 69 kV Line Rebuild

Project Description

Rebuild the Denham - Harry Maser 69 kV line. The total estimated cost of this project is \$5.11 million dollars. The expected in-service date for this project is May 2, 2025.

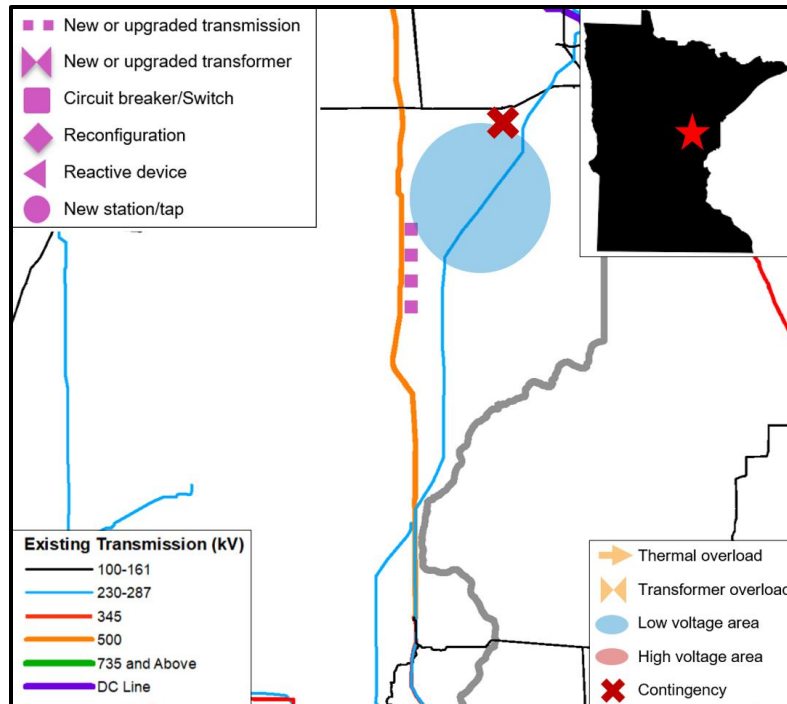


Figure 4.5.6-3: P22028 Geographic transmission map of project area

Project Need

Age and condition. For the loss of the Cromwell - Kettle River 69 kV line, there is a voltage violation at Sturgeon Lake but after rebuilding the existing line from 1/0 ACSR to 477 ACSR we will no longer have this voltage violation.

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
NB12	GRE-STURGEN8 69 kV	0.95	0.934	0.972

Table 4.5.6-3: P22028 Voltage performance drivers

Alternatives Considered

No alternatives were considered for this project.

Project #21065: Big Swan 5N18 115 kV Breaker Addition

Project Description

Add the Big Swan 5N18 breaker on the Crow River 115 kV line. The total estimated cost of this project is \$0.7 million dollars. The expected in-service date for this project is November 1, 2024.

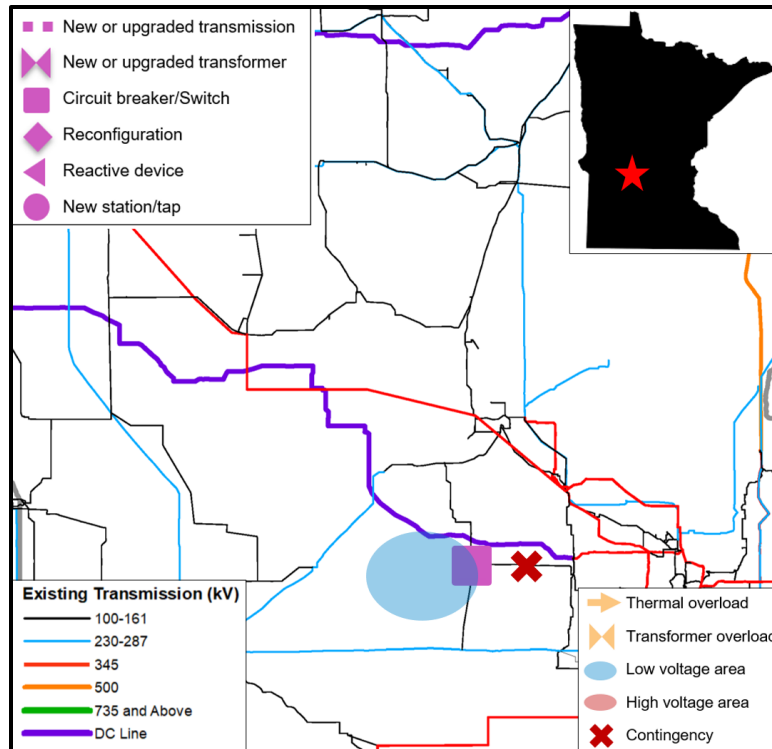


Figure 4.5.6-4: P21065 Geographic transmission map of project area

Project Need

Reduce over tripping for a fault on the Big Swan - Crow River 115 kV line. A fault on the Big Swan - Crow River 115 kV line trips the Big Swan 115/69 kV transformer causing low voltage problems in the 69 kV system.

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P6	GRE-LTCHFLD8 69 kV	0.95	0.920	1.007

Table 4.5.6-4: P21065 Voltage performance drivers

Alternatives Considered

No alternatives were considered for this project.

Project #22029: Walden 115/41.6 kV LTC

Project Description

Replace the existing Walden 115/41.6 kV transformer with an LTC transformer. The total estimated cost of this project is \$2.5 million dollars. The expected in-service date for this project is August 31, 2023.

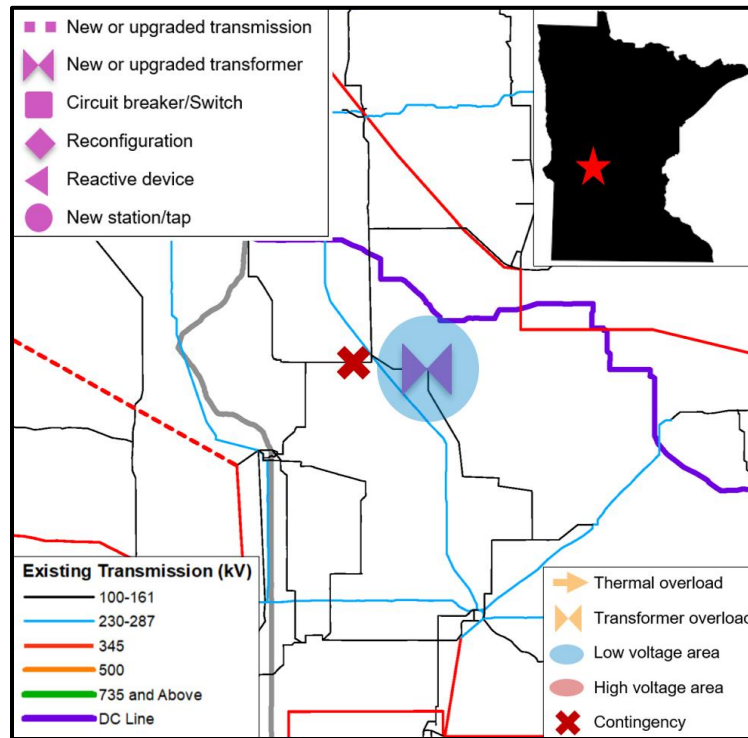


Figure 4.5.6-5: P22029 Geographic transmission map of project area

Project Need

This project is primarily needed to address low voltage concerns in the area. System planning studies with historical summer peak loads have shown that the Walden to Elbow Lake 41.6 kV system experiences marginal system intact voltage. The study also showed post contingent low voltage planning criteria violations in the Elbow Lake – Walden 41.6 kV system. The most critical contingencies for the area are the Morris to Morris OTP 115 kV line outage and Walden 115/41.6 kV transformer outage. For the loss of Morris to Morris OTP 115 kV line outage, all loads that are served from the Walden 115/41.6 kV transformer experience low voltage that are below GRE's post contingent low voltage criteria of 0.92 per unit.

Even though wide voltage swings are expected between different loading profiles throughout the year in the 41.6 kV system, historical voltage profile of the Walden 41.6 kV bus shows that the Walden 41.6 kV bus experiences high voltage problems that violates GRE's high voltage criteria of 1.05 per unit. The Walden LTC transformer is required to regulate both the low voltage and high voltage concerns in the area.

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P1-2	HOFFCOOP 41.6 kV	0.92	0.889	0.964

Table 4.5.6-5: P22029 Voltage performance drivers

Alternatives Considered

No alternatives were considered for this project.

Project #22025: Parkwood 115 kV Ring Bus

Project Description

Rebuild the 115 kV bus at Parkwood substation as a ring bus. The total estimated cost of this project is \$7.62 million dollars. The expected in-service date for this project is November 8, 2024.

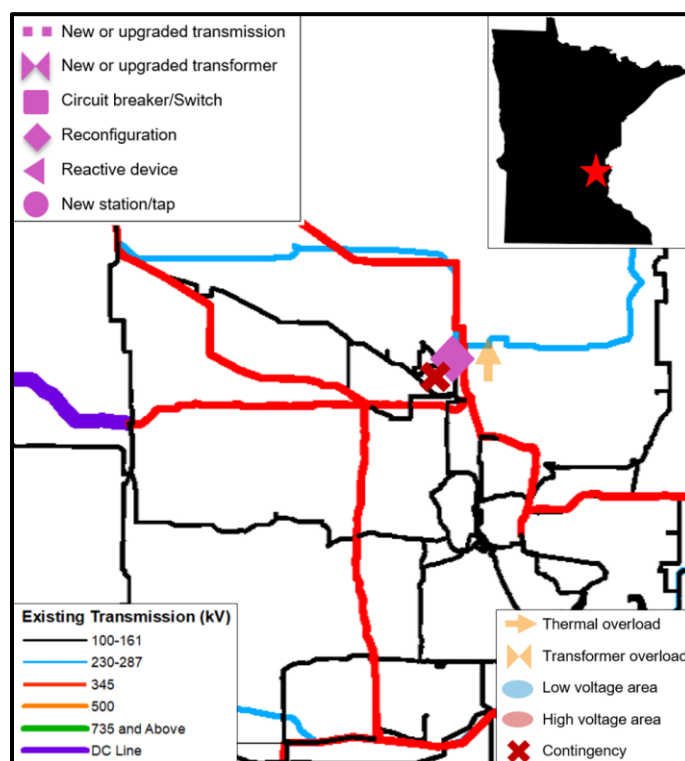


Figure 4.5.6-6: P22025 Geographic transmission map of project area

Project Need

115 kV line faults cause a trip of a Parkwood 115/69 kV transformer. Parkwood 12WB2 breaker failure causes overload of the Bunker Lake-Village Ten 69 kV line under peak loading conditions.

A Parkwood 115/69 kV transformer differential protection activation causes transfer tripping to the remote substations. For the Crooked Lake 115 kV line, this causes the loss of Connexus' load service at Crooked Lake, which serves a major commercial development.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
NB13	GRE-BUNKER 8 - GRE-VILGTEN8 69 kV Ckt 1	75.7	100	44

Table 4.5.6-6: P22025 Thermal loading drivers

Alternatives Considered

No alternatives were considered for this project.

Project #22030: Rost 161/69 kV Substation and Lorain - Rost 69 kV Line

Project Description

Construct the Rost 161/69 kV substation interconnection in the Heron Lake – Round Lake 69 kV transmission line. Construct approximately 9 miles of 69 kV transmission line from the Lorain substation to the Rost substation. The total estimated cost of this project is \$13.37 million dollars. The expected in-service date for this project is October 30, 2026.

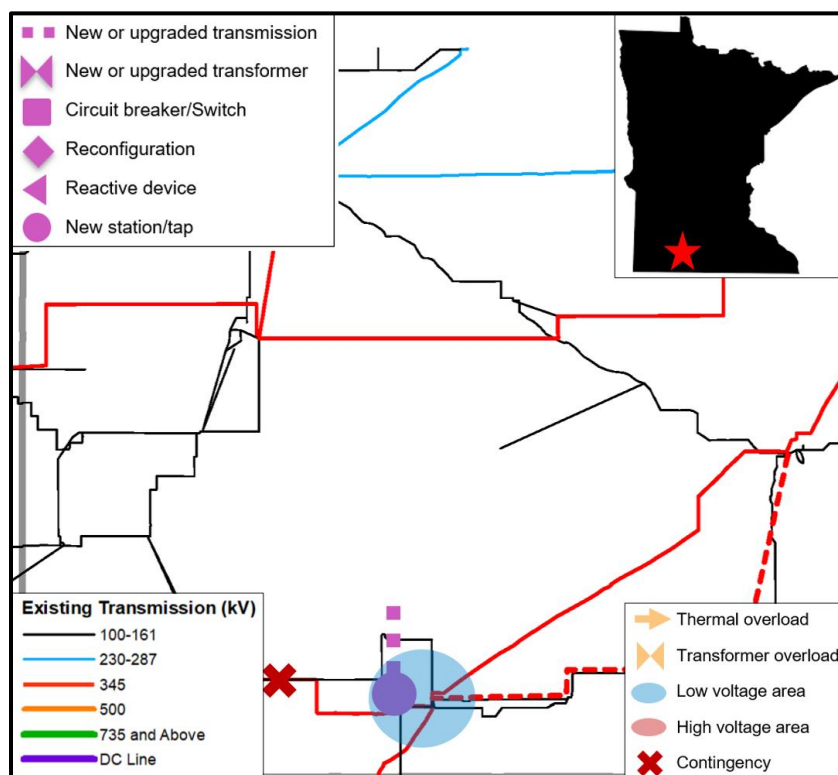


Figure 4.5.6-7: P22030 Geographic transmission map of project area

Project Need

Establish alternative source beyond Elk to serve the Worthington load center. Allow redundant 161/69 kV sources to be connected (Heron Lake and Elk). Lower reliability exposure to multiple 69 kV loads and wind farm connections on long 69 kV corridors. Establish new relaying point to enhance relay settings with consideration for the wind farm sources and allow for future wind connections into the substation.

Any outage on the 161 kV between Split Rock Substation (Xcel Energy) and Magnolia Substation leaves the system susceptible to low voltages for faults between Lakefield Junction and Elk 161 kV. Eliminate considerable operational concerns for prior-outage situation with a second event in the area. 161 kV prior outages of concern from Split Rock to Lakefield Junction including 161 kV substation on the corridor of Magnolia, Elk, Brewster, and Heron Lake. Reduce temporary operating guide and system reconfiguration due to voltage issues caused by a second contingency event. Eliminate significant risk of load loss within a broad area of southwestern Minnesota.

Cont. Type	Limiting Element	Voltage Limit (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P6	WRTHNGT8 69 kV	0.95	0.881	0.996

Table 4.5.6-7: P22030 Voltage performance drivers

Alternatives Considered

No alternatives were considered for this project.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
20666	Litchfield - Litchfield Tap Double Circuit Rebuild	Rebuild ~2.5 miles of the Litchfield - Litchfield 69 kV tap line to double circuit.	12/31/2026	\$2.28
21705	Mud Lake - Benton County - Monticello Storm Structure Addition	Add 5 storm resistant structures to the Mud Lake - Benton County - Monticello 230 kV line.	12/1/2022	\$1.82
21823	Benson - Morris Storm Structures	Install storm structures in the Benson - Morris 115 kV line.	12/1/2023	\$1.28
21824	Riverton - Wing River Storm Structures	Install storm structures in the Riverton - Wing River 230 kV line.	12/1/2023	\$0.66
21825	Cormorant - Pelican Rapids Storm Structures	Install storm structures in the Cormorant - Pelican Rapids 115 kV line.	6/28/2024	\$0.65
21826	Bunker Lake - Elk River Storm Structures	Install storm structures in the Bunker Lake - Elk River 230 kV line.	6/28/2024	\$0.6

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22026	Deer River - Jessie Lake - Wirt Tap 69 kV Line Rebuild	Rebuild the Deer River - Jessie Lake - Wirt Tap 69 kV line.	9/30/2025	\$20.7

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
20865	Shamrock 69 kV Substation	Construct an in/out line to the Wright Hennepin Electric Association's Shamrock substation. The Shamrock substation is a rebuild of the nearby Corcoran substation. The load served at Corcoran will be gradually moved to the new Shamrock substation.	2/21/2023	\$1.71
21829	South Dayton 115 kV Substation	Construct the new Connexus Energy (CE) South Dayton 115 kV in-and-out distribution substation in the Xcel 5522 line.	9/1/2023	\$1.5
21830	Shamaineau Lake 115 kV Substation	Construct the new Crow Wing Power Shamaineau Lake distribution substation.	11/21/2022	\$0.79
22445	Cedar 115 kV Substation	DEA building new in/out distribution substation. GRE will need to double circuit tap into sub off of DA-DA line. GRE is also responsible for the high-side structures and associated equipment.	6/1/2023	\$2.77
22528	Matter 69 kV Substation	Install a manual load break 3-way load break 69 kV/1200A switch on a steel pole and foundation on GRE's DX (Crow River to Victor) line. Install adjacent grading structures if needed. Build HS DE to 115 kV.	1/1/2024	\$0.4

4.5.7 ITC Midwest (ITCM)

ITC Midwest proposed 27 projects at an estimated cost of \$193 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the ITCM area, all of these projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, one is a Baseline Reliability Project, 16 are Other projects, and 10 are Generator Interconnection projects with a signed Generator Interconnection Agreement. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Baseline Reliability Projects:

Project 21929 Lakefield Corners to Rost 161 kV Circuit & Substations

Project Description

As part of a joint study to serve load expansion in Worthington, MN, ITC will construct a 161 kV circuit between new Lakefield Corners and Rost substations. The total estimated cost of this project is \$13.5 million dollars. The expected in-service date for this project is December 31, 2026.

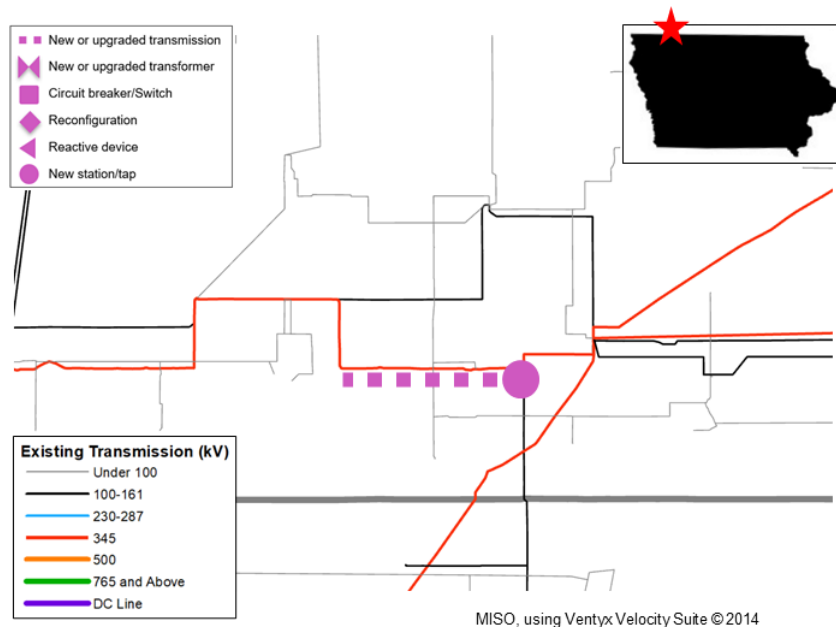


Figure 4.5.7-1: P21929 Geographic transmission map of project area

Project Need

Load addition on the Southeast side of the city of Worthington, MN (MTEP ID 22030 – Worthington Area Projects) requires transmission projects on GRE, ITCM, and MRES facilities. Outages on the 161 kV system between Split Rock and Lakefield Junction cause widespread voltage concerns and have resulted in long term operating guides that result in significant consequential load shedding.

Cont. Type	Limiting Element	Rating (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P6	WRTHNGT8 69 kV	0.95	0.881	0.996

Table 4.5.7-1: P21929 Voltage Performance drivers

Alternatives Considered

No alternatives were considered for this project.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21930	Millstream 69 kV Switching Station	ITC has been working with Amana Society on way to improve reliability to their substation, and ITC will be building a new 69 kV switching station called "Millstream" adjacent to the existing Amana Society substation to improve area transmission system reliability.	12/31/2024	\$7.74
21939	Cedar Rapids Hunter Field Substation Expansion	IPL has notified ITC that it will be adding a 3rd 161x69/24.9 kV transformer at the Hunter Field Substation. ITC will need to add an additional breaker and bus to connect the new IPL transformer at Hunter Field. All new ITC equipment to be designed to 161 kV specs to allow future conversion of the substation to 161 kV operation. Substation will initially be operated at 69 kV.	12/31/2024	\$1.28
21943	Rose Hollow Add 161 kV Terminal for North Gate line	MidAmerican has notified ITC that they will be building a new 161 kV line from their existing North Gate substation to the ITC Rose Hollow substation. Per the terms of the agreement with MidAmerican, ITC will need to install a new 161 kV terminal at Rose Hollow for the new line to North Gate.	12/31/2024	\$0.98

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21828	Agency Street New Control Enclosure & Relay Replacements	The relays at Agency Street are approaching their 20-year end of life. ITC will install a new control enclosure with all new relays to provide ITC a separate control enclosure from IPL and minimize required outages and outage times for the relay replacements. All necessary outages should be coordinated with IPL. Also coordinate with IPL on location of new control building as this is an IPL owned site.	12/31/2024	\$2.09
21842	Jasper County Control Enclosure and Relay Replacements	A new Control Enclosure is required at the Jasper County Substation due to age and condition. ITC will install a new Control Enclosure with all new relays.	12/31/2024	\$2.48

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21933	Fox Lake Relay, Breaker, & SCADA Replacements	ITC will replace all of its relays and SCADA equipment as well as remove the existing carrier equipment/wave trap at the Fox Lake substation. Circuit Breaker 736 will also be replaced at Fox Lake substation.	12/31/2025	\$2.65
21935	Hayward Transformer Replacement	ITC will replace the two 161/69 kV transformers at Hayward Substation with a single 150 MVA 161/69 kV unit to make room for the new 161 kV bus tie breaker. ITC will move the Glenworth – Hayward 161 kV terminal to the opposite side of the Hayward – Murphy Creek 161 kV terminal.	12/31/2025	\$6.74
21940	Turkey River to Guttenberg 69 kV Rebuild	The 9.04 mile Turkey River – Guttenberg 69 kV line is at the end of its useful life and needs to be rebuilt. New line should be rebuilt with T2-4/0 ACSR conductor and OPGW for future communications path between Turkey River and Monona.	12/31/2025	\$10.4
21941	Guttenberg to Garnavillo 69 kV Rebuild	The 11.23 mile Guttenberg - Garnavillo 69 kV line along with the 0.18 mile tap to Garnavillo are at the end of their useful life and need to be rebuilt. New line should be rebuilt with T2-4/0 ACSR conductor and OPGW for future communications path between Guttenberg and Monona.	12/31/2028	\$14.4
21942	Garnavillo to Elkader 69 kV Rebuild	The 7.63 mile Garnavillo - Elkader 69 kV line is at the end of its useful life and needs to be rebuilt. New line should be rebuilt with T2-4/0 ACSR conductor and OPGW for future communications path between Garnavillo and Elkader.	12/31/2028	\$10.0
21946	Pelican-Triboji 69 kV Rebuild	Two sections of the Pelican-Triboji 69 kV line are near end of life and in poor condition. The line between Pelican and structure 162 (~1.04 miles) and the line between Triboji and structure 72 (~4.45 miles) will be rebuilt to current 69 kV standards. All new line should be rated 600A or higher. All necessary outages should be coordinated with IPL and CBPC.	12/31/2023	\$6.46
21976	Jasper to Laurel 161 kV Line Rebuild	ITC will rebuild approximately 10 miles of the Jasper to Laurel 161 kV line. The 161 kV line conductor will be capable of 1600A or greater.	12/30/2027	\$18.8

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21980	ITC Midwest Asset Replacement Program 2024	Replace aging and outdated equipment on a cycle that will ensure each system is replaced near its expected end of life. Modern equipment can improve reliability, use state of the art technology, and will typically use longer maintenance intervals which reduces maintenance costs. Equipment is commonly equipped with better monitoring and alarming functionality giving improved remote supervision.	12/31/2024	\$37.1

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21936	Centerville Cooper Creek 69 kV Interconnection	IPL will be building a new 69/25x12.5 kV substation named "Centerville Cooper Creek" adjacent to the existing Centerville North substation on the location of the old combustion turbines. This new substation will replace the existing IPL/ITC Centerville North Substation and accommodate future area load growth as well as IPL's future conversion to 25 kV distribution. IPL will be installing two 12/20 MVA transformers at the new Centerville Cooper Creek substation. For the ITC owned portion of Centerville North, ITC will build a new 69 kV substation with two single boxes, two line breakers, and a bus tie breaker. Once the new Centerville Cooper Creek substation is in-service, the existing Centerville North substation will be retired.	6/30/2026	\$3.70
21937	Chariton North 69 kV Substation Interconnection	IPL will be building a new 69/26.2 kV Substation named Chariton North. IPL will be installing one 69/26.2 kV 20/33 MVA or 30/50 MVA transformer initially but is planning the substation to accommodate a second 69/26.2 kV 20/33 MVA or 30/50 MVA transformer when all the load is added. All the ITC 69 kV substation equipment will be capable of 1200A or greater. The 69 kV lines to the new substation will be T2-4/0 or larger and will be the standard 69 kV construction.	12/30/2024	\$6.83

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21982	Southern Dubuque 161/69 kV Source & Bunker Hill 69 kV Loop	ITC Midwest will construct a new 161/69 kV substation on the south side of Dubuque. Then a new right of way 69 kV line ~2.65 miles to the existing Bunker Hill substation in the center of Dubuque. The new line will provide the 2nd 69 kV line to the radial Bunker Hill 69 kV sub. Alliant Energy will expand the Bunker Hill site to allow for a 2nd distribution transformer and ITC to have two 69 kV line breakers and a 69 kV bus tie breaker. The new ITC 161/69 kV sub will be capable of two 150 MVA 161/69 kV transformers (one initially). The 161 kV will be constructed with the standard 161 kV ring convertible to breaker and 1/2 with just the single transformer and the two 161 kV line taps initially. The 69 kV will need to allow for a potential 2nd 150 MVA transformer and 69 kV bus tie breaker but will initially be just a single 69 kV line to feed the Bunker Hill loop.	12/31/2025	\$19.3

Generator Interconnection Projects:

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21977	J1050 Generator Interconnection - TOIF	The Interconnection Customer, Great Pathfinder Wind, LLC, has proposed connection of a 225 MW wind powered generating facility to the Doud 161 kV bus. A breaker position in the Doud 161 kV ring bus will be added to accommodate the interconnection. TOIF cost to include dead end, line disconnect switch, arresters, PT's, relay panel, and other associated equipment. The IC has requested ITCM to begin design under an E&P agreement to better accommodate their requested in service date while the MISO process is still ongoing.	8/22/2022	\$0.4
21978	J1050 Generator Interconnection - Network Upgrade	The Interconnection Customer, Great Pathfinder Wind, LLC, has proposed connection of a 225 MW wind powered generating facility to	8/22/2022	\$0.4

		the Doud 161 kV bus. A breaker position in the Doud 161 kV ring bus will be added to accommodate the interconnection. Network Upgrade cost to include the new breaker position, switch and associated equipment. The IC has requested ITCM to begin design under an E&P agreement to better accommodate their requested in service date while the MISO process is still ongoing.		
21987	J1181 GIA - TOIF Lasalle 345 kV Switching Station	ITC is entering into a Generator Interconnection Agreement with Invenergy for the interconnection of project J1181, a 200 MW wind-powered generating facility to a new, 3 breaker, 3 terminal 345 kV "Lasalle" switching station along the Hazleton to Mitchell 345 kV line. Invenergy's collector substation will be located in close proximity to the new Lasalle switching station. Invenergy has requested an October 31, 2022 in service date for the project. Under the terms of the GIA, Invenergy will transfer property for Lasalle 345 kV station to ITC, and Invenergy will also provide a topographical survey and soil borings for the ITC facilities. Lasalle switching station will be constructed with 3,000A capability.	8/31/2023	\$0.9
21989	J1181 GIA - TO Funded 345 kV Network Upgrades	ITC is entering into a Generator Interconnection Agreement with Invenergy for the interconnection of project J1181, a 200 MW wind-powered generating facility to a new, 3 breaker, 3 terminal 345 kV "Lasalle" switching station along the Hazleton to Mitchell 345 kV line. Invenergy's collector substation will be located in close proximity to the new Lasalle switching station. Invenergy has requested an October 31, 2022 in service date for the project. Under the terms of the GIA, Invenergy will transfer property for Lasalle 345 kV station to ITC, and Invenergy will also provide a topographical survey and soil borings for the ITC facilities. Lasalle switching station will be constructed with 3,000A capability.	8/31/2023	\$12.6

21991	J836 GIA - TOIF - Ledyard 345 kV	7-21-21: ITCM and Duke Energy have entered into a MISO Generator Interconnection Agreement. Initial sketch provided Design work for TOIF under an Engineering and Procurement Agreement. The work under the E&P Agreement included design of TOIF and Network Upgrades necessary for the expansion of Ledyard 345 kV to a 4-terminal, 4-breaker ring bus configuration.	8/26/2022	\$1.1
21992	J836 TO Funded Generator Interconnection - Ledyard 345 kV	ITCM and Duke Energy have entered into a MISO Generator Interconnection Agreement for upgrade of Ledyard 345 kV switching station. The work under the Agreement will include design of TOIF and Network Upgrades necessary for the expansion of Ledyard 345 kV to a 4-terminal, 4-breaker ring bus configuration. Design work for TOIF and Network Upgrade commenced under an Engineering and Procurement Agreement, and the GIA provides for continuation and completion of the work at Ledyard.	7/1/2022	\$1.8
21996	J1084 Generator Interconnection TOIF	TOIF includes facilities and equipment at the 345 kV substation dedicated to the 345 kV radial line and Generating Facility.	10/1/2023	\$0.7
21997	J1084 Generator Interconnection Network Upgrades	Rock Creek 345 kV ring bus will be expanded with a 345 kV breaker and other necessary equipment to interconnect the new terminal for project J1084.	10/1/2023	\$1.3
21998	J1164 Generator Interconnection TOIF	Magnolia 161 kV substation currently has no circuit breakers protecting the Magnolia to Elk 161 kV line and the Magnolia to Rock County 161 kV line. The Facilities Study for J1164 identified Network Upgrades and TOIF necessary for the interconnection of project J1164 to ITCM's Magnolia 161 kV substation. Project J1164's impact to the transmission system is currently being evaluated as part of the DPP 2018 April West Area Study, which has not been finalized pending results of an Affected System Study.	8/1/2024	\$0.5

21999	J1164 Generator Interconnection Network Upgrades	Magnolia 161 kV substation currently has no circuit breakers protecting the Magnolia to Elk 161 kV line and the Magnolia to Rock County 161 kV line. The Facilities Study for J1164 identified Network Upgrades and TOIF necessary for the interconnection of project J1164 to ITCM's Magnolia 161 kV substation. Project J1164's impact to the transmission system is currently being evaluated as part of the DPP 2018 April West Area Study, which has not been finalized pending results of an Affected System Study.	8/1/2024	\$8.8
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4.5.8 MidAmerican Energy Company (MEC)

MidAmerican Energy Company proposed 27 projects at an estimated cost of \$153 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the MEC area, all of these projects were recommended by MISO to be included in MTEP22 Appendix A at an estimated cost of \$153 million. Of these projects, three are Baseline Reliability and 24 are Other projects. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Baseline Reliability Projects:

Project 21874 Sub 17 Expansion to Add 345-161 kV Transformer

Project Description

Convert Sub 17 161 kV yard to a ring bus. Loop the Sub 49-Beaver Channel 161 kV line into Sub 17. Add a new short 345 kV line connecting Cordova and Sub 17 and a new 345-161 kV transformer at Sub 17. The total estimated cost of this project is \$23 million dollars. The expected in-service date for this project is December 31, 2024.



Figure 4.5.8-1: P21874 Geographic transmission map of project area

Project Need

Local 161 kV bus voltages identified as being below planning criteria minimum for NERC TPL-001 Category P6 N-1-1 contingencies in the MTEP21 reliability assessment.

Cont. Type	Limiting Element	Rating (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
P6	SUB 49 5	0.95	0.879	1.003
P6	SUB 17 5	0.95	0.881	1.011
P6	SUB17-49CAP5	0.95	0.881	1.011
P6	BVR CHN L1 5	0.93	0.907	.999
P6	BVR CHN L2 5	0.93	0.907	.999
P6	ALBANY 5	0.93	0.915	.992
P6	BELEVUM8	0.93	0.923	.956
P6	PARADIS8	0.93	0.925	.959
P6	ALBANY 6	0.93	0.925	.992
P6	BELLTAP8	0.93	0.925	.959

Table 4.5.8-1: 21874 Voltage performance drivers

Alternatives Considered

1) WPPI Proposed Alternatives

- A. New 161 kV circuit between Rock Creek and Sub 49 in open position on Quad Cities-Rock Creek 345 kV line; loop-in of Sub 49-Beaver Channel 161 kV circuit into Sub 17, adding one 161 kV terminal at Sub 17 and operating the two Sub 17-Sub 49 161 kV line segments as a single bifurcated circuit (which will require some short segments of new transmission near Sub 17 and Sub 49).

- B. New 161 kV circuit between Rock Creek and Sub 49 in open position on Quad Cities-Rock Creek 345 kV line
- C. New 161 kV circuit between Rock Creek and Sub 49 in open position on Quad Cities-Rock Creek 345 kV line. Loop in Sub 49-Beaver Channel 161 kV into Sub 17, with new 161 kV terminals at both Sub 49 and Sub 17
- D. 2nd 345:161 kV Transformer at ITCM's Rock Creek Substation

Alternative A was not selected as a bifurcated circuit would not adhere to MEC's construction standards. Alternatives B, C, and D were developed as part of the alternative study meeting, and none of these alternatives fully solved the identified voltage issues.

Project 21909 Sac City Switching Station & Rebuild of Sac County-Odebolt-Schaller 69 kV Line

Project Description

Construct new Sac City switching station to interconnect two MidAmerican 69 kV lines and one Corn Belt 69 kV line. The total estimated cost of this project is \$20 million dollars. The expected in-service date for this project is December 1, 2024.



Figure 4.5.8-2: P21909 Geographic transmission map of project area

Project Need

Low voltage occurs for loss of Sac County 161-69 kV Transformer.

Cont. Type	Limiting Element	Rating (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
NB13	LKVIEW MUNI8	0.95	0.946	.977
NB13	WALLLAK8	0.95	0.946	.977
NB13	LKVIEW TAP 8	0.95	0.948	.978

Table 4.5.8-2: 21909 Voltage performance drivers

Alternatives Considered

No alternatives were considered for this project.

Project 21914 Butler 161-69 kV Transformer

Project Description

Install 161-69 kV, 125 MVA Transformer at Butler Substation to accommodate a networked 69 kV system between MidAmerican's Butler, New Hartford and Shell Rock Substations and Corn Belt's Donald Feldman Switching Station. The project also includes upgrading 34.5 kV lines from Butler to New Hartford and New Hartford to Shell Rock to 69 kV lines. The total estimated cost of this project is \$25.7 million dollars. The expected in-service date for this project is December 1, 2024.

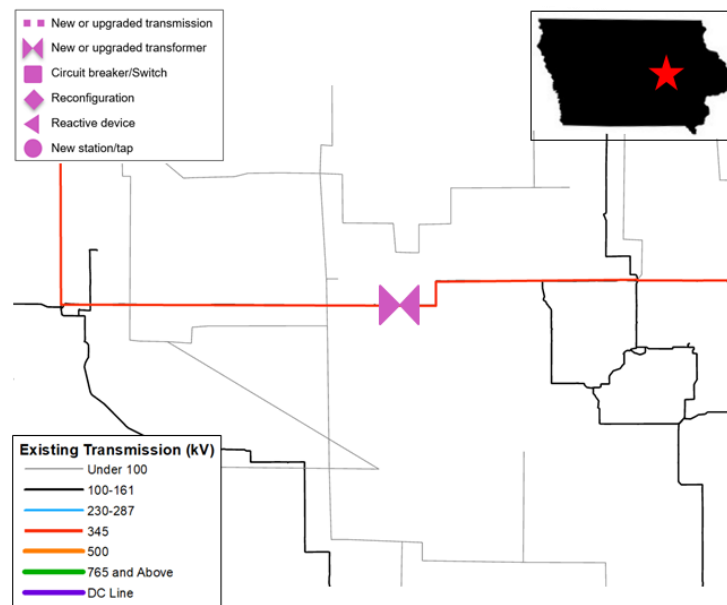


Figure 4.5.8-3: P21914 Geographic transmission map of project area

Project Need

To satisfy MidAmerican's 69 kV n-1-1 voltage reliability criteria as well as a 161 kV outage that causes the Butler Transformer to overload.

Cont. Type	Limiting Element	Rating (pu)	Pre-Project Voltage (pu)	Post-Project Voltage (pu)
NBP6	SHELLRCK8 69 kV	0.90	0.82	.98
NBP6	IANORTH8 69 kV	0.90	0.82	.98
NBP6	BAUMAN8 69 kV	0.90	0.82	.98
NBP6	FLINTHILLS8 69 kV	0.90	0.82	.98
NBP6	SHELLRCKJCT8 69 kV	0.90	0.82	.98
NBP6	WAVLY JC8 69 kV	0.90	0.82	.98
NBP6	IANORTHJCT8 69 kV	0.90	0.82	.98
NBP6	UNVERFRT8 69 kV	0.90	0.86	.98
NBP6	NEAL8 69 kV	0.90	0.89	.99

Table 4.5.8-3: 21914 Voltage performance drivers

Alternatives Considered

No alternatives were considered for this project.

Other Projects:

Projects Driven by Other Local Needs

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21872	Sub E East Moline-Sub 37 69 kV OPGW Installation	Install OPGW on the Sub E East Moline-Sub 37 69 kV Line	12/31/2024	\$1.0
21873	Sub 39-Sub 28 161 kV OPGW Installation	Install OPGW on the 161 kV line from Sub 39-Sub 28	12/31/2023	\$0.7
22305	Lakefield Jct to Obrien 345 kV Line Structure Replacements	The project will replace some line structure on MidAmerican owned part of Lakefield to Obrien 345 kV line. This project will increase the line rating of MidAmerican portion of Lakefield to Obrien 345 kV.	12/31/2022	\$2.2
22747	Interstate Containment Structures Installation Blanket	Based on recent experience from storms and its effects on transportation, installation of containment structures on either side of heavily traveled interstate and similar highway crossings is planned. The structures will reduce the likelihood of failure. If a severe storm occurs, the containment structures will reduce cascading and enable faster cleanup of the highway because the conductor can be cut free as it will not be under high tension.	12/31/2022	\$13.0

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21893	Neal North 161-69 kV Transformer Replacement	Replace Neal North 161-69 kV Transformer AC2 (83.3 MVA unit) which recently failed with a spare unit. Purchase a new 161-69 kV transformer at Neal North as a spare unit.	12/31/2021	\$1.6
21895	Sycamore 161-69 kV Transformer Replacement	Replace the failed 161-69 kV transformer at Sycamore substation with a spare unit. Purchase a new 161-69 kV spare transformer at Sycamore substation.	9/30/2022	\$3.0
21898	Increase Bus Withstand Capabilities	At Raun 161 kV, Neal North 161 kV, CBEC 161 kV, Sycamore 161 kV and Sycamore 69 kV, replace insulators, install new bus supports and/or replace bus to increase bus withstand capabilities.	12/1/2024	\$6.5
21907	Replace structures on Templeton-Audubon North-Exira 69 kV line to increase line rating	Replace terminal equipment at Audubon North and replace structures on Templeton-Audubon North-Exira 69 kV line to increase the line rating.	6/1/2023	\$1.6
21913	Northgate Substation: Ring bus expansion and 161 kV tie line to ITCM's Rose Hollow Substation	Expand Northgate Substation to a ring bus configuration and build 161 kV tie line from Northgate Substation to ITCM owned Rose Hollow Substation.	12/1/2024	\$3.5

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21871	Sub E East Moline 69 kV Upgrades	Replace aged equipment at Sub E East Moline. Project includes installing a new control building, replacing 69 kV oil circuit breakers, and replacing 69 kV cap-and-pin switches and insulators.	12/31/2024	\$7.0
21901	Raun Substation: 345 kV Switch Replacements	Replace 10 switches due to condition.	12/1/2022	\$0.9
21906	Sheldon-Boyden 69 kV Line Uprate	Replace structures and jumpers on Sheldon-Boyden 69 kV line to increase line rating.	12/1/2022	\$0.6
21908	Carroll County-Auburn-Sherwood 69 kV Line Rebuild & Terminal Upgrades	Rebuild 69 kV lines from Carroll County to Auburn and Auburn to Sherwood and replace line structures.	12/1/2024	\$7.7
21910	Hills 345 kV Equipment Replacements	Replace 345 kV Breakers and switches at Hills substation due to condition and operational issues. Replace relaying on two Hills lines and remote ends. The project will be completed in conjunction with MTEP Project 20121.	12/31/2022	\$5.4

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21911	IC Enron Substation: 161 kV Line Switch Additions	Add line switches at IC Enron Substation.	12/31/2022	\$0.3
21912	Substation J 161 kV Capacitor Bank Replacement	Replace 161 kV Capacitor due to condition and operational failures.	12/31/2022	\$0.5
22746	Overland Trail to Fallow Avenue 345 kV line structure replacements	Replace five line structures on Overland Trail to Fallow avenue 345 kV line.	12/31/2022	\$1.3
23480	Substation 18: Replace Failed 161/69 kV Transformer 1	Replace transformer that was failed in May 2022. Purchase replacement spare transformer for the unit being replaced at Sub. 18	12/31/2023	\$3.4

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21896	NE 94th Avenue 161-13.2 kV Substation	Construct NE 94th Ave 161-13.2 kV substation and taps	6/1/2023	\$6.7
21897	Berwick 161-13.2 kV Substation	Construct Berwick 161-13.2 kV substation and taps	6/1/2025	\$6.7
21902	Anthon Substation	Construct a new 69-12.47 kV distribution substation and associated 69 kV line work.	12/1/2023	\$1.5
21904	Sioux Center Substation	Construct a new 69-12.47 kV distribution substation and associated 69 kV line work.	12/1/2023	\$0.6
23366	Powell Creek 161-13 kV Substation	Construct a new Powell Creek 161-13 kV distribution substation tapping the Buena Vista-Sac County 161 kV line.	1/2/2024	\$5.5
23516	Enterprise Way South Substation	Construct new 161-13.2 kV distribution substation bisecting the existing Sub 56-Enterprise Way 161 kV line.	9/1/2024	\$3.5

4.5.9 Minnesota Municipal Power Agency (MMPA)

Minnesota Municipal Power Agency did not submit any new projects for MTEP22. MISO has not identified any issues in MMPA's area.

4.5.10 Minnesota Power (MP)

Minnesota Power proposed ten projects at an estimated cost of \$102 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the TO area, all of these projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, four Baseline Reliability Projects, and the remaining six are Other projects. All the projects' expected in-service dates and estimated costs are provided as of September 30, 2022.

Baseline Reliability Projects:

Project 17868 Duluth 115 kV

Project Description

Construct new 115 kV lines: Hilltop - Haines Road 115 kV and Haines Road - Ridgeview 115 kV.

Combine existing Arrowhead - Haines Road 115 kV and Haines Road - Swan Lake 115 kV lines to create Arrowhead - Swan Lake 115 kV Line. Expand Ridgeview Substation to a ring bus which will allow two additional transmission line connections and add a 115 kV line entrance at Hilltop and add a breaker at Haines Road. This project fixes thermal overloads that occur from P3 and P6 contingencies. The estimated cost for this project is \$47 million dollars, with an expected in-service date of 12/31/2026.

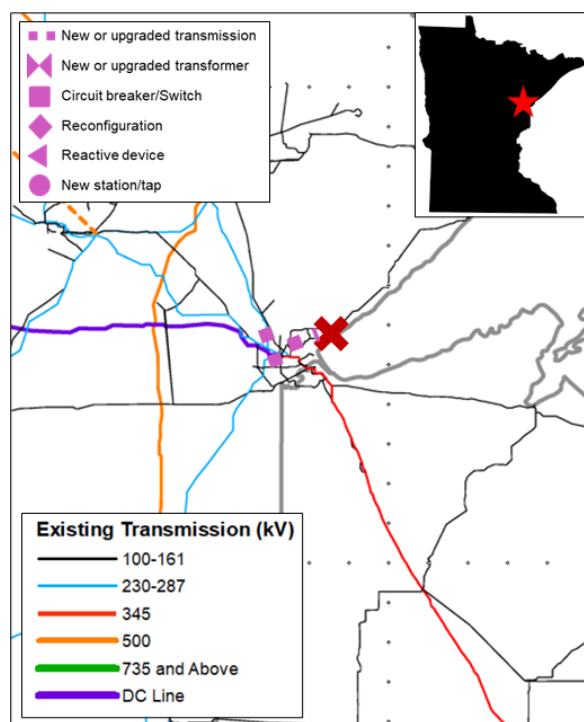


Figure 4.5.10-1: P17868 Geographic transmission map of project area

Project Need

This project is resolving many voltage collapse issues that are being masked by Taconite Harbor Generation being online and issues that occur from the loss of connection to Arrowhead Substation.

Project fixes thermal overloads on Arrowhead - Colbyville 115 kV and several other thermal overloads that occur from P3 and P6 contingencies in that area.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	608673 'ARROWHD7' 115.000 - 608688 'COLBYVL7' 115.000 1	115	109	91
P6	608673 'ARROWHD7' 115.000 - 608674 'HANESRD7' 115.000 1	115	108	87

Table 4.5.10-1: P17868 Thermal loading drivers

Alternatives Considered

No alternatives were considered for this project.

Project 20077 Duluth 115 kV

Project Description

Upgrade Hilltop 1TR to 373 MVA base 230/115 kV transformer. Add 230 kV breaker to high side of transformer. Disconnect Hilltop Tap from 3-Terminal 230 kV line and extend this tap 0.7 Miles to a new line entrance at Arrowhead 230 kV.

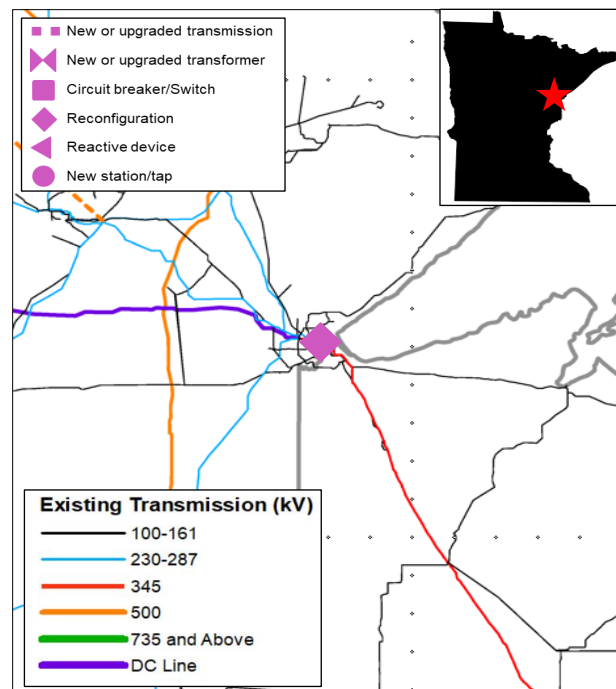


Figure 4.5.10-2: P20077 Geographic transmission map of project area

Project Need

Project eliminates Arrowhead and hilltop transformer thermal overloading issues for breaker failure contingencies that occur when Taconite Harbor generation is turned off.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	608615 'ARROWHD4' 230.000 - 608624 'FORBES 4' 230.000 1	230	139	56
P6	608673 'ARROWHD7' 115.000 - 618001 'GRE-BERGNT7' 115.000 1	115	139.8	56
P6	608673 'ARROWHD7' 115.000 - 608679 'GARY 7' 115.000 1	115	145	61

Table 4.5.10-2: P20077 Thermal loading drivers

Alternatives Considered

No alternatives were considered for this project.

Project 21608 105 & 106 Line Upgrade

Project Description

Rebuild and upgrade the two short parallel Blackberry - Iron Range 230 kV Lines

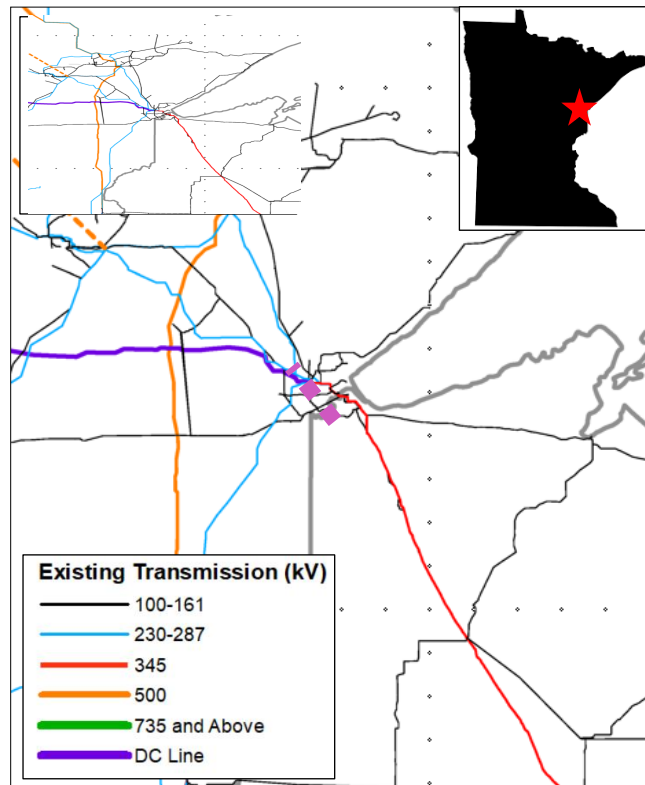


Figure 4.5.10-3: P21608 Geographic transmission map of project area

Project Need

105 & 106 Line upgrades resolve P12 and P6 issues from the MTEP21 results and fix many thermal overloads. This project solves real time operation issues that occurred due to summer drought conditions in Manitoba loss of either IRR-BBY 230 kV Line.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P12	608622 'IRONRNG4' 230.000 - 608625 'BLCKBRY4' 230.000 2	230	99	62
P12	608622 'IRONRNG4' 230.000 - 608625 'BLCKBRY4' 230.000 2	230	99	62
P24	608622 'IRONRNG4' 230.000 - 608625 'BLCKBRY4' 230.000 2	230	108	62
P6	608622 'IRONRNG4' 230.000 - 608625 'BLCKBRY4' 230.000 2	230	114	89

Table 4.5.10-3: P21608 Thermal loading drivers

Alternatives Considered

No alternatives were considered for this project.

Project 21686 Mesaba Junction 137 Line Extension

Project Description

Add breaker and transmission line entrance to Mesaba Junction Switching Station. Extend existing Embarrass - Babbitt 115 kV Line to switching station and establish a normal open between the Babbitt Tap and the Argo Tap.

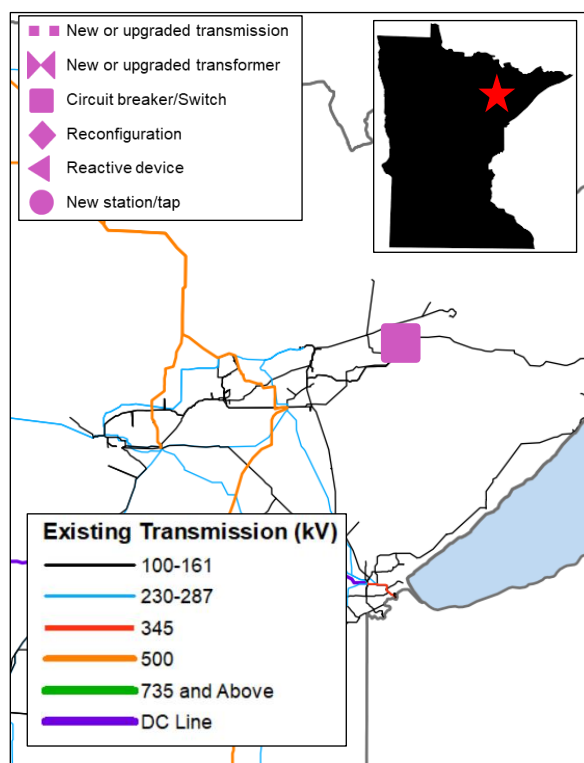


Figure 4.5.10-4: P21686 Geographic transmission map of project area

Project Need

Project provides redundancy to an industrial load pocket that requires near-constant availability. Improves reliability by providing to independent sources to the Babbitt area. Enables asset renewal by allowing the existing Embarrass-Babbitt 115kV Line to be taken out of service and improved. Project solves P2-1 and P6 low voltage issues on Argo and Babbitt substations.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Voltage (MVA)	Post-Project Voltage (MVA)
P2-1	608972 'ARGOSUB7' 115.000	225	1.290508	1.092
P2-1	608973 'NSM1SUB7' 115.000	115	1.290391	1.091
P2-1	608974 'NSM2SUB7' 115.000	115	1.290093	1.091
P2-1	608705 'BABBITT7' 115.000	115	1.265539	1.087
P6	608970 'BBT TAP7' 115.000	115	1.257186	1.072
P6	608971 'ARGOTAP7' 115.000	115	1.255853	1.068

Table 4.5.10-4: P21686 Thermal loading drivers

Alternatives Considered

No alternatives were considered for this project.

Other Projects:

Projects Driven by Other Local Needs

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21766	West Cohasset Project	Add new 115-23 kV transformer at existing Boswell SES substation and rename substation	12/31/2024	\$0.5

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
18058	Square Butte - Arrowhead HVDC Line Hardening	Structure replacements to improve HVDC line resiliency and restorability at critical locations, including transportation and infrastructure crossings.	12/31/2023	\$14

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
20074	Two Islands 115 kV Project	Construction of new 115 kV switching station near the existing Tac Harbor Substation. Route North Shore-Tac Harbor and Mesaba Junction-Tac Harbor Ckt #1 & Ckt #2 into the new switching station. One capacitor bank will be added to the switching station and it will connect to a new GRE 115/69 kV Two Islands Substation.	12/31/2024	\$14.1

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21763	North Shore Transformer Addition	Add 115/14 kV transformer to existing North Shore Switching Station. Relocate 13K cap bank to make room for transformer.	12/31/2022	\$0.2
21766	126 Line Asset Renewal	Replace transmission line structural components on Little Fork - I. Falls 115 kV Line, replace Little Fork 126L breaker, and replace relay panels on both ends of 126 Line.	12/31/2023	\$0.9
21605	Hibbing Transmission Exits	Rebuild two 115 kV lines coming into Hibbing Substation on double circuit structures and realign entrances to coordinate with Hibbing Substation Modernization Project (MTEP #18064).	12/31/2025	\$2

4.5.11 Missouri River Energy Services (MRES)

Missouri River Energy Services proposed two projects at an estimated cost of \$8.38 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the TO area, the projects were recommended by MISO to be included in MTEP22 Appendix A. Both of these projects are classified as Other projects. Missouri River Energy Services did not propose any Baseline Reliability projects or Generator Interconnection projects. The projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Other Projects:

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21938	Worthington Sub 2 Breaker	A breaker will be added to Worthington Public Utilities (WPU) Sub #2. The breaker will protect for a fault between Sub #2 and Elk.	4/1/2023	\$0.38

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
23346	Hutchinson Substation	This project will add a second 115/69 kV transformer to the existing City of Hutchinson MN, Hutchinson Substation. The project will also rebuild the Hutchinson 115 kV into a four position ring bus.	12/31/2025	\$8

4.5.12 Montana-Dakota Utilities Co. (MDU)

Montana-Dakota Utilities Co. proposed three projects at an estimated cost of \$22.1 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the TO area, the projects were recommended by MISO to be included in MTEP22 Appendix A. All three of these projects are classified as Other projects. Montana-Dakota Utilities Co. did not propose any Baseline Reliability projects or Generator Interconnection projects for MTEP22. The projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Other Projects:

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21889	Hettinger 115/69/41.6	Rebuild Hettinger Substation 115/69/41.6 due to age and condition.	12/31/2024	\$6.5
21894	Retire Tioga 1	Retire Tioga 1, expand Tioga 2 to move the 60 kV lines from Tioga 1 to Tioga 2 along with a 115/60 kV transformer, move 115 kV lines position in station and add another 115 kV breaker.	12/31/2024	\$7.05

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
20946	Poplar2 115/60 kV	Build a Poplar Oil Field 115/60 kV sub and a 115 kV line from the Poplar Trans sub to the Poplar Oil Field sub to tie into the 60 kV system near MDU's Poplar Oil Field load.	12/31/2024	\$8.53

4.5.13 Muscatine Power and Water (MPW)

Muscatine Power and Water did not submit any new projects for MTEP22. MISO has not identified any issues in MPW area.

4.5.14 Northwestern Wisconsin Electric (NWECE)

Northwestern Wisconsin Electric did not submit any new projects for MTEP22. MISO has not identified any issues in NWECE area.

4.5.15 Otter Tail Power Company (OTP)

Otter Tail Power proposed five projects at an estimated cost of \$12.5 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the Otter Tail Power area, all of these projects were recommended by MISO to be included in MTEP22 Appendix A categorized as two MPFP and three Other projects. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Other Projects:

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21427	OTP Max Water Treatment Plant 41.6 kV	This project consists of adding a one 41.6 kV circuit breaker at Max.	8/31/2022	\$0.3

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
22606	OTP Jamestown, ND Load Addition	OTP will install a new 115/41.6 kV 24 MVA transformer at the Jamestown 345 kV substation. The 115 kV bus will be expanded to allow for the new 115/41.6 kV transformer. The existing 41.6 kV load will be moved to be served off the new 115/41.6 kV 24 MVA transformer.	11/1/2023	\$2

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
20385	OTP Buffalo to Alice 115 kV Rebuild	Rebuild 7 miles of 115 kV line due to poor performance.	11/1/2024	\$2
20565	Forman 230/115 kV uprate (EPR)	Increase the capacity of the Forman 230/115 kV transformer and the Forman - Forman WAPA 115 kV line.	10/9/2024	\$6
23148	OTP Fergus Falls & Canby Area MPFPs	OTP has been requested by a Market Participant (Market Participant Funded Project - MPFP) to increase the capacity of the following Facilities: Canby - Granite Falls 115 kV, Hoot Lake - Fergus Falls 115 kV, and Hoot Lake - Fergus Falls SE 115 kV.	6/30/2023	\$1.7

4.5.16 Rochester Public Utilities (RPU)

Rochester Public Utilities did not submit any new projects for MTEP22. MISO has not identified any issues in RPU area.

4.5.17 Southern Minnesota Municipal Power Agency (SMMPA)

Southern Minnesota Municipal Power Agency did not submit any new projects for MTEP22. MISO has not identified any issues in SMMPA area.

4.5.18 Wilmar Municipal Utilities (WMU)

Wilmar Municipal Utilities did not submit any new projects for MTEP22. MISO has not identified any issues in WMU area.

4.5.19 WPPI Energy (WPPI)

WPPI Energy did not submit any new projects for MTEP22. MISO has not identified any issues in WPPI area.

4.5.20 Xcel Energy (Northern States Power)

Xcel Energy proposed 19 projects at an estimated cost of \$169.4 million targeting MTEP22 Appendix A. After MISO's independent reliability analysis of the Xcel Energy area, all of these projects were recommended by MISO to be included in MTEP22 Appendix A. Of these projects, two are Baseline Reliability projects and 17 are Other projects. All the projects' expected in-service dates and estimated costs are provided as of September 15, 2022.

Baseline Reliability Projects:

Project 20505 - Blue Earth-South Bend Area Upgrades

Project Description

Rebuild approx. 30 miles of 161 kV line and upgrade South Bend TR6 to a 336 MVA transformer to resolve thermal overloads under a P7 contingency. This project has an estimated in-service date of April 15, 2025.

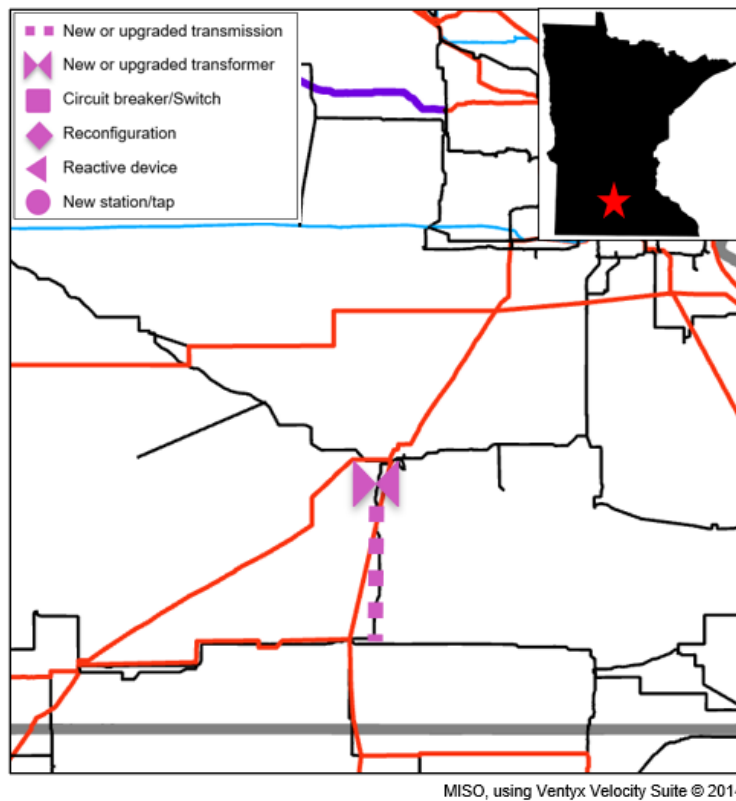


Figure 4.5.20-1: P20505 Geographic transmission map of project area

Project Need

This project resolves thermal overloads on the upgraded transmission elements for a nearby P7 contingency on the 345 kV system. The violations occur in the shoulder high wind and light load high wind cases.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P7-1	South Bend Transformer 161/115 kV	215	143	79
P7-1	South Bend - Blue Earth 161 kV	254	119	57
P7-1	Blue Earth - Huntley 161 kV	254	122	71

Table 4.5.20-1: P20505 Thermal loading drivers

Alternatives Considered

No alternatives were considered for this project.

Project 22686 - Briggs Road - LaCrosse Upgrade

Project Description

Reconductor of Briggs Road - Mayfair - LaCrosse line for age and condition. Upgrade to larger size to remediate N-1-1 thermal overloads. This project has an estimated in-service date of May 15, 2023.

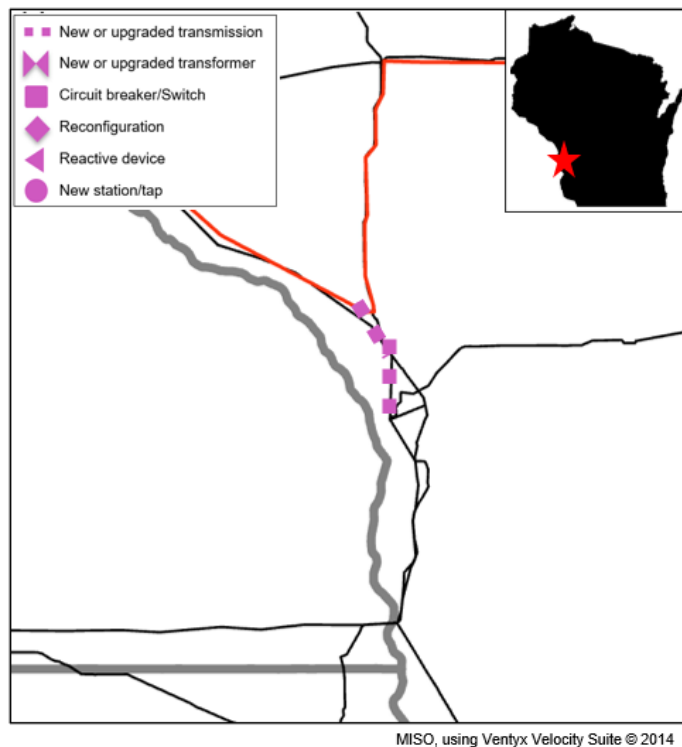


Figure 4.5.20-2: P22686 Geographic transmission map of project area

Project Need

This project resolves thermal overloads on the upgraded transmission elements for several P6 contingencies in the summer 2024 case. The violations appear in the case when the suspended Genoa 3 coal unit is turned off and the model is re-dispatched per MTEP22 tier order.

Cont. Type	Limiting Element	Rating (MVA)	Pre-Project Loading (%)	Post-Project Loading (%)
P6	Briggs Rd - Mayfair 161kV	256	116	89
P6	Mayfair - LaCrosse 161kV	218	115	76

Table 4.5.20-2: P22686 Thermal loading drivers

Alternatives Considered

No alternatives were considered for this project.

Other Projects:

Projects Driven by Other Local Needs

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21866	Sleepy Eye Cap Bank Retirement	This cap bank is no longer needed to support the area and will be retired to prevent unnecessary maintenance costs on the system.	6/1/2023	\$0.1
21868	Canistota Junction Cap Bank Retirement	This cap bank is no longer needed to support the area and will be retired to prevent unnecessary maintenance costs on the system.	6/1/2023	\$0.1
21869	BRI CHB 34.5 kV Cap Bank Retirement	Retire 34.5 kV cap banks at Buffalo Ridge/Chanarambe feeders BRI311, BRI312, and CHB341.	6/1/2023	\$0.25
22629	Wheaton Sub Tie	Project will tie Wheaton generation bus 1 to bus 2, effectively moving TR3 to the same bus as TR1 and TR2.	6/1/2023	\$1.5

Projects Driven by Local Reliability

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21285	Elm Creek TR10	Install second 345/115 kV transformer at Elm Creek sub. Route SHC-CNC #1 345 kV Line Into Elm Creek Sub. Add 50 MVAR reactor to existing TR9.	6/1/2023	\$9.75
21834	Boyd Sub Removal	Retire end of life Boyd sub and install two 5 MVAR capacitor banks at nearby Bateman sub.	6/12/2023	\$0.5
21846	Hyland Lake TR1 and TR2 Upgrade	Upgrade TR1 and TR2 to 115/13.8 kV 70 MVA transformers at Hyland Lake Substation. Install high-side line breaker. Install new feeder bay.	10/15/2022	\$6.67
21848	Willmar 230/115 kV Interconnection	Build new dead-end structure to re-terminate the 0823 line to a southern position at the Willmar sub to accommodate GRE's substation reconfiguration.	July 9, 2022	\$0.8
21853	North Fork Sub Expansion	Retire the Flambeau breaker station and will add MOD's to the switches at North Fork substation.	12/1/2023	\$1.5
21877	Coon Creek Substation 345 kV Breaker Additions	Install three 345 kV circuit breakers at the Coon Creek (CNC) substation to provide a better isolation between the Main Bus 1 & 2 to the line 1 & 2.	12/31/2023	\$2.7
21887	Rogers Lake Breaker Addition	Install two breakers at Rogers Lake to separate Airport - Rogers Lake 115 kV line and Rogers Lake TR2.	2/1/2022	\$1.9

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
23452	Brookings – Lyon, Hampton – Helena 2 nd 345kV Circuits	Install second 345 kV circuit between the Brookings County and Lyon County substations. Install second 345 kV circuit between the Hampton Corner and Helena substations. Perform substation upgrades associated with installation of line.	9/1/2025	\$48.7

Projects Driven by Age and Condition

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21833	Mallard Facility Rebuild	Joint NSP - Central Power facility rebuild. NSP substation grading and fencing, 3 new box structures, control and wiring modifications to accommodate Central Power equipment moves.	5/15/2023	\$2.3
21845	NSPM Metro Steel Pole Replacement	Address painted poles concerns between Riverside Sub and Main Street Sub. Approximately 4-miles of triple circuit structure(35-structures) to be either replaced, painted or a combination of replace/paint.	12/15/2024	\$20
21888	Line 0761 Rebuild	Wreck-out 19.7 miles of existing 69 kV on line 0761 rebuild to current 69 kV Standards.	12/15/2022	\$8.5
21891	Line 0726 Rebuild	NSM 0726-69 kV Line Major line rebuild from Pipestone to Tracy Switching Station.	12/21/2024	\$17.1

Projects Driven by Load Growth

Project ID	Project Name	Project Description	ISD	Estimated Cost (\$M)
21892	West Shakopee Interconnection	City of Shakopee T-L Interconnection Request. In-and-out off NSP's Scott County - Dean Lake 115 kV transmission line. Joint substation, NSP will own the high side.	6/1/2023	\$3

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